



**BASE24 IMNi/IMNsc
Interface Manual**

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Preface

This manual describes the IMN interface processes “IMNi” and “IMNsc” for both POS and ATM systems version 6.0, respectively. These interfaces are used to establish a communications link between a BASE24 network and the Inter-Member Network (IMN). This manual is intended as a reference for resources responsible for the interface between BASE24 and IMN. This manual provides a comprehensive description of all the functions performed by the IMNi/IMNsc Interface process involving incoming and outgoing message traffic for both acquirer and issuer processors.

Reading Prerequisites

While no firm prerequisites exist for reading this manual, it is helpful if readers are somewhat knowledgeable about BASE24 and BASE24 files maintenance.

Document Location

This document may be printed using Microsoft Word and copied as needed by ACI customers who have purchased the IMNi/IMNsc Interface software. The document may also be ordered, at a cost, through ACI Publications.

Section 1

Introduction

The IMNi/IMNsc is a switch interface application that is responsible for formatting all messages according to the Inter-Member Network (IMN) standard – ISO 8583 version 87 format. In addition, an IMN header must be placed on all messages sent to the other member application.

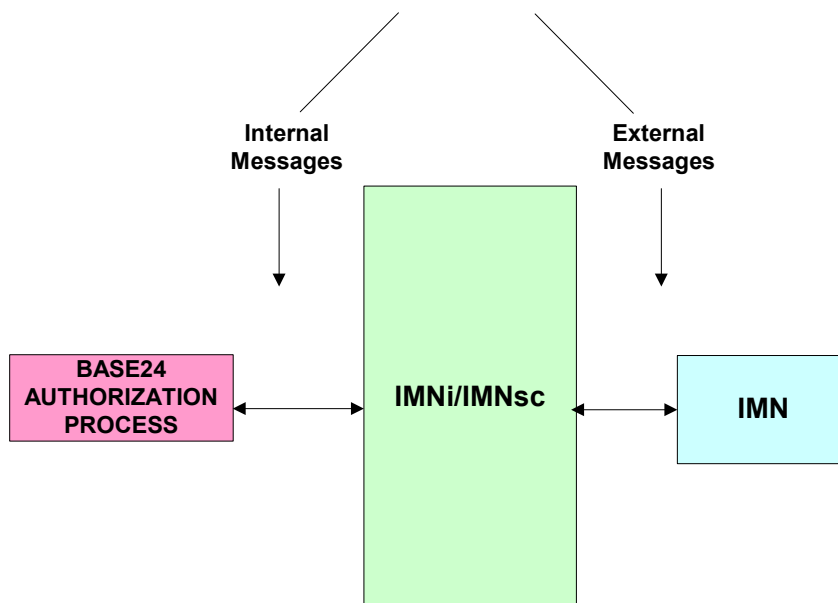
IMNi is a BASE24-pos interchange facility that allows connection to the Interac Direct Payment (IDP) service of the Interac Inter-Member Network (IMN).

IMNsc is a BASE24-atm interchange facility that allows connection to the Shared Cash Dispensing (SCD) service of the Interac Inter-Member Network (IMN).

This section introduces the BASE24-pos/BASE24-atm software IMNi/IMNsc and provides an overview of their functionality. This section also identifies the BASE24 files used by IMNi/IMNsc, and the message types supported.

IMNi/IMNsc Overview

The IMNi/IMNsc processes, shown in the illustration, are responsible for the online message transfer between BASE24 and the IMN node.



IMNi performs the following functions:

- Facilitates message exchange between BASE24-pos and the IMN node.

When BASE24-pos receives a transaction from the IMN node, IMNi translates the message into the appropriate internal message format (PSTM) and sends the message to a BASE24-pos Authorization process. When BASE24-pos sends a transaction to the IMN node to authorize, IMNi translates the message from the internal message format to the ISO external message format (ISO 8583 Standard) and forwards the message to the IMN node for delivery to the remote node.

- Provides store-and-forward capabilities.

IMNi stores certain messages (0220, 0420, 0422, 06xx) intended for the IMN node in its Store-And-Forward file (SAF) to ensure delivery. IMNi attempts delivery of these messages from the SAF file, at intervals defined in LNCF parameters, until a satisfactory response arrives from the remote node.

- Performs key management and security functions.

Along with the IMNM process, IMNi manages security keys exchanged with other POS nodes on the IMN network. IMNi performs all security functions, including translation of encrypted PIN blocks message authentication using MACs and Encryption Message Key.

- Manages communications between BASE24-pos and the IMN node.

IMNi sends and receives network management messages to determine the status of the link between BASE24-pos and the IMN node. Network management messages include pacing messages.

IMNsc performs the following functions:

- Facilitates message exchange between BASE24-atm and the IMN node.

When BASE24-atm receives a transaction from the IMN node, IMNsc translates the message into the appropriate internal message format (STM) and sends the message to a BASE24-atm Authorization process. When BASE24-atm sends a transaction to the IMN node, IMNsc translates the message from the internal message format to the ISO external message format (ISO 8538 standard) and forwards the message to the IMN node for delivery to the remote node.

- Provides store-and-forward capabilities.

IMNsc stores certain messages (0420, 06xx) intended for the IMN node in its Store-And-Forward file (SAF) to ensure delivery. IMNsc attempts delivery of these messages from the SAF file, at intervals defined in LNCF parameters, until a satisfactory response arrives from the remote node.

- Performs key management and security functions.

Along with the IMNM process, IMNsc manages security keys exchanged with other SCD nodes on the IMN network. IMNsc performs all security functions, including translation of encrypted PIN blocks and message authentication using MACs.

- Manages communications between BASE24-atm and the IMN node.

IMNsc sends and receives network management messages to determine the status of the link between BASE24-atm and the IMN node. Network management messages include pacing messages.

IMN Manager (IMNM)

The IMNM is a BASE24 satellite process. The IMNM switch is mainly responsible for managing the transaction flow to multiple IMNi/IMNsc and manages key change. IMNi/IMNsc opens the IMNM by using the Inter-Process Communication (IPC) method and the IMNM also communicates with all the IMNi/IMNsc processes in the environment by using the IPC method. The IMNM also performs the following functions:

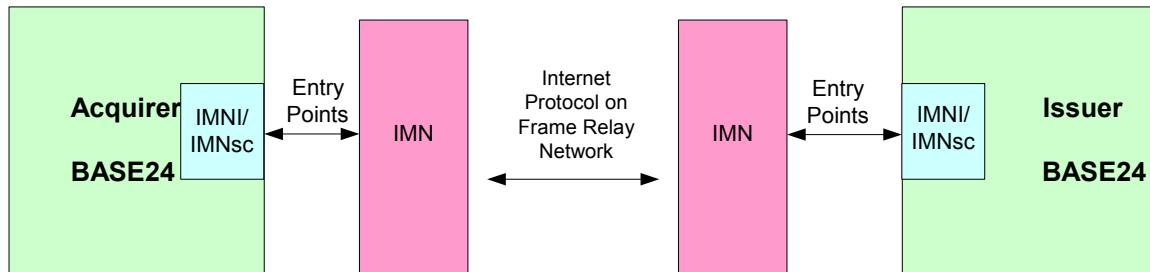
- Process cutover function.
- Set timer for daily key exchange, keep track of key usage counter and manage all the key exchange activities.
- Control Store-and-Forward messages and keep track of the SAF file usage size.
- Handle IMN Administrative Message.
- Process NCS operator commands.

Inter-member Network (IMN) Interface

The BASE24-atm IMNsc connects a BASE24-atm with the IMN node. The BASE24-pos IMNi connects a BASE24-pos with the IMN node. An entry point (EP) defines an interface between IMNi/IMNsc and an IMN node. Both IMNi/IMNsc use the Inter-Process Communication (IPC) method to EP for any request and response messages.

The Inter-Member Network (IMN) provides the Interac Association Members with a network for the exchange of electronic financial transactions. All messages sent to a remote IMN node are forwarded on the entry points defined for the local IMN node. Likewise, all messages received from a remote IMN node arrive at IMNi/IMNsc via the entry points. IMN obtains the information in the IMN headers to determine how to transport the transaction to proper node via the router-based

frame relay network. However, IMN will not inspect or modify the body of these messages being transported, only the IMN header.



Each IMN node is identified on the Inter-Member Network by an assigned 6-digit ID number (4-digit Member number and 2-digit node number) and by an equivalent 6-character node name.

To ensure the transaction path is valid for messages transportation, it is important to monitor the connection and entry point status using the Connection Summary (CNXNSUM) screen of the IMN node management sub-system. The IMN transaction path is made up of two main components: Communications Interface Layer (CIL) and Communications Control Layer (CCL). CIL routes messages to and from the entry points and the appropriate CCLs, gathers statistics and handles other control functions. CCL manages connection setup and communications over the multiple TCP/IP sessions (sockets).

Basic Definitions

Connection

A connection is a logical link between a requester and a responder, for a message class. A connection is defined as follows:

<requester node> <message class> <responder node>

IMN carries all requests from the requester to the responder, and all responses back from the responder to the requester.

Requester node

- Sends request into IMN.
- Usually, but not always, the acquirer end.

Message Class

- identifies the application type:

00	-	All message classes
01	-	IDP Debit

02	-	Test IDP Debit
03	-	SCD
04	-	Test SCD
05	-	Credit Transaction
06	-	Test Credit

Responder node

- Receives request from the IMN and returns response.
- Usually, but not always, the issuer end.

Entry point

An Entry Point (EP) defines an interface between the IMNi/IMNsc and the adjacent IMN node. The IMNi/IMNsc and IMN node communicate with each other through HP IPC, using a pair of inter-process OPENs. All messages sent to a remote IMN node are forwarded on the entry points defined for the local IMN node. Likewise, all messages received from a remote IMN node arrive at IMNi/IMNsc via the entry points.

Entry points are defined using the Entry Point Definition (EPDEFN) Screen in the IMN. They are monitored and managed from the Entry Point Summary (EPSUM) and Entry Point Control (EPCNTL) screens of the IMN node management subsystem. Both the IMNi/IMNsc and IMN use the same definition of the entry point. During initialization, IMNi/IMNsc gets the location of the Entry Point Configuration File (EPCF) from the LNCf assign "EPCF".

For information on defining entry points, please refer to the IMN Operations Manual.

IMN IIN routing table

The IMN Institution Identification Number (IIN) routing table contains card prefixes that uniquely identify card issuer institutions. These 10-digit prefixes are associated with particular institutions and used in card routing. IMNi/IMNsc uses this file to determine the corresponding issuer node number and assign outbound requests to the specific appropriate IMN connection. During initialization, IMNi/IMNsc gets the location of the IMN IIN routing table from the LNCf assign "IMN-BIN". For new node installation, member can download the IMN IN routing table from IMN Control Center (IMNCC).

Please refer to Interac's IMN Functional Specifications or ACI's IMN Manuals for more details on the IMN application.

Issuer and Acquirer

In the financial industry, the terms issuer and acquirer define the parties on opposite ends of the electronic message exchanges required to authorize and process transactions.

Issuers are card-issuing or account-owning institutions or their agents. These institutions or agents are responsible for authorizing cardholder transactions and keeping track of cardholder accounts.

Acquirers are the parties initiating transactions on behalf of the card acceptors.

An IMNi/IMNsc process operates as either an Issuer and Acquirer or both, depending on the entry points that are defined for the IMNi/IMNsc process. Entry points defined with message class 01 serves the IDP traffic and those defined with message class 03 serves the SCD traffic.

Terms

BASE24-atm Authorization - BASE24 atm authorization process

BASE24-pos Router/Auth - BASE24 point of sale system router and authorization process

HISO - Host ISO Interface process

IMN - Inter-member Network

IMNi - IMNi Interface process

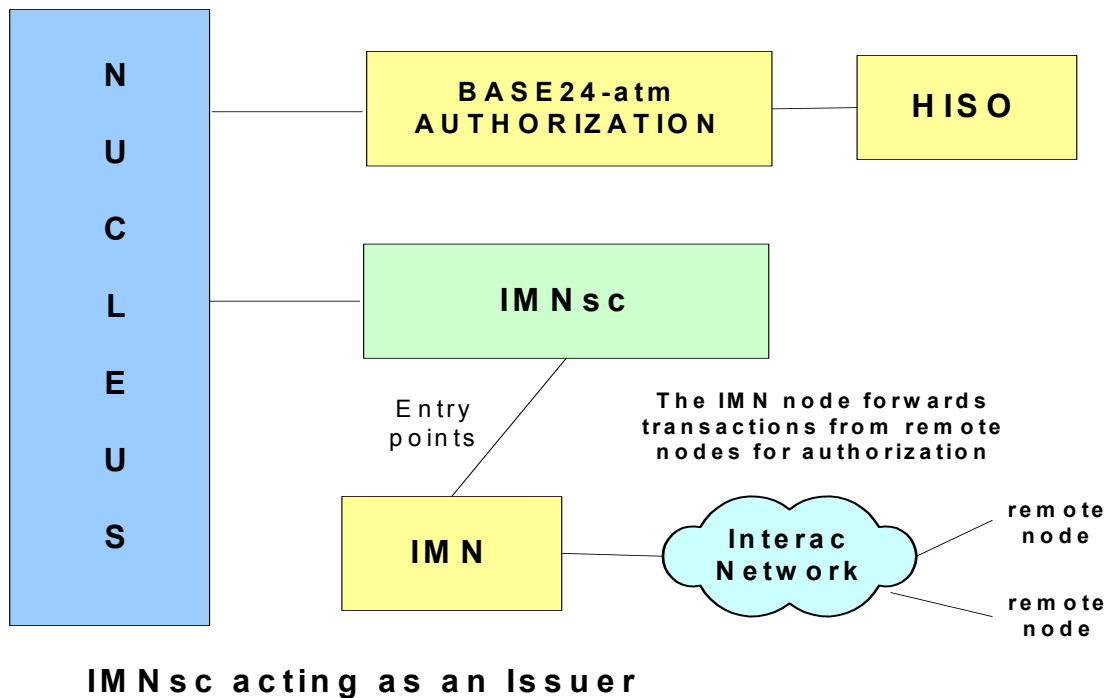
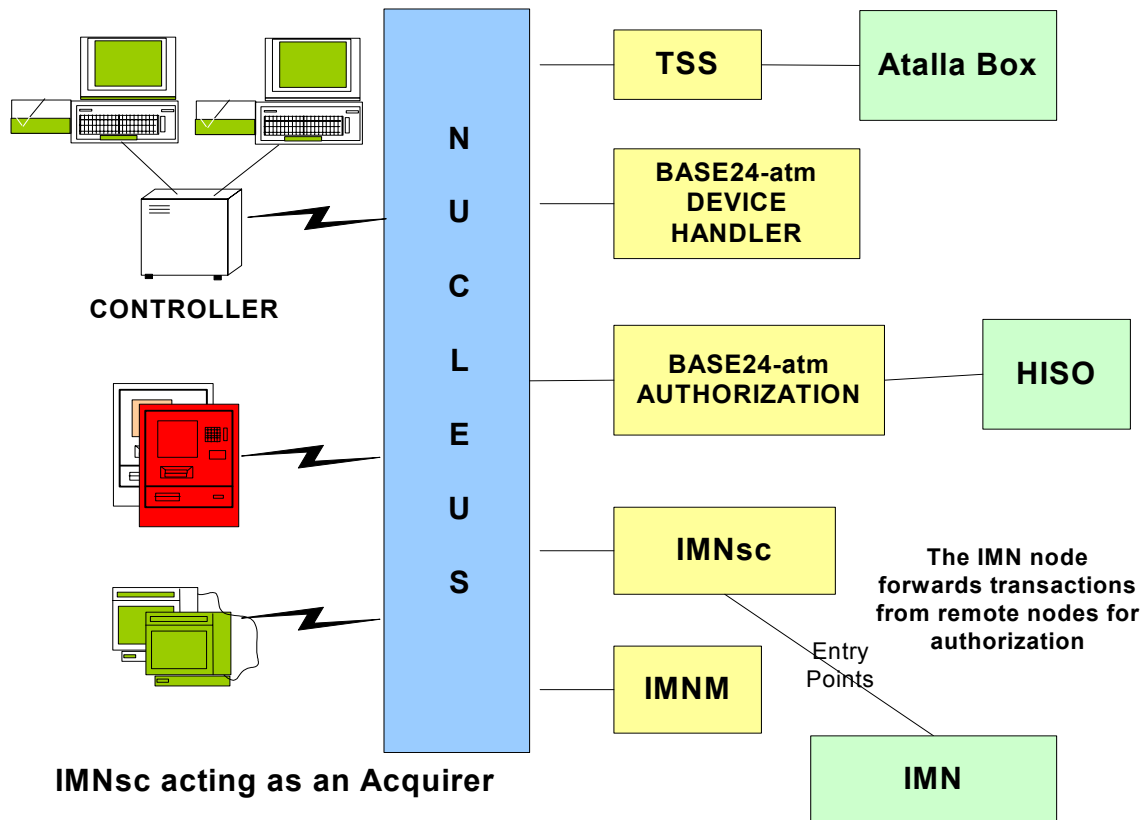
IMNM - IMN Manager process

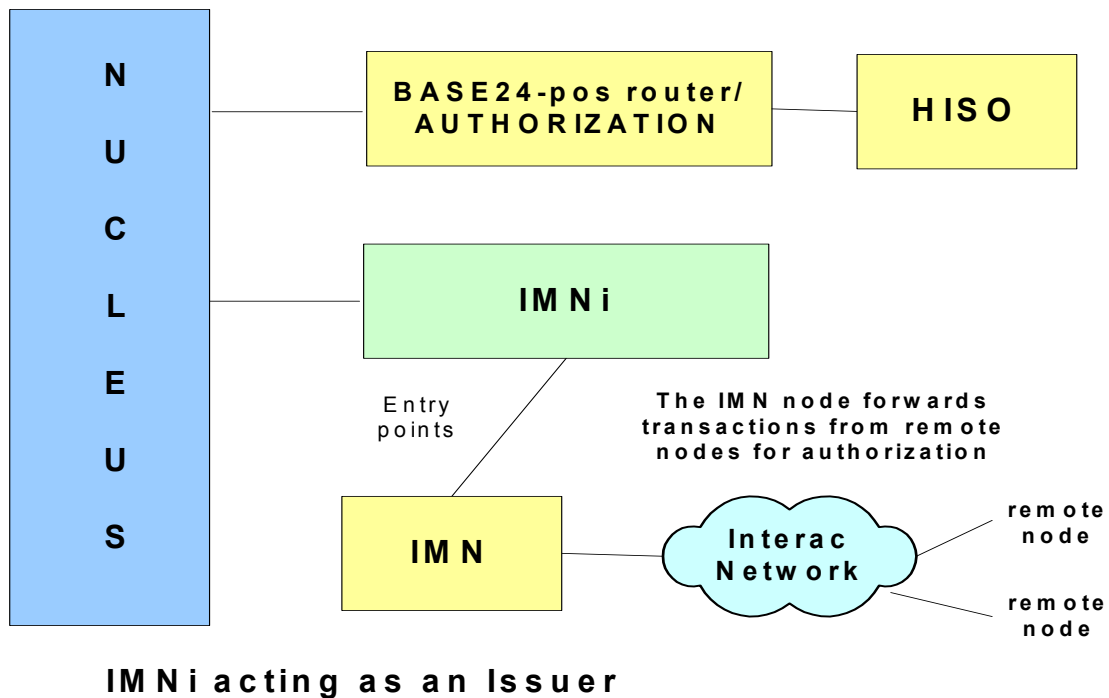
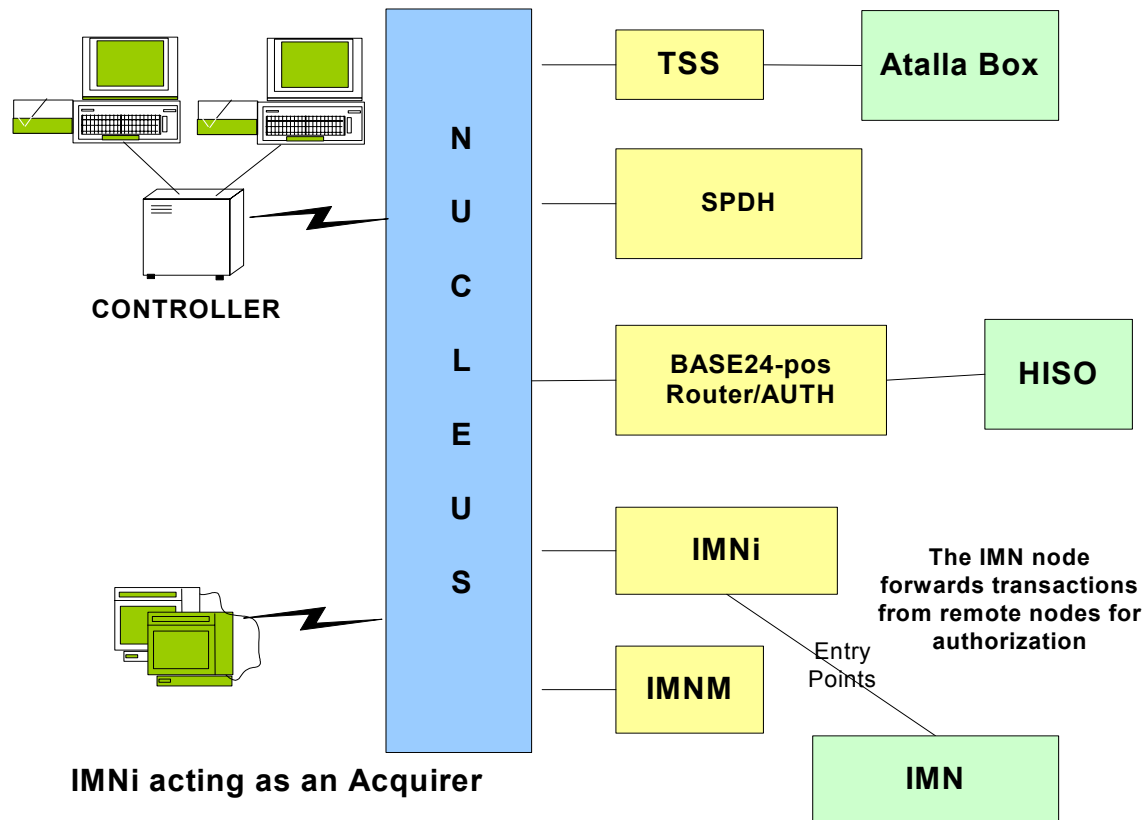
IMNsc - IMNsc Interface process

NUCLEUS - XPNET node

SPDH - Standard pos Device Handler

TSS - Transaction Security System





IMNi/IMNsc Message Format And Processing

Internal Message Format

The internal message format is used whenever information is routed between the IMNi/IMNsc process and the rest of BASE24-pos/BASE24-atm. Depending on the BASE24 product receiving the message, the internal message is commonly referred to as the BASE24-atm Standard Internal Message (STM) or BASE24-pos Standard Internal Message (PSTM).

For more information on the STM, please refer to the BASE24-atm Transaction Processing Manual. For more information on the PSTM, please refer to the BASE24-pos Transaction Processing Manual.

External Message Format

The structure and content of the external message used by the IMNi/IMNsc process to communicate with the IMN node is defined by Interac. The Interac IDP/SCD ISO external message (ISO 8538 standard, please refer to Appendix B for the message format) is a variable-length, variable-content message defined in the following structure:

IMN Header	Message Type Identifier	Primary Bit Map	Secondary Bit Map	Primary Data Fields	Secondary Data Fields
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Specifications for the IMNi external message can be found in the Interac IDP Application Interface Layer Functional Specifications, or IMN-based Interac IDP Service Operating Regulations. Specifications for the IMNsc external message can be found in the Interac SCD Application Interface Layer Functional Specifications, or IMN-based Interac SCD Service Operating Regulations.

Store-and-Forward Function

A store-and-forward function is incorporated into the IMNi/IMNsc process to manage transaction messages that are marked as “must complete”.

IMNi/IMNsc stores SAF messages for a node in Store-and-Forward (SAF) files, and attempts delivery at specified intervals until IMNi/IMNsc receives satisfactory responses. For more information on store-and-forward processing, see the section on Store and Forward Processing, later in this document.

Message Processing

For more information on how messages flow between the IMN node and IMNi/IMNsc, please refer to the following sections, based on the type of message:

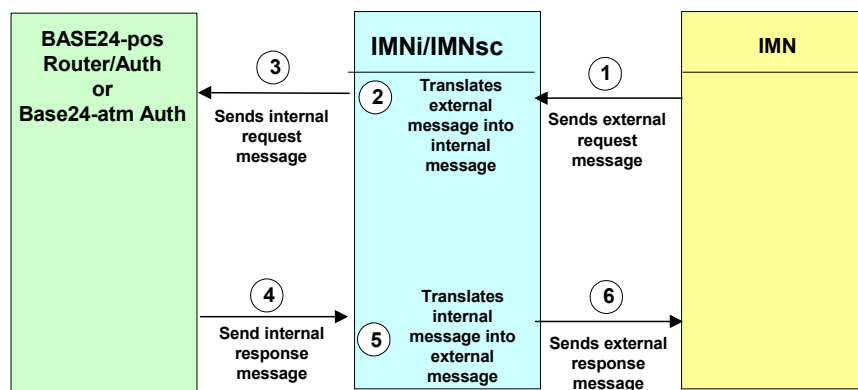
- Network management message flows, including those for dynamic key management, are documented in Section 4.

- Store-and-forward message flows are documented in Section 3.
- BASE24-pos/BASE24-atm transaction flows are documented in Section 4.

Normal Message Flow

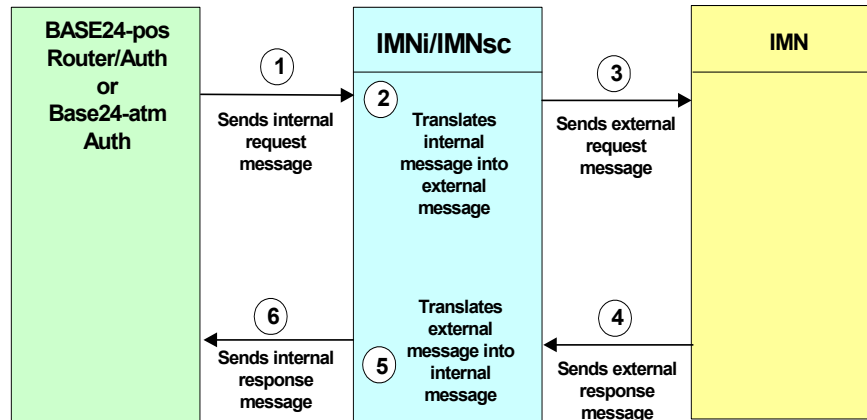
Issuer Processing

After receiving a message from the IMN node, IMNi/IMNsc translates this request message from the external format into the appropriate internal format. IMNi/IMNsc then sends the internal message to the appropriate BASE24-pos/BASE24-atm Authorization process. When the message has been processed, the authorizing process sends an internal response back to IMNi/IMNsc. Using information from the internal message, IMNi/IMNsc formats an external response message and sends this response message back to the IMN node. The message processing flow is described in the diagram below.



Acquirer Processing

After receiving a message from a BASE24-pos router/auth or BASE24-atm authorization process, IMNi/IMNsc translates this request message from the internal format into the external format, builds the IMN message header and determines the message destination using the IIN table. IMNi/IMNsc sends the external message to the IMN node to be forwarded to its destination for authorization. When the message has been processed, the IMN node returns an external response back to IMNi/IMNsc. Using information from the external message, IMNi/IMNsc formats an internal response message and sends the response message back to the BASE24-pos router/auth or BASE24-atm Authorization process. The message processing flow is described in the diagram below.



File Usage

The IMNi process uses a number of BASE24-pos files and the IMNsc process uses a number of BASE24-atm files. These files can be divided into control files and transaction files, and are identified below.

Control Files

Control files contain configuration information and parameters that determine how the IMNi/IMNsc process operates. The control files that contain configuration information for IMNi/IMNsc include:

- Entry Point Configuration File (EPCF)
- Interchange Configuration File (ICF)
- Key File (KEYF)
- Logical Network Configuration File (LNCF)
- Token File (TKN)
- IMN BIN Routing Table (IMNBIN)
- Log Message Configuration File (LMCF)

For information on how these files are used to configure the IMNi/IMNsc processes, please refer to the next section of this document.

Transaction Files

Transaction files are used to store information about individual transactions processed through IMNi/IMNsc.

The transaction files used by the IMNi/IMNsc include:

- Interchange Log File (ILF)
- Store-and-Forward File (IMNSAF)

- Administrative Messages File (IMNADMN)

For information on the contents of the IMNi/IMNsc ILF and SAF, please refer to the corresponding Data Definition Libraries (DDLs). The ILF and SAF DDLs can be accessed via DDLDOC.

Supported Message Types

IMNi/IMNsc supports the message types indicated below, between BASE24-pos/BASE24-atm and the IMN node.

Message Type	Description	IMNi	IMNsc
0100	Authorization Request	x	
0110	Authorization Request Response	x	
0120	Authorization Advice		
0121	Authorization Advice Repeat		
0130	Authorization Advice Response		
0200	Financial Transaction Request	x	x
0210	Financial Transaction Request Response	x	x
0220	Financial Transaction Advice	x	
0221	Financial Transaction Advice Repeat		
0230	Financial Transaction Advice Response	x	
0402	Card Issuer Reversal Request		
0412	Card Issuer Reversal Request Response		
0420	Acquirer Reversal Advice	x	x
0421	Acquirer Reversal Advice Repeat		
0422	System Generated (Issuer) Reversal Request	x	
0430	Reversal Advice Response	x	x
0432	Issuer Reversal Response	x	
0500	Acquirer Reconciliation Request		
0502	Issuer Reconciliation Request		
0510	Acquirer Reconciliation Request		

	Response		
0512	Issuer Reconciliation Request Response		
0520	Acquirer Reconciliation Advice		
0521	Acquirer Reconciliation Advice Repeat		
0522	Issuer Reconciliation Advice		
0523	Issuer Reconciliation Advice Repeat		
0530	Acquirer Reconciliation Advice Response		
0532	Issuer Reconciliation Advice Response		
0600	Administrative Acquirer Request	x	x
0602	Administrative Issuer Request	x	x
0610	Administrative Acquirer Response	x	x
0612	Administrative Issuer Response	x	x
0620	Administrative Acquirer Adjustment Advice Request	x	x
0622	Administrative Issuer Adjustment Advice Request	x	x
0630	Administrative Acquirer Adjustment Advice Response	x	x
0632	Administrative Issuer Adjustment Advice Response	x	x
0800	Network/Key Management Request	x	x
0810	Network/Key Management Request Response	x	x
0820	Network Management Advice		
0821	Network Management Advice Repeat		
0830	Network Management Advice Response		

Dynamic Key Management

IMNi/IMNsc uses working keys to protect transaction data moving between IMN nodes. Working keys are used to maintain the proper security to protect Personal Identification Numbers (PINs) and verify Message Authentication Codes (MACs). Working keys should be changed regularly to ensure security. The monitoring and changing of these keys is known as key management and can be performed manually using text commands or automatically using dynamic key management.

Dynamic key management allows IMN nodes to automatically change working keys based on thresholds set for time, number of transactions, and number of single or consecutive errors. One or more working keys can be changed when any one of the defined thresholds is surpassed.

Dynamic key management requires the use of hardware security modules to perform PIN encryption, PIN verification, message encryption (for IMNi only) and message authentication.

Fields in the Key File (KEYF), Interchange Security File (ISECD) and the Interchange Configuration File (ICF) must be set to specific values to allow for dynamic key management. Please refer to Section 2 for a listing of ICF, KEYF and ISECD settings.

Section 2

Configuration and Control

This section describes the configuration and control of the IMNi/IMNsc processes and includes the following:

- Installation and compilation of the IMNi/IMNsc switch interface software
- Descriptions of files that IMNi/IMNsc uses
- Descriptions of timers that IMNi/IMNsc uses and the processing performed when each timer expires
- Process configuration information
- Description of institution-level and IMN node-level controls

Installation

To Install BASE24 Release 6.0 IMNi and IMNsc switch interface software, add and configure the following BASE24 6.0 components:

- IMNi/IMNsc files
- IMNA & ICF Pathway requesters and servers
- IMNi, IMNi Manager, IMNsc and IMNsc manager XPNET Satellite processes
- LNCF parameters and assigns
- Data files
- ICF records
- KEY records
- ISECD

For more details on installation, please refer to BASE24 IMNi/IMNsc Installation Guide.

IMNi Software

SW60IMNI

IMNACM	IMNI Admin Cleanup MAKE
IMNACS	IMNI Admin Cleanup Source
IMNEXTEM	IMNI MAKE for External File Generation
IMNIARM	IMNI Alternate Routing MAKE

IMNIARS	IMNI Alternate Routing Source
IMNICMDM	IMNI NCS Operator Command Processor MAKE
IMNICMDS	IMNI NCS Operator Command Processor Source
IMNID	IMNI EMV Dummy Source file
IMNIDCLS	IMNI Global Data Declarations
IMNIDDL	IMNI DDL MAKE
IMNIDDL	IMNI DDL Source
IMNIEMVD	IMNI EMV Dummy Source file
IMNFIOM	IMNI File I/O MAKE
IMNFIOS	IMNI File I/O Source
IMNIINIM	IMNI Initialization Procedure MAKE
IMNIINIS	IMNI Initialization Procedure Source
IMNILFXG	ILF Extract Global
IMNILFXM	ILF Extract MAKE
IMNILFXS	ILF Extract Source
IMNIM	IMNI MAKE
IMNIMLM	IMNI Mainline MAKE
IMNIMLS	IMNI Mainline Source
IMNIOPTS	IMNI Compile Options
IMNIPSTM	IMNI PSTM Processor MAKE
IMNIPSTS	IMNI PSTM Processor Source
IMNISCNS	IMNI Screen Source
IMNISEMM	IMNI SEM Processor MAKE
IMNISEMS	IMNI SEM Processor Source
IMNIUTLM	IMNI Utility Procedures MAKE
IMNIUTLS	IMNI Utility Procedures Source
IMNMCMDM	IMNM NCS Operator Command Processor MAKE
IMNMCMDS	IMNM NCS Operator Command Processor Source
IMNMDCLS	IMNM Global Data Declarations
IMNMINIM	IMNM Initialization Procedures MAKE
IMNMINIS	IMNM Initialization Procedures Source
IMNMM	IMNM MAKE
IMNMMLM	IMNM Mainline MAKE
IMNMMLS	IMNM Mainline Source
IMNMOPTS	IMNM Compile Options
IMNMUTLM	IMNM Utility Procedures MAKE
IMNMUTLS	IMNM Utility Procedures Source
MAKEIMNI	IMNI Local MAKE File
RQCGFM	CGF Requester MAKE
RQCGFS	CGF Requester Source
RQIMNAM	IMNI Admin Requester MAKE
RQIMNAS	IMNI Admin Requester Source
RQIMNIM	ICF-IMN Requester MAKE
RQIMNIS	ICF-IMN Requester Source
SCRNCGFS	CGF Screen Source
SVCGFM	CGF Server MAKE

SVCGFS	CGF Server Source
SVIMNAM	IMNI Admin Server MAKE
SVIMNAS	IMNI Admin Server Source
SWIMNIFM	IMNI MAKE Configuration
SWIMNIM	IMNI MAKE File
SWIMNIMM	IMNI File Macros (Defines)

SW60IIMN (EMV code)

IIMNFM	IMNi EMV MAKE Configuration
IIMNMM	IMNi EMV File Macros
IMNIEMVM	IMNi EMV MAKE
IMNIEMVS	IMNi EMV Source
SWIIMNM	IMNi MAKE

IMNsc Software**SW60IMSC**

IMNACM	IMNSC Admin Cleanup MAKE
IMNACS	IMNSC Admin Cleanup Source
IMNMCMDs	IMNM NCS Operator Command Processor Source
IMNMDCLS	IMNM Global Data Declarations
IMNMINIS	IMNM Initialization Procedures Source
IMNMM	IMNM MAKE
IMNMMLS	IMNM Mainline Source
IMNMOPTS	IMNM Compile Options
IMNMUTLS	IMNM Utility Procedures Source
IMSCARS	IMNSC Alternate Routing Source
IMSCCMDs	IMNSC NCS Operator Command Processor Source
IMSCD	IMNSC EMV Dummy Source file
IMSCDCLS	IMNSC Global Data Declarations
IMSCDDLm	IMNSC DDL MAKE
IMSCDDLs	IMNSC DDL Source
IMSCEMVD	IMNSC EMV Dummy Source file
IMSCFIOS	IMNSC File I/O Source
IMSCINIS	IMNSC Initialization Procedure Source
IMSCLFxG	ILF Extract Global
IMSCLFxM	ILF Extract MAKE
IMSCLFxS	ILF Extract Source
IMSCM	IMNSC MAKE
IMSCMLS	IMNSC Mainline Source
IMSCOPTS	IMNSC Compile Options
IMSCSCNS	IMNSC Screen Source
IMSCSEMS	IMNSC SEM Processor Source
IMSCSTMS	IMNSC STM Processor Source
IMSCUTLS	IMNSC Utility Procedures Source
MAKEIMSC	IMNSC Local MAKE File

RQCGFM	CGF Requester MAKE
RQCGFS	CGF Requester Source
RQIMSAM	IMNSC Admin Requester MAKE
RQIMSAS	IMNSC Admin Requester Source
RQIMSCM	ICF-IMN Requester MAKE
RQIMSCS	ICF-IMN Requester Source
SCRNCGFS	CGF Screen Source
SVCGFM	CGF Server MAKE
SVCGFS	CGF Server Source
SVIMSAM	IMNSC Admin Server MAKE
SVIMSAS	IMNSC Admin Server Source
SWIMSCFM	IMNSC MAKE Configuration
SWIMSCM	IMNSC MAKE File
SWIMSCMM	IMNSC File Macros (Defines)

SW60IIMS (EMV code)

IIMSFM	IMNsc EMV MAKE Configuration
IIMSM	IMNsc EMV File Macros
IMSCEMVM	IMNsc EMV MAKE
IMSCEMVS	IMNsc EMV Source
SWIIMSM	IMNSC MAKE

Compilation

Run the global MAKE and perform the following steps.

MAKE IMNi object files:

- Set the IMNI_ON flag in your CUSTMACS to “true”. Please ensure the related product flag is set properly i.e. CLASSIC_ON, POS_ON. For more details on the product flag set up, please refer to the BASE24-CLASSIC 6.0 Install Guide.
- If you have purchased the EMV modules, set the IMNI_EMV_ON flag in your CDNMACS to “true”.
- After executing MAKESCAN, please verify the SWIMNIMM and SWIMNIFM to ensure that all the MAKE definitions are pointing to the correct locations. If you have purchased the IMNi EMV modules also verify IIMNFM and IIMNMM.
- Run the global MAKE using the OBEY file B24C. Change the =allmacs define to point to the CDNMACS file instead of the standard CUSTMACS file.
- Verify all the objects are created.
- Set the tss_key_idx_on flag to TRUE in CUSTMACS (needed for TSS).

MAKE IMNsc object files:

- Set the IMNSC_ON flag in your CUSTMACS to true. Please ensure the related product flag is set properly i.e. CLASSIC_ON, ATM_ON. For more details on the product flag set up, please refer to the BASE24-CLASSIC 6.0 Install Guide.
- If you have purchased the EMV modules, set the IMSC_EMV_ON flag in your CUSTMACS to “true”.
- After executing MAKESCAN, please verify the SWIMNSCMM and SWIMNSCFM to ensure that all the MAKE definitions are pointing to the correct locations. Any other “M” files to check for EMV If you have purchased the IMNsc EMV modules also verify IIMSFM and IIMSMM.
- Run the global MAKE using the OBEY file B24C. Change the =allmacs define to point to the CDNMACS file instead of the standard CUSTMACS file.
- Verify all the objects are created.
- Set the tss_key_idx_on flag to TRUE in CUSTMACS (needed for TSS).

LNCF Assigns and Parameters

IMNi/IMNsc processes use a number of Logical Network Configuration File (LNCF) assigns and parameters in their processing. These assigns and parameters are described below. For more information on configuring IMNi/IMNsc via these assigns and parameters, please refer to the BASE24 Logical Network Configuration File Manual and IMNi/IMNsc Installation Guide.

Assigns

IMNi and IMNsc LNCF Assigns

Assign Name	IMNi Usage	IMNsc Usage	Assign Value	Description
CGF	X	X	\<sys>.\$<vol>.<subvol>. IMCGF	This assign specifies the filename of the Connection Group File (CGF), which the Acquirer IMN Switch Interface uses to route transactions to an issuer node. This assign is required if Automatic Routing is enabled (i.e. IMN-AUTO-ROUTING set to “Enabled”) and configured. Otherwise, this assign is not required.

Assign Name	IMNi Usage	IMNsc Usage	Assign Value	Description
EPCF	X	X	\<sys>.\$<vol>.<subvol>. EPCF	This assign specifies the filename of the Entry Point Configuration File (EPCF), which contains the Entry Point (EP) information for connecting to the IMN node. This file is a non-BASE24 file maintained via the IMN node EPCNTL and EPDEFN screens.
ICF	X	X	\<sys>.\$<vol>.<subvol>. ICF	This assign specifies the filename of the Interchange Configuration File (ICF), where the IMNi Switch Interface finds its configuration information. Configuration information includes IMN node ID, switch date, timer and SAF file information.
ILF	X		\<sys>.\$<vol>. <subvolDATA>. ILYYMMDD \<sys>.\$<vol>. <subvolTPLT>. ILYYMMDD	This assign specifies the filename of the Interchange Log File (ILF), where the IMNi Switch Interface logs transactions that are sent to the switch and received from the switch. Specify both the data location of the file and the template file location. NOTE: Each IMN Switch Interface Process is assigned a unique ILF file and template location.
ILF-SC		X	\<sys>.\$<vol>. <subvolDATA>. ILYYMMDD \<sys>.\$<vol>. <subvolTPLT>. ILYYMMDD	This assign specifies the filename of the Interchange Log File (ILF), where the IMNsc Switch Interface logs transactions that are sent to the switch and received from the switch. Specify both the data location of the file and the template file location. Please ensure

Assign Name	IMNi Usage	IMNsc Usage	Assign Value	Description
				the ILF data location or filename for IMNsc is distinctive from IMNi's ILF data location/filename NOTE: Each IMN Switch Interface Process is assigned a unique ILF file and template location.
IMN-ADMN	X	X	\<sys>.\$<vol>.<subvol>. IMNADMN	This assign specifies the filename of the IMN-ADMN file that contains IMN administrative transactions that are entered by an operator from the IMNA screen.
IMN-BIN	X	X	\<sys>.\$<vol>.<subvol>. MNBIN	This assign specifies the filename of the IMN-BIN file that contains card prefixes that the IMN Switch Interface uses to route not-on-us transactions to the appropriate card issuer. This file is a non-BASE24 file maintained via the IMN node.
IMN-IMNM	X		\<sys>.\$<process_id>	This assign specifies the process name of the IMNi's Manager process.
IMN-IMNM		X	\<sys>.\$<process_id>	This assign specifies the process name of the IMNsc's Manager process.
IMN-SAF- TEMPLATE	X		\<sys>.\$<vol>. <subvolDATA>.IMNSAF \<sys>.\$<vol>. <subvolTPLT>.IMNSAF	This assign specifies the filename of the IMNi Switch Interchange process SAF file. The IMNM process uses the template file location to create the SAF file for the IMN Switch Interchange process.

Assign Name	IMNi Usage	IMNsc Usage	Assign Value	Description
IMN-SAF-TEMPLATE-SC		X	\<sys>.\$<vol>. <subvolDATA>.IMNSAF \<sys>.\$<vol>. <subvolTPLT>.IMNSAF	This assign specifies the filename of the IMNsc Switch Interchange process SAF file. The IMNM process uses the template file location to create the SAF file for the IMN Switch Interchange process. Please ensure the SAF template, data location or filename for IMNsc is distinctive from IMNi's SAF as the record size of IMNsc SAF is different to IMNi's SAF.
KEYF	X	X	\<sys>.\$<vol>.<subvol>. KEYF	This assign specifies the filename of the KEY File, which contains the key management information.
LMCF	X	X	\<sys>.\$<vol>.<subvol>. LMCF	This assign specifies the filename of the Log Message Configuration File (LMCF), which contains log message information. This is a BASE24 product file.
PATHMON-NAME	X	X	\$<process_id>	This assign specifies the BASE24 Pathway process name used by IMN Admin server.

Parameters

IMNi and IMNsc LNCF Parameters

Parameter Name	IMNi Usage	IMNsc Usage	Parameter Default Value	Description
ATM-LN-CUTOVER		X	00:00	This parameter indicates the logical network cutover time for BASE24-atm.
EVALUATE-SWITCH-DATE	X	X	A	This parameter indicates what action the IMN Switch Interface should take if the switch settlement date on the ICF differs from the expected switch settlement date. The expected switch date is calculated using the current system date and the IMN cutover times. Valid values are: “A” – Abend. “B” – Use ICF switch date even if different from the expected date and log a warning. “E” – Use expected date and log a warning.
IMN-ACQ-DISCR-DATA-SUPPRS		X	N	This parameter indicates to the IMNsc Switch Interface process whether Track 2 discretionary data for Credit Card Cash Advance Acquirer transactions should be suppressed when logging to ILF.
IMN-ATM-RESP-TIME-BLOCKED-10 TH		X	100	This parameter indicates the response time (in tenths of seconds) at which the IMN Switch Interface process will declare the ATM product blocked and will notify the IMN node of this condition. Min=10, Max=32766

Parameter Name	IMNi Usage	IMNsc Usage	Parameter Default Value	Description
IMN-ATM-RESP-TIME-CONGESTED-10 TH		X	500	This parameter indicates the response time (in tenths of seconds) at which the IMN Switch Interface process will declare the ATM product congested and will notify the IMN node of this condition. Min=10, Max=32766
IMN-AUTO-ROUTING	X	X	DISABLED	This parameter enables or disables the Auto Routing feature. If running IMNI and IMSC in the same logical network, this parameter should be set to DISABLED.
IMN-BAL-AND-CUTOVER-START	X	X		This parameter indicates the start time of the daily cutover period for the IMN network. This is used to check the transaction capture date. Must be a valid time value using a 24-hour format (HH:MM). Recommended time: 21:00 (Interac mandates cutover window 21:00-21:30)
IMN-BAL-AND-CUTOVER-END	X	X		This parameter indicates the end time of the daily cutover period for the IMN network. This is used to check the transaction capture date. Must be a valid time value using a 24-hour format (HH:MM). Recommended time: 21:30 (Interac mandates cutover window 21:00-21:30)
IMN-DAILY-KEYCHANGE-TIME	X	X		This parameter indicates the time of the day to perform daily keychanges. Interac requires all working keys to be exchanged automatically once a day. Must be a valid time value using a 24-hour format (HH:MM).

Parameter Name	IMNi Usage	IMNsc Usage	Parameter Default Value	Description
				Please note that the IMNi and IMNsc can be configured to have a different time of the day to perform daily exchanges by specifying a symbolic name of an IMNi or IMNsc Manager process in the "READ BY" field.
IMN-FIL-ID	X			This parameter indicates the acquirer AIL (FIL) ID of the BASE24 resource node. This parameter is used by the IMNi Switch Interchange Process to find a valid entry point in the EPCF. Valid values are "1" to "9".
IMN-FIL-ID-SC		X		This parameter indicates the acquirer AIL (FIL) ID of the BASE24 resource node. This parameter is used by the IMNsc Switch Interchange Process to find a valid entry point in the EPCF. Valid values are "1" to "9".
IMN-HANDSHAKE-TIMEOUT	X	X	60	This parameter indicates the amount of time (in seconds) to wait for a handshake response message from the IMN Commonitor process.
IMN-ILF-NOTIFY-INTERVAL	X		50	This parameter indicates the number of writes to the IMNi's ILF between each file full percentage check. Min=10, Max =32766.
IMN-ILF-NOTIFY-INTERVAL-SC		X	50	This parameter indicates the number of writes to the IMNsc's ILF between each file full percentage check. Min=10, Max =32766.

Parameter Name	IMNi Usage	IMNsc Usage	Parameter Default Value	Description
IMN-ILF-NOTIFY-PERCENTAGE	X		70	This parameter indicates the threshold at which the operator is notified of the IMNi ILF's size as a percentage. Min=10, Max=99
IMN-ILF-NOTIFY-PERCENTAGE-SC		X	70	This parameter indicates the threshold at which the operator is notified of the IMNsc ILF's size as a percentage. Min=10, Max=99
IMN-ISS-DISCR-DATA-SUPPRS		X	N	This parameter indicates to the IMNsc Switch Interface process whether Track 2 discretionary data for Credit Card Cash Advance Issuer transactions should be suppressed when logging to ILF.
IMN-KEYCHANGE-DELAY	X	X	5	This parameter indicates the interval (in seconds) between changing keys to a destination and the next. The delay avoids simultaneous keychanges of all keys (MAC, PIN and MSG) to all destinations. The delay is used during startup and daily keychange time.
IMN-MIN-PACING-STAGGER-THRESH	X		5	This parameter is used by IMNi process to indicate the rate at which pacing messages are sent. This avoids sending pacing messages all at once.
IMN-MIN-PACING-STAGGER-THRESH-SC		X	5	This parameter is used by IMNsc process to indicate the rate at which pacing messages are sent. This avoids sending pacing messages all at once.

Parameter Name	IMNi Usage	IMNsc Usage	Parameter Default Value	Description
IMN-NOTIFY-COUNTER	X	X	7	This parameter specifies a percentage of the key usage counter limit, which is used by the IMN Switch Interface to notify the IMNM of a key change.
IMN-PACING-ABNORMAL-SECS	X		30	This parameter indicates the interval (in seconds) between pacing messages sent out by the IMNi Switch Interchange Process to the Commonitor when the state of a connection is blocked. Min=10, Max=32766. Suggested value: 60
IMN-PACING-ABNORMAL-SECS-SC		X	30	This parameter indicates the interval (in seconds) between pacing messages sent out by the IMNsc Switch Interchange Process to the Commonitor when the state of a connection is blocked. Min=10, Max=32766. Suggested value: 60
IMN-PACING-NORMAL-SECS	X		90	This parameter indicates the interval (in seconds) between pacing messages sent out by the IMN Switch Interchange Process to the Commonitor when the state of a connection is normal. Min=10, Max=32766. Suggested value: 120
IMN-PACING-NORMAL-SECS-SC		X	90	This parameter indicates the interval (in seconds) between pacing messages sent out by the IMNsc Switch Interchange Process to the Commonitor when the state of a connection is normal. Min=10, Max=32766. Suggested value: 120

Parameter Name	IMNi Usage	IMNsc Usage	Parameter Default Value	Description
IMN-PACING-RECENT-WEIGHTING	X		70	This parameter indicates how recent the weighting is that is used to calculate the average response time from the POS product. A value of 100 indicates current response time with no consideration of past performance. Valid values are 0 to 100.
IMN-PACING-RECENT-WEIGHTING-SC		X	70	This parameter indicates how recent the weighting is that is used to calculate the average response time from the ATM product. A value of 100 indicates current response time with no consideration of past performance. Valid values are 0 to 100.
IMN-PACING-TIMER-SECS	X		30	The interval time (in seconds) that the IMN Switch Interface process will use to re-evaluate POS response times for pacing changes. The two IMN pacing parameters work in the following manner: On every transaction sent successfully to an entry point, the IMNi Switch Interface process increments a throughput counter. On every response from BASE24, the IMNi Switch Interface process keeps track of the response times. Every N seconds the IMN Switch Interface process will examine these throughput counters and the average BASE24 response time over the period. The N seconds is the IMN-PACING-TIMER-SECS parameter. Min=10, Max=32766.

Parameter Name	IMNi Usage	IMNsc Usage	Parameter Default Value	Description
IMN-PACING-TIMER-SECS-SC		X	30	The interval time (in seconds) that the IMNsc Switch Interface process will use to re-evaluate ATM response times for pacing changes. The two IMN pacing parameters work in the following manner: On every transaction sent successfully to an entry point, the IMNsc Switch Interface process increments a throughput counter. On every response from BASE24, the IMNsc Switch Interface process keeps track of the response times. Every N seconds the IMN Switch Interface process will examine these throughput counters and the average BASE24 response time over the period. The N seconds is the IMN-PACING-TIMER-SECS parameter. Min=10, Max=32766.
IMN-POS-RESP-TIME-BLOCKED-10 TH	X		100	This parameter indicates the response time (in tenths of seconds) at which the IMN Switch Interface process will declare the POS product blocked and will notify the IMN node of this condition. Min=10, Max=32766
IMN-POS-RESP-TIME-CONGESTED-10 TH	X		500	This parameter indicates the response time (in tenths of seconds) at which the IMN Switch Interface process will declare the POS product congested and will notify the IMN node of this condition. Min=10,max=32766

Parameter Name	IMNi Usage	IMNsc Usage	Parameter Default Value	Description
IMN-PRODUCTION-TEST	X	X	TEST	This parameter indicates the operating mode of the IMN Switch Interface process (production or test). The interface process sets its message classes accordingly (e.g. "01" versus "02"). Although valid values are either "Prod"uction or "Test", only "PROD" is supported.
IMN-SAF-DEST-FORWARD-PERCENTAGE	X	X	50	This parameter indicates the percentage of the SAF file size for a particular destination at which the SAF mode changes to forward-only mode.
IMN-SAF-DEST-NORMAL-PERCENTAGE	X	X	30	This parameter indicates the percentage of the SAF file size for a particular destination at which the SAF mode changes to normal mode.
IMN-SAF-DEST-NOTIFY-PERCENTAGE	X	X	40	This parameter indicates the percentage of the SAF file size for a particular destination when a warning message is issued.
IMN-SAF-FAST-THRESHOLD	X		2	This parameter is used by IMNi switch interface to indicate the number of consecutive transactions that meet the criteria of switching to the SAF fast mode.
IMN-SAF-FAST-THRESHOLD-SC		X	2	This parameter is used by IMNsc switch interface to indicate the number of consecutive transactions that meet the criteria of switching to the SAF fast mode.

Parameter Name	IMNi Usage	IMNsc Usage	Parameter Default Value	Description
IMN-SAF-FORWARD-PERCENTAGE	X	X	80	This parameter indicates the percentage of the SAF file size for the whole system at which the SAF mode changes to forward-only mode.
IMN-SAF-NORMAL-PERCENTAGE	X	X	60	This parameter indicates the percentage of the SAF file size for the whole system at which the SAF mode changes to normal mode.
IMN-SAF-NOTIFY-INTERVAL	X		50	This parameter indicates the interval (number of SAF writes) that the IMNi Switch Interface processes check the size of the SAF file.
IMN-SAF-NOTIFY-INTERVAL-SC		X	50	This parameter indicates the interval (number of SAF writes) that the IMNsc Switch Interface processes check the size of the SAF file.
IMN-SAF-NOTIFY-PERCENTAGE	X		70	This parameter is used by IMNi switch interface to indicate the threshold, as a percentage of the SAF file size, at which the operator is notified of the SAF file size.
IMN-SAF-NOTIFY-PERCENTAGE-SC		X	70	This parameter is used by IMNsc switch interface to indicate the threshold, as a percentage of the SAF file size, at which the operator is notified of the SAF file size.

Parameter Name	IMNi Usage	IMNsc Usage	Parameter Default Value	Description
IMN-SAF-RETRANSMIT-INTERVAL	X		30	This parameter is used by IMNi switch interface to indicate the interval between SAF messages that are resent to a particular destination. Used when the SAF messages have not been acknowledged.
IMN-SAF-RETRANSMIT-INTERVAL-SC		X	30	This parameter is used by IMNsc switch interface to indicate the interval between SAF messages that are resent to a particular destination. Used when the SAF messages have not been acknowledged.
IMN-SAF-SLOW-INTERVAL	X		60	This parameter is used by IMNi switch interface to indicate the interval between SAF messages sent to any destination and the destination is not responding successfully to the SAF messages.
IMN-SAF-SLOW-INTERVAL-SC		X	60	This parameter is used by IMNsc switch interface to indicate the interval between SAF messages sent to any destination and the destination is not responding successfully to the SAF messages.
IMN-SAF-SLOW-THRESHOLD	X	X	3	This parameter is used by IMNi switch interface to indicate the number of consecutive transactions that meet the criteria of switching into the SAF slow mode.
IMN-SAF-SLOW-THRESHOLD-SC	X	X	3	This parameter is used by IMNsc switch interface to indicate the number of consecutive transactions that meet the criteria of switching into the SAF slow mode.

Parameter Name	IMNi Usage	IMNsc Usage	Parameter Default Value	Description
IMN-TRANSLATE-ACCOUNT-BALANCE	X		F	This parameter determines how the IMNi Switch Interface process converts the account balance field from CBC (Cipher Block Chaining) to ECB (Electronic Code Book). Valid values are: "T" – Translate Interac CBC to ECB for BASE24. "F" – Leave as CBC for BASE24.
IMN-TRANSLATE-ACCOUNT-BALANCE-SC		X	F	This parameter determines how the IMNsc Switch Interface process converts the account balance field from CBC (Cipher Block Chaining) to ECB (Electronic Code Book). Valid values are: "T" – Translate Interac CBC to ECB for BASE24. "F" – Leave as CBC for BASE24.
INTERNAL-BIN-TABLE-ENTRIES	X	X	32000	This parameter indicates the maximum number of entries allowed in the internal BIN table. Maximum is 128000
INTERNAL-FKT-TABLE-ENTRIES	X	X	100	This parameter indicates the maximum number of entries allowed in the AIL (FIL) Key Table internal table.
INTERNAL-DOWN-VERSION-LEVEL	X		0	This parameter is used by IMNi switch interface to indicate the allowable range below the current Version-level set in the IMNsc Switch Interface that a COMMONITOR must have to allow communication to occur.
INTERNAL-DOWN-VERSION-LEVEL-SC		X	0	This parameter is used by IMNsc switch interface to indicate the allowable range below the current Version-level set in the IMNi Switch Interface that a

Parameter Name	IMNi Usage	IMNsc Usage	Parameter Default Value	Description
				COMMONITOR must have to allow communication to occur.
INTERNAL-RECOVERY-TIMER-SECS	X	X	60	This parameter determines the timer interval (in seconds) that the IMN Switch Interface uses to check the state of the entry points and the IMNM. Any entry points that are closed are re-started at this time.
INTERNAL-SAF-MAX-TRANS-OUT	X		10	This parameter is used in IMNi switch interface to indicate the maximum number of SAF transactions that are outstanding (in-flight).
INTERNAL-SAF-MAX-TRANS-OUT-SC		X	10	This parameter is used in IMNsc switch interface to indicate the maximum number of SAF transactions that are outstanding (in-flight).
INTERNAL-TRACE-BITS	X		0000000000 000000	This parameter is used in IMNi switch interface to indicate the initial trace bit values for logging operator trace messages. A string of 16 ones & zeroes represents each of the trace bits. A "0" means the trace bit is OFF and a "1" means the trace bit is ON. The first trace bit (bit number 0) is the right-most (last) position in the string.
INTERNAL-TRACE-BITS-SC		X	0000000000 000000	This parameter is used in IMNsc switch interface to indicate the initial trace bit values for logging operator trace messages. A string of 16 ones & zeroes represents each of the trace bits. A "0" means the trace bit is OFF and a "1" means the trace bit is ON. The first trace bit (bit number 0) is the right-most (last) position in the string.

Parameter Name	IMNi Usage	IMNsc Usage	Parameter Default Value	Description
				position in the string.
INTERNAL-UP-VERSION-LEVEL	X		0	This parameter is used in IMNi switch interface to indicate the allowable range above the current Version-level set in the IMNi Switch Interface that a Commonitor must have to allow communication to occur.
INTERNAL-UP-VERSION-LEVEL-SC		X	0	This parameter is used in IMNsc switch interface to indicate the allowable range above the current Version-level set in the IMNsc Switch Interface that a Commonitor must have to allow communication to occur.
POS-LN-CUTOVER	X		00:00	This parameter indicates the logical network cutover time for BASE24-pos.
POS-SCD-IMNA	X		N	This parameter is used by IMN Admin server to indicate if SCD admin transactions are allowed to be sent from the POS environment.
SAF-LOG-THRESHOLD	X		20	This parameter is used in IMNi switch interface to indicate the threshold at which the operator is notified of the number of SAF records in the SAF file. See TO-CHECK-SAF.
SW-ILF-FILE-RETENTION	X		7	This parameter indicates the number of days the ILF will remain on the system before being purged by the IMNi Switch Interface process.

Parameter Name	IMNi Usage	IMNsc Usage	Parameter Default Value	Description
SW-ILF-FILE-RETENTION-SC		X	7	This parameter indicates the number of days the ILF will remain on the system before being purged by the IMNsc Switch Interface process.
SW-ILF-PURGE-ATTEMPT	X		7	This parameter is used in IMNi switch interface to indicate the number of purge attempts on older ILFs excluding those that are under the protection of the SW-ILF-FILE-RETENTION value. For example if the SW-ILF-PURGE-ATTEMPT is set at 7 and SW-ILF-FILE-RETENTION is set at 4, IMNi will retain 4 days of ILFs and begin purging ILFs that are 5 to 11 days old.
SW-ILF-PURGE-ATTEMPT-SC		X	7	This parameter is used in IMNsc switch interface to indicate the number of purge attempts on older ILFs excluding those that are under the protection of the SW-ILF-FILE-RETENTION value. For example if the SW-ILF-PURGE-ATTEMPT is set at 7 and SW-ILF-FILE-RETENTION is set at 4, IMNsc will retain 4 days of ILFs and begin purging ILFs that are 5 to 11 days old.
SW-IMNI-PROC-NAME _{nn}	X			This parameter defines the symbolic name of an IMN Switch Interface Process that the IMNM process manages. Add one parameter per IMN Switch Interface process, up to a maximum of 50. <u>Note:</u> Ensure the symbolic process name matches the IMNI process name in the XPNET configuration.

Parameter Name	IMNi Usage	IMNsc Usage	Parameter Default Value	Description
TO-CHECK-SAF	X		60	This parameter is the timer interval that the IMNM uses to check the SAF record count thresholds. The timer interval is specified in hundredths of seconds. See also SAF-LOG-THRESHOLD.

Configuring Timers

The IMNi/IMNsc process uses internal timers to control IMN node message traffic. The length of each of these timers is specified in ICF fields or LNCf parameters.

Network Management Message Timer

IMNi/IMNsc starts a Network Management Message timer each time it sends a 0800 key exchange request to the IMN node. The Network Management Message timer limits the amount of time that the IMNi/IMNsc process waits for a response after transmitting a 0800 key exchange request. The number of seconds to use for this timer is taken from the NETWORK MANAGEMENT field on ICF screen 3.

When this timer expires before IMNi/IMNsc receives a response to a 0800 key exchange request, IMNi/IMNsc notifies the IMNM process that the key exchange attempt failed. IMNM will select an IMNi/IMNsc process to re-attempt the key exchange.

Outbound Request Timer

IMNi/IMNsc sets an Outbound Request timer each time it sends a request message to the IMN node. The Outbound Request timer limits the amount of time that IMNi/IMNsc waits for a response from the IMN node after transmitting a request message. The number of seconds to use for this timer is taken from the outbound timer limit field on the product-specific ICF screen.

If this timer expires before IMNi/IMNsc receives a response to a 0200 financial request, IMNi/IMNsc returns a decline response to the BASE24-pos/BASE24-atm authorization process, generates a 0420 reversal request for the remote node, and adds the reversal request to the SAF file for the remote node.

If this timer expires before IMNi/IMNsc receives a response to a SAF request, IMNi/IMNsc sets a SAF Re-Transmission timer for the request and increments

the number of consecutive SAF failures. If the SAF mode for the destination is not SLOW, and the number of consecutive SAF failures has reached the IMN-SAF-SLOW-THRESHOLD parameter value, IMNi/IMNsc changes the SAF mode for the destination to SLOW.

Inbound Request Timer

IMNi/IMNsc sets an Inbound Request timer each time it sends a 0200 message to a BASE24-pos/BASE24-atm Authorization process. The Inbound Request timer limits the amount of time that IMNi/IMNsc waits for a response from a BASE24-pos/BASE24-atm Authorization process after transmitting a request message. The number of seconds to use for this timer is taken from the inbound timer limit field on the product-specific ICF screen.

If an Inbound Request timer expires before IMNi/IMNsc receives a response message, IMNi/IMNsc returns a 0210 decline response to the IMN node, generates a 0420 reversal advice request, and sends this 0420 reversal to the BASE24-pos/BASE24-atm Authorization process.

Store-and-Forward Interval Timer

IMNi/IMNsc maintains a SAF Interval timer for a destination whenever there are SAF requests for that destination in the SAF file and the SAF mode is not STORE-ONLY. If the SAF mode of the destination is SLOW, the timer is set from the IMN-SAF-SLOW-INTERVAL parameter in LNCF. If the SAF mode of the destination is FAST, the timer is set from the SAF FAST interval on page 12 of the ICF record for the destination.

When the SAF Interval timer expires, IMNi/IMNsc sends the next SAF request in sequence from the SAF file, if any, and sets a new SAF Interval timer.

Store-and-Forward Re-Transmission Timer

Whenever an outbound SAF request times out, or when IMNi/IMNsc receives an unsatisfactory response to an outbound SAF request, IMNi/IMNsc starts a SAF Re-Transmission timer for the request. The IMN-SAF-RETRANSMIT-INTERVAL parameter defines the interval for this timer.

When the timer expires, IMNi/IMNsc attempts to send the SAF request again.

ICF Screens

This section contains a listing of ICF fields that IMNi/IMNsc uses. Screens 1 through 11 are shared by multiple interchanges. Screens 12 to 14 are interchange-specific, meaning they are used exclusively for IMNi/IMNsc.

Tables in the following sub-sections identify the ICF fields used by IMNi/IMNsc and their expected values. Values in quotation marks (“ ”) are literals that must be input as quoted in the corresponding fields. IMNi/IMNsc does not use all the fields on all the ICF screens. These fields are denoted with “not used” in the Value column.

For documentation on screens 1 through 11 and their corresponding field descriptions, please refer to the BASE24 Base Files Maintenance Manual. Screens 12 to 14 and description of their fields are provided later in the corresponding sub-sections.

Note: The information contained in the following tables does not indicate whether or not a field is required, but rather if a field can be used if you so choose. The field may or may not be required, depending on your selected configuration options.

There are two types of ICF records required to configure the IMNi/IMNsc process:

- Control Record
- Destination Record

ICF Screen 1

The Control Record contains information about the local IMN node. There is only one Control Record in the file. The Control Record has "IMN" in the Interchange FIID field and sixteen (16) zeroes in the Process field. An example of a Control Record is shown below.

```

BASE24-BASE INTERCHANGE CONFIG      QF60      03/10/29 16:11 01 OF 13
INTERCHANGE FIID: IMN                PROCESS: 0000000000000000

      SWITCH TYPE: IMN
INTERCHANGE LOGICAL NET: IMNI
      REPORTING NAME: $$.#IMN

      INSTITUTION ID: CONTROL-NETWORK
      SWITCH ID: IMN
      STATION 1:
      STATION 2:

      SIC CODE: 0
      CURRENCY CODE: 124 (CAN)
      DEFAULT TERM NUM:
      DEFAULT ACQUIRER ID NUM: 00000777762
      CUSTOMER BALANCE DISPLAY: 0 (DON'T DISPLAY OR PRINT)

RECORD LAST CHANGED: 03/10/22 11:18 BY USER: 0255 , 00000255 CHANGE
***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                F12-HELP

```

The Destination Record contains information about the remote IMN node. There can be several Destination Records in the file, one for each remote IMN node. The Destination Record has "IMN" in the Interchange FIID field and node name and message class, separated by a space, in the Process field. An example of a Destination Record for IMNI is shown below.

```

BASE24-BASE INTERCHANGE CONFIG      QF60      03/10/29 16:11 01 OF 13
INTERCHANGE FIID: IMN                PROCESS: ACITST 01

      SWITCH TYPE: IMN
INTERCHANGE LOGICAL NET: IMNI
      REPORTING NAME: $$.#IMN

      INSTITUTION ID: 777761
      SWITCH ID: IMN
      STATION 1:
      STATION 2:

      SIC CODE: 0
      CURRENCY CODE: 124 (CAN)
      DEFAULT TERM NUM:
      DEFAULT ACQUIRER ID NUM: 00000777761
      CUSTOMER BALANCE DISPLAY: 0 (DON'T DISPLAY OR PRINT)

RECORD LAST CHANGED: 03/10/22 11:18 BY USER: 0255 , 00000255 CHANGE
***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                F12-HELP

```

For field descriptions of screens 1 to 11; please refer to the BASE24 Base File Maintenance Manual.

ICF Screen 12

ICF screen 12 contains IMNi/IMNsc processing information. An example of the screen is illustrated below, followed by the description of the fields.

```

BASE24-IMNI  ICF FILE MAINTENANCE      QF60      03/11/13  17:59  12 OF 14
INTERCHANGE FIID: IMN                      PROCESS: ACITST 01

                                IMNI PROCESS NAME


DESTINATION-NODE NUMBER:  777761
ACQUIRER CNXN STATE:      3 (UP)
ISSUER CNXN STATE:        2 (KEY EXCHANGE)
GMT OFFSET:               300 (MINUTES)
SAF FILENAME:             $CDN60.QA60DATA.SAF
SAF FAST INTERVAL:        0 (100th's OF SECONDS)


***** BASE24 *****
NEW PAGE:                FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                               F12-HELP

```

IMNi/IMNsc Process Name

This field contains the IMNsc process symbolic names for a given IMN node ID only.

Note: Please be aware that the IMNi process name is not supported on this screen. The IMNi process name is defined in the LNCf parameter. Please refer to the IMNi LNCf Parameters for more details.

Destination-Node Number

This field contains the Interac Node ID for this destination. The Control Record contains the local Node ID. The Destination Record contains the responder Node ID.

Acquirer CNXN State

This field contains the current connection status information. The possible values are:

- | | |
|-------------|---|
| “0” (Down) | The Issuer AIL is not available for receiving messages from the Acquirer AIL. |
| “1” (Ready) | The Issuer AIL is initialized, but the issuer AIL has not received key exchanges from the Acquirer AIL. |
| “3” (Up) | Working keys have been received and normal traffic is flowing (default value). |

Issuer CNXN State

This field contains the current connection status information. The possible values are:

- | | |
|--------------------|---|
| “0” (Down) | The Issuer AIL is marked down by the Acquirer AIL. |
| “2” (Key Exchange) | The Acquirer AIL is attempting to send a set of working keys to the Issuer AIL (default value). |
| “3” (Up) | The Issuer AIL has received all working keys and transactions can be sent to the IMN node. |

GMT Offset

This field contains the GMT offset in minutes.

SAF Filename

This field contains the physical location of the SAF file for this destination node/message class.

SAF Fast Interval

This field indicates the SAF fast interval (in 100th seconds) of outbound SAF messages for this destination node/message class combination.

ICF Screen 13

ICF screen 13 contains the Key change thresholds and counters. An example of the screen is illustrated below, followed by descriptions of its fields.

BASE24-IMNI ICF FILE MAINTENANCE		QF60	03/11/13	18:00	13 OF 14
INTERCHANGE FIID: IMN		PROCESS: ACITST 01			
KEY CHANGE TRANSACTION COUNTERS KEY STATES:					
PIN ENCRYPTION KEY (KPE):	100	KPE: 0	0	OK	
MAC ENCRYPTION KEY (KMAC):	100	KMAC: 0	1	KEY CHANGE PENDING	
MESSAGE ENCRYPTION KEY:	100	KME: 0	2	KEY CHANGE REQUIRED	
MAC FAILURES:	5		3	KEY CHANGE DISABLE	
MAX BAD PIN BLOCK ERRORS:	5			(KME only)	
MESSAGE ENCRYPTION FAILURES:	5				
MACING ENABLED:	Y				
MAC FAILURE ALARM:	4				
KEY SYNCHRONIZATION ALARM:	4				
KEY EXCHANGE ALARM:	4				
KEY EXCHANGE TIMEOUT ALARM:	1				
KEY EXCHANGE REJECTION ALARM:	1				
PACING NORMAL INTERVAL:	120	(SECONDS)			
PACING CONGEST INTERVAL:	60	(SECONDS)			
***** BASE24 *****					
NEW PAGE:	FILE DESTINATION:	NEW LOGICAL NETWORK ID:			
	F12-HELP				

PIN Encryption Key (KPE)

This field indicates the maximum number of transactions that should occur before the Acquirer IMNi/IMNsc initiates a 0800 PIN key change (KPE).

Key States

This field indicates the state of the given keys (KPE, KMAC and KME). The valid values are:

"0"	No key change in progress.
"1"	Key change in progress.
"2"	Key change required.
"3"	Key change disable.

Please note that the KME is supported in IMNi only.

MAC Encryption Key (KMAC)

This field indicates the maximum number of transactions that should occur before the Acquirer IMNi/IMNsc initiates a 0800 Message Authentication key change (KMAC).

Message Encryption Key

This field indicates the maximum number of transactions that should occur before the Acquirer IMNi initiates a 0800 Message Encryption key change (KME). The Message Encryption Key is only supported in the IMNi.

MAC Failures

This field indicates the maximum number of consecutive MAC verification failures allowable for an Issuer FI before the Acquirer IMNi/IMNsc should initiate a 0800 KMAC change.

Max Bad PIN Block Errors

This field indicates the maximum number of consecutive PIN encryption failures that can occur at the Issuer FI before the Acquirer IMNi/IMNsc should initiate a 0800 KPE change.

Message Encryption Failures

This field indicates the maximum number of consecutive message encryption failures that can occur at the Issuer FI before the Acquirer IMNi should initiate a 0800 KME change.

MACing Enabled

This field specifies whether MACing will be enabled or disabled. The valid values are:

- “Y” MACing is enabled. (Default value)
- “N” MACing is disabled.

MAC Failure Alarm

This field indicates the number of times a MAC error may occur before a message is sent to the operator logger (acquirer and issuer).

Key Synchronization Alarm

This field indicates the number of times a key synchronization error may occur before a message is sent to the operator logger (acquirer and issuer).

Key Exchange Alarm

This field indicates the number of times a key conveyance error may occur before a message is sent to the operator logger (acquirer and issuer).

Key Exchange Timeout Alarm

This field indicates the number of times a key change timeout error may occur before a message is sent to the operator logger (acquirer).

Key Exchange Rejection Alarm

This field indicates the number of times a key change rejection may occur before a message is sent to the operator logger (acquirer).

Pacing Normal Interval

This field indicates the number of seconds between pacing normal messages to the IMN node.

Pacing Congest Interval

This field indicates the number of seconds between pacing abnormal messages to the IMN node.

ICF Screen 14

Select the transactions you want logged to the ILF by entering a **Y** next to the message type. Enter a **N** if you don't want the IMNi/IMNsc to log the transaction to the ILF.

BASE24-IMNI	ICF FILE MAINTENANCE	QF60	03/11/13	18:00	14 OF 14
INTERCHANGE FIID: IMN		PROCESS: ACITST 01			
TRANSACTION LOGGING CONFIGURATION					
0110	Y	0210	Y		
0230	Y	0430	Y		
0432	Y	0610	Y		
0612	Y	0630	Y		
0632	Y	0810	Y		
TRANSACTIONS MARKED '*' ARE ALWAYS LOGGED ON OUTBOUND REQUESTS					
***** BASE24 *****					
NEW PAGE:	FILE DESTINATION:	NEW LOGICAL NETWORK ID:			
	F12-HELP				

Transaction Logging Configuration

This field determines whether a transaction of the message type indicated will be logged to the ILF.

KEYF Screens

When using the Transaction Security Service (TSS) subsystem, hardware-encrypted key information is maintained in the TSS database. The hardware-encrypted key fields in the BASE24 database are set to a value known as a key locator value. The TSS process uses the key locator value as the primary key value to retrieve the actual hardware-encrypted key information from its database.

In BASE24 Release 6.0, the actual keys are no longer stored in the KEYF record. Instead, a reference to the actual key (Key Locator) is entered into the key field. The actual keys are kept in the Interchange Security File (ISECD). Please refer to the BASE24 Transaction Security Manual for more details.

Data in the ISECD is maintained by the Interface Security Configuration window. Also, after running a KEYF file conversion, the remaining data in the KEYF, such as the PIN and MAC encrypt types, PIN block format and all dynamic key information is maintained by the BASE24 KEYF Configuration window on the ACI Desktop user interface. This allows all security data to be maintained from a single user interface.

If you need to add KEYF and ISECD records after conversion, you must add a new KEYF record from the BASE24 KEYF Configuration window before adding a new ISECD record from the Interface Security Configuration window. Please refer to the BASE24 TSS Processing Guide for more details.

The key in the KEYF record is similar to the key in the ICF record. The DPC/FIID is always "IMN" and the Interface Process name is the concatenation of the node-name, message class and AIL ID ("1" as issuer and "A" as acquirer). In BASE24 6.0 however there are NO spaces in the Interface Process Name field. Basically, all the Destination Records (remote node) in the ICF must be added in the KEYF to set up all of the key information.

The following tables indicate the KEYF fields used by IMNi/IMNsc and their expected values. Values in quotation marks (" ") are literal values that must be placed in the corresponding fields. IMNi/IMNsc does not use all the fields on all the KEYF screens. These fields are noted with "Not used" in the Values column.

For documentation on the KEYF screens and their corresponding field descriptions, please refer to the BASE 24 Files Maintenance Manual.

For information on BASE24-pos or BASE24-atm security, please refer to the BASE 24 Transaction Security Manual.

The information contained in the following tables does not indicate whether or not a field is required, but rather if a field can be used if you so choose. The field may or may not be required, depending on your configuration options, as initiated by the Acquirer AIL.

KEYF Screen 1

KEYF screen 1 defines the number of PIN and MAC keys, the length of the Key Exchange Keys (KEKs), and the PIN and MAC exchange keys currently being used by IMNi and IMNsc. In addition, screen 1 contains security information used to set up the PIN and MAC parameters associated with the IMNi and IMNsc process. The fields on KEYF screen 1 used by IMNi and IMNsc and their expected values are indicated in the following table.

Note: Other than the information on Screen 1, data on the other screens need not be entered. When the IMN is started, key changes are performed.

BASE24-BASE KEY FILE		QA60	07/05/25	13:48	01 OF 04
DPC/FIID: IMN		INTERFACE PROCESS: ACIT0103A			
ENCRYPT TYPE: 1 (SECURE DEVICE)		BASE24 ENCRYPT TYPE: 1 (SECURE DEVICE)			
PIN BLOCK FORMAT: 1 (ANSI)		ANSI PAN FORMAT: 0 (12 RIGHT/NO CHK)			
MAC ENCRYPT TYPE: 1 (SECURE DEVICE)		PIN PAD CHARACTER: F			
MAC DATA TYPE: 0 (ASCII)		NUMBER OF KEYS: 1 (COMBINED)			
KEY LENGTH: 1 (SINGLE)		FULL MESSAGE MAC: N (SELECTED FIELDS)			
INTERMEDIATE KEYS					
CLEAR: 0000000000000000		CHECK DIGITS: 0000			
ENCRYPTED: 0000000000000000					
EXCHANGE KEYS					
PIN KEY: DA00000000000014		0000000000000000	CHECK DIGITS: 0000		
MAC KEY: DA00000000000014		0000000000000000	CHECK DIGITS: 0000		
RECORD LAST CHANGED: 03/09/18 16:56 BY USER: 0255 , 00000255 CHANGE					
***** BASE24 *****					
NEW PAGE:		FILE DESTINATION:		NEW LOGICAL NETWORK ID:	
F12-HELP					
RECORD RETRIEVED FROM \ACCIS1.\$CDN60.QA60DATA.KEYF 0000					

KEYF SCREEN 1	
Field	Value
DPC/FIID	Host DPC number or financial institution identifier - "IMN"
INTERFACE PROCESS	Node Name (6 characters) from ICF, message class (2 characters, 01 = IDP, 03 = SCD), AIL ID (1 character); AIL ID is 'A' for outbound key records and '1' for inbound records.
ENCRYPT TYPE	0 for "No encryption" 1 for "Secure device" 2 for "DES encryption" This field must contain a 1 if using dynamic PIN key management.
BASE24 ENCRYPT TYPE	0 for clear PINs, 1 for Security module PIN

KEYF SCREEN 1	
Field	Value
	management
MAC ENCRYPT TYPE	0 for no MAC processing, 1 for Security module MAC processing, 2 for Software MAC processing. This field must contain a 1 if using dynamic MAC key management.
ANSI PAN FORMAT	0 for the last 12 digits of the PAN, excluding the check digit; 1 for the last 12 digits of the PAN, including the check digit; or 2 for the first 12 digits of the PAN.
PIN BLOCK FORMAT	0 for clear PINS, 1 for ANSI (PIN/ PAD) PIN block, 3 for PIN/PAD PIN block. This field must contain 1 or 3 if using dynamic PIN key management. For Interac, this field must be 1 (ANSI)
PIN PAD CHARACTER	A–F
MAC DATA TYPE	0 or 1, not used
NUMBER OF KEYS	1 if the same key is used for inbound and outbound messages, or 2 if different keys are used for inbound and outbound messages or if the KEY PROCESSING TYPE field on KEYF screen 3 is set to C. not used
KEY LENGTH	1 for single-length key exchange keys (KEKs), or 2 for double-length KEKs, not used
FULL MESSAGE MAC	N for authentication of selected fields, or Y for full message authentication, not used
INTERMEDIATE KEYS: CLEAR	Clear version of Intermediate key, not used
INTERMEDIATE KEYS: ENCRYPTED	Encrypted version of Intermediate key, not used
INTERMEDIATE KEYS: CHECK DIGITS	0–9, A–Z, not used
EXCHANGE KEYS: PIN KEY	Two sets of hexadecimal characters
EXCHANGE KEYS: CHECK DIGITS	0–9, A–Z
EXCHANGE KEYS: MAC KEY	Two sets of hexadecimal characters
EXCHANGE KEYS: CHECK DIGITS	0–9, A–Z

KEYF Screen 2

KEYF screen 2 contains inbound and outbound working key information for PINs and MACs. The fields on KEYF screen 2 are used by IMNi and IMNsc and their expected values are indicated in the following table.

Outbound keys are used for 'A' records; inbound keys are used for AIL ID '1' records.

BASE24-BASE KEY FILE		QA60	07/05/25 13:49	02 OF 04
DPC/FIID: IMN		INTERFACE PROCESS: ACIT0103A		
PIN KEY INFORMATION				
OUTBOUND KEYS:				
PIN KEY1:	DA000000000000014	CHECK DIGITS1:	0000	CURRENT INDEX: 1
PIN KEY2:	DA000000000000014	CHECK DIGITS2:	0000	KEY COUNTER: 000000
INBOUND KEYS:				
PIN KEY1:	DA000000000000014	CHECK DIGITS1:	0000	CURRENT INDEX: 1
PIN KEY2:	DA000000000000014	CHECK DIGITS2:	0000	KEY COUNTER: 000000
MAC KEY INFORMATION				
OUTBOUND KEYS:				
MAC KEY1:	DA000000000000014	CHECK DIGITS1:	0000	CURRENT INDEX: 1
MAC KEY2:	DA000000000000014	CHECK DIGITS2:	0000	KEY COUNTER: 000000
INBOUND KEYS:				
MAC KEY1:	DA000000000000014	CHECK DIGITS1:	0000	CURRENT INDEX: 1
MAC KEY2:	DA000000000000014	CHECK DIGITS2:	0000	KEY COUNTER: 000000
RECORD LAST CHANGED: 03/09/18 16:56 BY USER: 0255 , 00000255 CHANGE				
***** BASE24 *****				
NEW PAGE:	FILE DESTINATION:	NEW LOGICAL NETWORK ID:		
F12-HELP				

KEYF SCREEN 2	
Field	Value
OUTBOUND KEYS: PIN KEY1	16 hexadecimal characters
OUTBOUND KEYS: CHECK DIGITS1	4 hexadecimal characters
OUTBOUND KEYS: CURRENT INDEX 1	1 for the value in the PIN KEY1 field, or 2 for the value in the PIN KEY2 field.
OUTBOUND KEYS: PIN KEY2	16 hexadecimal characters
OUTBOUND KEYS: CHECK DIGITS2	4 hexadecimal characters
OUTBOUND KEYS: KEY COUNTER	6 hexadecimal characters
INBOUND KEYS: PIN KEY1	16 hexadecimal characters
INBOUND KEYS: CHECK DIGITS1	4 hexadecimal characters
INBOUND KEYS: CURRENT INDEX 1	1 for the value in the PIN KEY1 field, or 2 for the value in the PIN KEY2 field.

INBOUND KEYS: PIN KEY2	16 hexadecimal characters
INBOUND KEYS: CHECK DIGITS2	4 hexadecimal characters
INBOUND KEYS: KEY COUNTER	6 hexadecimal characters
OUTBOUND KEYS: MAC KEY1	16 hexadecimal characters
OUTBOUND KEYS: CHECK DIGITS1	4 hexadecimal characters
OUTBOUND KEYS: CURRENT INDEX 1	1 for the value in the MAC KEY1 field, or 2 for the value in the MAC KEY2 field.
OUTBOUND KEYS: MAC KEY2	16 hexadecimal characters
OUTBOUND KEYS: CHECK DIGITS2	4 hexadecimal characters
OUTBOUND KEYS: KEY COUNTER	6 hexadecimal characters
INBOUND KEYS: MAC KEY1	16 hexadecimal characters
INBOUND KEYS: CHECK DIGITS1	4 hexadecimal characters
INBOUND KEYS: CURRENT INDEX 1	1 for the value in the MAC KEY1 field, or 2 for the value in the MAC KEY2 field.
INBOUND KEYS: MAC KEY2	16 hexadecimal characters
INBOUND KEYS: CHECK DIGITS2	4 hexadecimal characters
INBOUND KEYS: KEY COUNTER	6 hexadecimal characters

KEYF Screen 3

KEYF screen 3 contains information on the timer, threshold, and processing parameters.

IMNi and IMNsc do not access any parameters on Screen 3.

BASE24-BASE KEY FILE		QA60	07/05/25 13:52	03 OF 04
DPC/FIID: IMN		INTERFACE PROCESS: ACIT0103A		
PIN KEY VARIANT: 1		MAC KEY VARIANT: 3		
LIMITS				
PIN KEY TIMER VALUE: 90		MAC KEY TIMER VALUE: 90		
PIN KEY TIMER INTERVAL: M (MINUTES)		MAC KEY TIMER INTERVAL: M (MINUTES)		
PIN KEY TRAN: 50		MAC KEY TRAN: 50		
PIN KEY ERROR: 10		MAC KEY ERROR: 10		
CONSECUTIVE PIN KEY ERROR: 10		CONSECUTIVE MAC KEY ERROR: 10		
		KMAC SYNCHRONIZATION ERROR: 10		
CLEAR OLD KEY TIMER VALUE: 0 (SECONDS)				
ORIGINATING ID:				
RECEIVING ID:				
KEY PROCESSING TYPE: N (NONE)				
NOTARIZATION SUPPORTED: 0 (NOT SUPPORTED)				
RECORD LAST CHANGED: 03/09/18 16:56 BY USER: 0255 , 00000255 CHANGE				
***** BASE24 *****				
NEW PAGE:		FILE DESTINATION:		NEW LOGICAL NETWORK ID:
F12-HELP				

KEYF SCREEN 3	
Field	Value
PIN KEY VARIANT	0 for PIN translation directly from the PIN KEK parts, or 1 for translation that includes the variant 1 constant with the PIN KEK parts, not used
MAC KEY VARIANT	0 for translation directly from the MAC KEK parts, or 3 for translation that includes the variant e constant with the MAC KEK parts. This field must be set to 3 when using dynamic key management, not used.
PIN KEY TIMER VALUE	0 or 5–1500 if timer interval is minutes; 0–1000 if timer interval is hours; or 0–180 if timer interval is days, not used.
MAC KEY TIMER VALUE	0 or 5–1500 if timer interval is minutes; 0–1000 if timer interval is hours; or 0–180 if timer interval is days, not used.
PIN KEY TIMER INTERVAL	D for days, H for hours, or M for minutes, not used.
MAC KEY TIMER INTERVAL	D for days, H for hours, or M for minutes, not used.

PIN KEY TRAN	0 if a total transaction limit is not used or 50–100000, not used.
MAC KEY TRAN	0 if a total transaction limit is not used or 50–100000, not used.
PIN KEY ERROR	0 if an error limit is not used or 5–9999, not used.
MAC KEY ERROR	0 if an error limit is not used or 5–9999, not used.
CONSECUTIVE PIN KEY ERROR	0 if a consecutive error limit is not used or 5–9999, not used.
CONSECUTIVE MAC KEY ERROR	0 if a consecutive error limit is not used or 5–9999, not used.
KMAC SYNCHRONIZATION ERROR	0 if a MAC working key synchronization error is not used or 3–9999, not used.
CLEAR OLD KEY TIMER VALUE	0 if the old key is not cleared or 1–9999 (seconds), not used.
ORIGINATING ID	ID of sender of dynamic key management messages, not used.
RECEIVING ID	ID of receiver of dynamic key management messages, not used.
KEY PROCESSING TYPE	C if each is only responsible for generating its own outbound keys, M if the is responsible for all key generation, S if the receives keys but does not generate them, or N if dynamic key management is not used, not used.
NOTARIZATION SUPPORTED	“0” , not used.

KEYF Screen 4

KEYF screen 4 contains the message encryption parameters used to support message encryption in IMNi process. The BASE24 Transaction Security Services process is required to support message encryption.

```

BASE24-BASE KEY FILE          QA60          07/08/31  10:54  04 OF 04
DPC/FIID:                     INTERFACE PROCESS:

                                MSG KEY INFORMATION

FULL MSG ENCRYPTION: 0 (NO ENCRYPTION)
MSG ENCRYPT TYPE: 0 (NO ENCRYPTION)

CHECK DIGITS: 0000
OUTBOUND KEY COUNTER: 000000    INBOUND KEY COUNTER: 000000

LIMITS
MSG KEY TIMER VALUE:      0
MSG KEY TIMER INTERVAL: M (MINUTES)
MSG KEY TRAN:            0
MSG KEY ERROR:           0
CONSECUTIVE MSG KEY ERROR: 0

RECORD LAST CHANGED: 06/04/20 08:41 BY USER: 0255 , 00000255 ADD
***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
F12-HELP

```

KEYF SCREEN 4	
Field	Value
FULL MSG ENCRYPTION	0 = No message encryption 1 = Full message encryption 2 = Encryption on specific field. This option is not currently support.
MSG ENCRYPT TYPE	0 = Clear Data 1 = Security module message encryption
CHECK DIGITS	4 hexadecimal characters
OUTBOUND KEY COUNTER	6 hexadecimal characters
INBOUND KEY COUNTER	6 hexadecimal characters
MSG KEY TIMER VALUE	0 or 5–1500 if timer interval is minutes; 0–1000 if timer interval is hours; or 0–180 if timer interval is days, not used.
MSG KEY TIMER INTERVAL	D for days, H for hours, or M for minutes, not used.
MSG KEY TRAN	0 if a total transaction limit is not used or 50–100000, not used.
MSG KEY ERROR	0 if a total transaction limit is not used or 50–9999, not used.
CONSECUTIVE MSG KEY ERROR	0 if a consecutive error limit is not used or 5–9999, not used.

Interchange Security File (ISECD)

ISECD contains the encryption/decryption keys referenced by key locators and the parameters required by the IMNi/IMNsc process for PIN encryption, PIN translation, message authentication, and data encryption. This file will be created when you run the KEYF conversion program or records can be added, deleted or modified manually from the ACI Desktop.

To support data encryption for the IMNi Interface process, you must configure the data encryption keys and data encryption key exchange keys on the Interface Security Configuration window of the ACI Desktop user interface. This record must correspond to an existing record in the KEYF. If a record does not already exist in the KEYF for the interface, you must first add the record in the BASE24 KEYF Configuration window using the ACI Desktop user interface. Data in the ISECD is maintained by the Interface Security Configuration window.

The Interface Security Configuration window enables you to configure keys and parameters required by IMNi/IMNsc interfaces for PIN encryption, message authentication, and data encryption. Key exchange keys for PIN and MAC keys are entered on the Exchange Information tab. Key exchange keys for data encryption keys are entered on the Data Encr Exch Intf tab.

In Exchange Information tab, the key exchange key (import key) must set to use the same value for both decrypting keys received from the IMNi/IMNsc interface and sending keys to the IMNi/IMNsc interface. PIN, MAC, and data encryption working keys are entered on the PIN Information, MAC Information, and Data Encr Key Intf tabs, respectively. Static master PIN and MAC keys used to encrypt and decrypt transport (key exchange) and session (working) keys for dynamic online distribution are entered on the Master Keys tab. The Data Encr Exch Intf and Data Encr Key Intf tabs are currently used for the IMNI interface only.

Please refer to Transaction Security Service Processing Guide for details.

Section 3

Store-and-Forward Processing

The purpose of the Store-and-Forward (SAF) subsystem is to ensure that a message is delivered to its destination. Therefore, to protect against a failure at the approximate same time that a SAF request message is received, the message must be written to the SAF file before the reply is sent to the originator. The message may be sent to the destination at any time, except when the System Generated Reversal is being generated by the local AIL, in which case this reversal must be written to the SAF file before being transmitted. Any malfunction in the SAF logging function must be treated as a critical error, and the connection to a destination must be disabled if the associated SAF file is not available.

IMNi/IMNsc uses its Store-And-Forward mechanism for transactions that must complete for financial integrity, and for administrative messages that are not time-critical. These messages include message types 0220, 0420, 0422, 0600, 0602, 0620, and 0622. Please note that message types 0220 and 0422 are applicable to IMNi only. IMNi/IMNsc records these messages in a Store-And-Forward file and attempts delivery until IMNi/IMNsc receives an appropriate response or the operator issues a command to delete the request.

This section describes store-and-forward processing and includes information on what messages can be placed in the SAF, when messages are placed in the SAF, and when messages from the SAF are sent to the destination.

The Store-and-Forward File (SAF)

The Store-and-Forward File (SAF) holds “must complete” messages bound for a destination when it is not possible to deliver the messages in real time. These Store-and-Forward requests are held and will be deleted when the responses arrives.

The store-and-forward cycle used by IMNi/IMNsc is first-in/first-out, so the oldest SAF records are sent to the destination first. SAF messages are sent from the SAF file without waiting for the reply to a previous transaction.

Messages are stored in the SAF formatted in the way they are to be sent to the destination.

Messages Placed in the SAF

The IMNi/IMNsc process places the following message types in the SAF as soon as they arrive from BASE24:

0220 — Financial completion
0420 — System Generated Reversal
0422 — Issuer System Generated Reversal
0600 — Administrative Acquirer Request
0602 — Administrative Issuer Request
0620 — Administrative Acquirer Adjustment Advice Request
0622 — Administrative Issuer Adjustment Advice Request

Messages Not Placed in the SAF

The IMNi/IMNsc process never places the following message types in the SAF:

0100 — Authorization Request
0110 — Authorization Request Response
0200 — Financial Transaction Request
0210 — Financial Transaction Request Response
0230 — Financial Completion Response
0430 — Reversal Advice Response
0610 — Administrative Acquirer Response
0612 — Administrative Issuer Response
0630 — Administrative Acquirer Adjustment Advice Response
0632 — Administrative Issuer Adjustment Advice Response

0100, 0110, 0200 and 0210 messages are interactive messages and their timing is crucial. If they are not processed within a given period of time, the transactions they represent are declined. Therefore, these messages are not placed in the SAF.

0230, 0430, 0610, 0612, 0630, and 0632 messages are acknowledgments or responses to SAF requests. If IMNi/IMNsc does not receive a SAF response within a given time, IMNi/IMNsc assumes the request was never received and retransmits it.

IMNi/IMNsc does not place messages received from the IMN node in the SAF file. In particular, IMNi/IMNsc forwards 0420 requests to a BASE24-pos/BASE24-atm Authorization process and does not expect a response, and IMNi/IMNsc stores incoming 0600, 0602, 0620 and 0622 in the IMN Admin file and does not forward them further.

Deleting Messages from the SAF

Records are deleted from the SAF under the following circumstances:

1. IMNi/IMNsc receives a successful response, with “00” in the ISO response code (BM-39) and in the IMN status code.

2. IMNi/IMNsc receives a response with an edit error (Response Code 06, 12, 13, 14, 81, 82, Q0), routing error (RC 92), or invalid capture date (RC 77), returned in the ISO response code (BM39).
3. The NCS operator delivers a SAFDELETE command to the IMNi/IMNsc process. In this case, the IMNi/IMNsc process deletes the oldest outstanding SAF request for the specified destination.

No messages are deleted from the SAF at cutover.

Forwarding SAF Messages to a Destination

The NCS operator can select between the SAF modes listed below for a destination.

- NORMAL - IMNi/IMNsc will store new SAF requests in the SAF file, and forward SAF requests from the SAF file.
- STORE - IMNi/IMNsc will store new SAF requests in the SAF file, but will not forward SAF requests from the SAF file.
- FORWARD - IMNi/IMNsc will forward existing SAF requests from the SAF file, but will not store new SAF requests in the SAF file. IMNi/IMNsc will decline all requests from BASE24, including 0200 financial requests.
- SLOW- IMNi/IMNsc will forward SAF requests from the SAF file based on Slow timer interval, which is typically 60 seconds.
- FAST - IMNi/IMNsc will forward SAF requests from the SAF file based on Fast timer interval. The FAST mode must be capable of emptying the SAF quickly without interfering with existing traffic. The Fast timer interval must be capable of a value as small as .01 second.

The NCS operator delivers SAFMODE command to the IMNi/IMNsc Manager process for a destination and message class to control the SAF processing.

The NCS operator can also select between SAF FAST mode or SAF SLOW mode for a destination. If the NCS operator has selected SAF FAST mode, IMNi/IMNsc may automatically switch to SAF SLOW mode as described below.

IMNi/IMNsc reads and sends SAF requests from the SAF file at the SAF FAST or SLOW Transmission Interval depending on the current SAF mode of the destination. If SAF request times out or receives an unsatisfactory response, IMNi/IMNsc retransmits the request after the SAF retransmission interval.

If the current SAF mode is FAST, IMNi/IMNsc will automatically switch to SAF SLOW mode for that destination after IMN-SAF-SLOW-THRESHOLD consecutive SAF request failures. If the current SAF mode is SLOW, and the SAF mode was not set to SLOW through an NCS operator command, IMNi/IMNsc will automatically switch the SAF mode back to FAST after IMN-SAF-FAST-THRESHOLD consecutive successful SAF requests.

Section 4

Transaction Handling

This section describes how IMNi/IMNsc handles various types of transactions. This section describes the basic flow of transaction processing and provides transaction flow diagrams illustrating how messages to and from the IMN node are handled.

Introduction

The following transaction flows are described in this section. In most cases, these flows are described twice: once when the message flow is initiated by BASE24, and once when the message flow is initiated by the IMN node.

- Requests
- Reversals
- Timeouts

Logging to the ILF

IMNi/IMNsc logs messages to the Interchange Log File (ILF) under the following circumstances:

- Response messages flagged for logging on page 14 of the ICF control record.
- Messages relating to security error conditions or other exceptional conditions
- SAF transactions

Whenever a reversal occurs, the IMNi/IMNsc process attempts to update the ILF record already logged for the transaction. Please refer to section 7 for more detail on ILF.

Message Flows to Remote IMN Nodes

The following pages describe the message flows for the following transactions outbound from BASE24 to the IMN node.

- Normal Transaction Requests (i.e. 100, 200, 6xx)
- Reversals of 0210 Responses (i.e. 0420)
- Timeouts of outbound 0200 Requests to the remote IMN node.

For the IDP and SCD message flows diagram communicated between the acquirer and issuer node, please refer to IDP or SCD Functional Specification.

Request Message to Remote IMN Nodes

The procedure below explains the message flow when a transaction is initiated by the BASE24 network and must be sent to a remote IMN node for authorization.

1. A BASE24-pos/BASE24-atm Authorization process sends the request to IMNi/IMNsc in the internal message format.
2. IMNi/IMNsc reformats the message into the external message format (ISO 8538 standard), translating the encrypted PIN block and generating a Message Authentication Code; uses the PAN in the request to select a destination node from the IMN IIN table; builds the IMN message header; sends the external message to the IMN node for delivery to the destination node; and starts an outbound request timer to wait for a response from the IMN node.
3. The IMN node routes the transaction to the appropriate authorizer. After the transaction has been approved or declined, the IMN node returns the response, in external message format, to IMNi/IMNsc.
4. IMNi/IMNsc deletes the outbound request timer, verifies the Message Authentication Code in the external format response, translate the account balance (IMNi only), formats a internal response message, sends the response to the BASE24-pos/BASE24-atm Authorization process to be delivered to the POS/ATM device, and if required logs a copy of the transaction to the Interchange Log File (ILF).

Reversals to Remote IMN Nodes

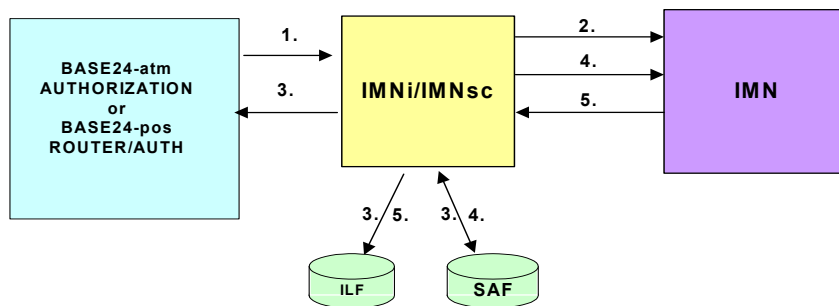
When a POS/ATM device connected to the local BASE24-pos/BASE24-atm network fails to complete a transaction as authorized, a reversal message must be sent to the IMN node. In this case, the POS device/ATM process will generate a 0420 reversal and forwards this reversal message to IMNi/IMNsc via BASE24-pos router process/BASE24-atm authorization process. IMNi/IMNsc

stores this request in its SAF file, and forwards this message according to the requirements for SAF processing.

When IMNi/IMNsc detects a problem in a 0210 response, such as an edit error or MAC failure, IMNi/IMNsc logs a 0210 transaction in the Interchange Log File (ILF) before returning a 0210 decline response to the BASE24 Authorization process, generates a 0420 reversal, stores the reversal in its SAF file, and forwards the reversal to the IMN node according to the requirements for SAF processing.

Timeouts of outbound 0200 Requests to the IMN Node

Each time IMNi/IMNsc sends a 0200 financial transaction request to the IMN node for authorization, IMNi/IMNsc starts an outbound request timer. If the outbound request timer expires before a response is received from the IMN node or the IMN node returns a timeout response with IMN status code 81, the transaction is timed out. The diagram below illustrates message routing when a financial transaction times out. The steps in the diagram are described below.



1. The BASE24 Authorization process sends the 0200 request to IMNi/IMNsc in internal message format.
2. IMNi/IMNsc reformats the 0200 message into external message format, sends the external message to the IMN node for authorization, and starts an outbound request timer to wait for the response from the IMN node.
3. When the timer expires before a response is received from the IMN node or the IMN node returns a response with IMN status code 81, IMNi/IMNsc formats a 0210 internal response message denying the transaction, and sends the 0210 response to the BASE24 Authorization process. IMNi/IMNsc then generates a 0420 reversal, logs a copy of the reversal transaction to the Interchange Log File (ILF), and stores the reversal on the SAF file for the destination.
4. IMNi/IMNsc sends the 0420 request to the IMN node for delivery to the remote IMN node according to the requirements for SAF processing.

The IMN node may return a 0210 late response to IMNi/IMNsc. IMNi/IMNsc logs the late response to the ILF. No reversal is sent because a reversal is always generated on timeout. This is an Interac requirement to always send the reversal on timeout and NOT if/when a late response is received.

Rejected Messages to the IMN node

When the IMNi/IMNsc process receives any request message from BASE24-pos/BASE24-atm process and this message is possible duplicated or this message cannot be processed or reformatted for internal use (i.e. invalid message type), IMNi/IMNsc will issue the alarm and drop the request message.

Message Flows from the IMN Node

The following pages describe the message flows for the following transactions inbound to BASE24 from the IMN node.

- Normal Transaction Requests (i.e. 100, 200, 6xx)
- 0420 Reversal Requests
- Timeouts of inbound 0200 Requests

For the IDP and SCD message flows diagram communicated between the acquirer and issuer node, please refer to IDP or SCD Functional Specification.

Request Messages from the IMN Node

The procedure below explains message routing when a transaction is sent from the IMN node to BASE24 for authorization.

1. The IMN node sends the request to IMNi/IMNsc in the external message format.
2. IMNi/IMNsc verifies the Message Authentication Code, reformats the message into the internal message format, sends the internal message to a BASE24 Authorization process for authorization, and starts an inbound request timer to wait for the response.
3. The BASE24 Authorization process sends the response to IMNi/IMNsc in the internal message format.
4. IMNi/IMNsc deletes the inbound request timer, formats an external response message with a Message Authentication Code, returns the response to the IMN node for delivery to the remote IMN node, and if required logs a copy of the transaction to the Interchange Log File (ILF).

Reversals from the IMN Node

When IMNi/IMNsc receives a 0420 reversal from the IMN node, IMNi/IMNsc verifies the Message Authentication Code in the external format request, formats a 0420 request in internal format, forwards the 0420 request to a BASE24 Authorization process, formats a 0430 response in external format with a Message Authentication Code, and returns the 0430 response to the IMN node for delivery to the remote IMN node.

Timeouts of Request Messages from the IMN Node

Each time IMNi/IMNsc sends a financial transaction to a BASE24 Authorization process for authorization, IMNi/IMNsc starts an inbound request timer. If the inbound request timer expires before a response is received from

BASE24-pos/BASE24-atm, the transaction is timed out. The procedure below explains message routing when a financial transaction times out.

1. The IMN node sends the request to IMNi/IMNsc in the 0200 external message format.
2. IMNi/IMNsc reformats the message into the 0200 internal message format, sends the internal message to a BASE24 Authorization process for authorization, and starts an inbound request timer to wait for a response from the Authorization process.
3. When the inbound request timer expires, IMNi/IMNsc formats a 0210 external response message denying the transaction and sends the 0210 message to the IMN node for delivery to the remote IMN node. IMNi/IMNsc formats a 0420 reversal in internal format, and sends the reversal to a BASE24 Authorization process.
4. If the BASE24 Authorization process sends a late 0210 response to IMNi/IMNsc, IMNi/IMNsc logs the late response to the ILF.

Rejected Messages from the IMN node

When the IMNi/IMNsc process receives an external (inbound) request message that this message cannot be processed or reformatted for internal use (i.e. non supported message class, invalid message length), IMNi/IMNsc builds a corresponding response message and an appropriate response code in IMN header portion with the original ISO request message, logs the response message in the ILF, and returns the response to the IMN node for delivery to the requester node.

Network Management and Operations

Network management messages can be divided into two categories: those used to manage the operational status of the communications links between BASE24 and the IMN node, and those used to manage PIN and message authentication keys between BASE24 and the IMN node.

IMNi/IMNsc supports two types of network management messages:

- Pacing messages
- Dynamic key management (DKM) messages

Network management messages are sent as 0800 messages. Key exchange requests require 0810 messages in response. This section describes the message flows that occur when network management messages are sent.

Issuer CNXN State

The Acquirer IMNi/IMNsc maintains the status of each Issuer connection. The status is displayed on Page 12 of the ICF screen. The possible Issuer status conditions are:

- **Down** - The IMNi/IMNsc will not send requests to the Issuer. This condition is used only in exceptional conditions. This state is set using the NCS "SET" command.
- **Up** - The issuer has received working keys and transactions can be sent to the Issuer. This state can be set using the "SET" NCS command.
- **Key Exchange** - The Acquirer AIL is attempting to send a set of working keys to the Issuer AIL (default value).

Refer to Section 6 for more detail on the "SET" NCS command.

Acquirer CNXN State

The Issuer IMNi/IMNsc maintains the status of each Acquirer connection. The status is displayed on Page 12 of the ICF screen. The possible Acquirer status conditions are:

- **Down** - The Issuer IMNi/IMNsc is not available for receiving messages from the Acquirer. Any messages received over the connection from the Acquirer is returned with response code "91". (Issuer not available) This state can be set using the NCS 'SET' command.

- **Up** - This Issuer IMNi/IMNsc is available for receiving messages from the Acquirer.
- **Ready** - The issuer AIL is initialized but it has not received key exchanges from the acquirer AIL.

The Issuer can set the state of the connection to “DOWN” to be able to operate as an acquirer only. For example, Institution A may mark its issuer connection to Institution B as “DOWN”. As such Institution A would respond to institution B requests with a ‘91’ response code (issuer not available), but institution B could still return approval responses to Institution A requests on Institution A acquirer connection.

Pacing Messages

On every transaction sent successfully to an entry point, IMNi/IMNsc increments a throughput counter. On every response from BASE24, IMNi/IMNsc keeps track of the response times. At the interval defined by IMN-PACING-TIMER-SECS, IMNi/IMNsc will examine these throughput counters and the average BASE24 response time over the period.

For the BASE24 pacing status, IMNi/IMNsc calculates a new average response time from the prior BASE24 average response time and the response time collected only in the last period. The new overall average is calculated with a variable amount of weighting given to the most recent period. This weight is the IMN-PACING-RECENT-WEIGHTING parameter given above. If the new overall average response time for BASE24 has crossed one of the parameter-driven thresholds for congested or blocked, IMNi/IMNsc changes its pacing status, and sends appropriate pacing messages to the IMN node. The higher the weighting parameter value (Max 100), the more sensitive the pacing mechanism will be to temporary problems.

A similar procedure is used for the entry points, except that the figures being manipulated are the entry point throughputs, not response times. The prior throughput is averaged with that of the current period, using the recent weighting. The resulting overall throughput is used, in combination with the queue sizes for the entry points, to select the entry point to be used for each transaction. Again, a high recent weighting factor will result in a highly sensitive routing mechanism. To high a weighting will result in an over-reaction to temporary problems.

Dynamic Key Management Messages

IMNi maintains three types of working security keys: message authentication (MAC) keys, PIN encryption (PIN) keys and Message Encryption Key (MSG) keys. IMNsc maintains two types of working security keys: message authentication (MAC) keys and PIN encryption keys. For each remote Issuer node, IMNsc maintains two keys of each type, which are generated and sent to the remote node by the acquirer. In addition, for each AIL ID at each remote

node, IMNsc maintains two keys of each type, which the remote node uses for communication with the IMNsc.

For more information please refer to section 7-Dynamic Key Management.

Section 5

Cutover and Settlement

Settlement is the process by which the claims that Interac members have against each other are paid. Members are required to make payment by passing payment items through the IDP and SCD systems operate with one cutoff point daily. Cutoff is at 2100 hours EST/EDT daily, seven days a week. Settlement totals and associated logs and reports are produced daily. Based on the 2100 hours cutoff time, the settlement period is (21:00:00 to 20:59:59). In order to minimize "in-flight" discrepancies, log scanning should not be done until at least thirty minutes after cutoff time.

Each Acquirer inserts the capture-date into all request financial messages based on the time at the Acquirer AIL (not local time at terminal), which forms the basis upon which both the Acquirer and Issuers establish which transactions are included in each period's settlement. Capture date and settlement date have the same meaning. An initial capture date is entered into the IMNi/IMNsc by switch date parameter in ICF. If no date is entered, the system date is used, adjusted for the time of day.

Since 2100 hours is the settlement point for each day, the capture-date must, therefore, be changed to the next calendar date by the IMNi/IMNsc acquirer, at 2100 hours EST/EDT daily, for subsequent transaction requests. An IMNi/IMNsc issuer also change its settlement (capture) date at 2100 hours, but because of clock discrepancies, the IMNi/IMNsc issuer still accept messages with the capture date set to the current and next day. For the period from 2100 to 2130 hours, messages from the current (new) and previous day must be accepted. SAF messages, which were originally created before 2100 hours continue to use the old capture date until 2129 hours.

Interac Cutover Processing

At the beginning of the cutover window, IMNM updates the switch posting date on the control record of the ICF. All IMNi/IMNsc processes use the new date in the capture date of all requests sent to the IMN node.

At the end of the cutover window, all IMNi/IMNsc processes close the ILF file for the old switch posting date, and create and open an ILF file for the day after the new switch posting date. The ILF file for the new switch posting date is already open, and remains open. After the end of the cutover window, IMNi/IMNsc rejects 0200 requests from the IMN node with response code 77, invalid capture date.

After the end of the cutover window, users can extract transactions from the ILF or TLF/PTLF logs for settlement with other Interac institutions.

LNCF parameters IMN-BAL-AND-CUTOVER-START and IMN-BAL-AND-CUTOVER-END define the beginning and end of the Interac cutover window.

After each cutover point, the IMNi/IMNsc acquirer sends the 0200, original 0220, or 0620/0622 messages with the new capture date (BM17). However, the IMNi/IMNsc issuer still accepts these transactions with the old capture date. For the SAF messages after 2100 hours must contain the same capture date and reconciliation slot number as the original transaction request which was sent or which is being reversed, until 2129 hours.

Acquirer IMNi/IMNsc Tasks

- Change capture date at 2100 hours.
- Insert current capture date on all 0100, 0200, original 0220, and 0620/0622 messages.
- On 0220 and 0620/0622 messages sent from the SAF, remains the same capture date as the original until 2129 hours.
- 0420 and 0422 messages have the same capture date as the message being reversed until 2129 hours.
- Do not send 0420 reversal for timeout after 2129 hours.
- Do not log messages in ILF until the response is received and log to the same day's ILF file as the capture date in the response.

Issuer IMNi/IMNsc Tasks

- Log to the same day's ILF file as the capture date in the request until the old log ILF file is closed.
- Accept message with the old capture date until the old ILF file is closed.
- After the old ILF file has closed, SAF message is logged to the new ILF file. IMNi/IMNsc sends the response message with the new capture date and set reconciliation state number to 1. The non SAF messages with the old capture date and 0420 reversals message for timeout are declined with response code 77 (Invalid Capture date).

BASE24 Cutover Processing

When an IMNi/IMNsc process receives a cutover message from a BASE24-atm Settlement Initiator process, IMNi/IMNsc updates its current POS/ATM business date. IMNi/IMNsc includes this date in inbound request messages before sending to BASE24-pos/BASE24-atm.

SAF Cutover Processing

At the end of the SAF cutover, the capture date is set to the new date for 0620 and 0622 messages. The capture date does not change for 0420 reversal messages.

Section 6

Process Setup and Usage

This section provides information on the configuration of the IMNi/IMNsc process. This section also describes the processing involved for initialization and network control.

Process Configuration and Control

Startup

IMNi/IMNsc and IMNM processes should be assigned a startup logic of AUTOMATIC in the Network Environment Source File (NEFS), or simply set the STARTUP attribute to "AUTOMATIC" via the Network Control Supervisor (NCS) screen. AUTOMATIC means that the process will be started automatically when the resource node is initiated. This ensures that the IMNi/IMNsc process is ready when it receives its first transaction.

The priority of the IMNM process should be set higher than that of the IMNi/IMNsc processes.

Log Messages

The IMNi/IMNsc process generates log messages to assist network operators in performing their daily tasks. Log messages are unsolicited messages generated by BASE24-pos/BASE24-atm processes that provide information regarding the internal condition of the system. They are a network operator's primary link to what is actually occurring in the system. Log messages provide operators with information ranging from system status, such as the completion of a procedure, to more serious circumstances requiring operator intervention.

For more information on the log messages, please refer to the BASE24 IMNi/IMNsc Interface Log Messages Manual.

IMNi/IMNsc Data Control Blocks

All global variables controlling process execution are defined and organized into a set of control blocks. The data structures are as follows:

ATM Control Block (ATMCb) and POS Control Block (POSCb)

These tables contain the variables specific to BASE24-atm or BASE24-pos authorization process i.e process name, pacing state, and transaction allowed. These tables are initialized during startup based on the information in LNCf and ICF. ATM Control Block and POS Control Block are used by IMNsc and IMNi respectively.

Configuration Control Block (ConfigCb)

This control block contains variables controlling process configuration and operation that are dynamically obtained i.e. SAF notify interval. Fields are set during initialization from various LNCf and ICF parameters.

Destination Control Table (Dct)

This table contains the secondary DCT fields i.e. keys information, connection state which are allocated in the extended segment. This table must be maintained in parallel with the PriDct.

Entry Point Table (Ept)

This table contains the definition, state (i.e. pacing, handshake) and other processing related fields for each Commonitor entry point available to the IMNi/IMNsc. This table is build during startup based on the information in the EPCF file (Entry Point Configuration File).

File Control Table (Fct)

This table contains disk files information accessed during execution i.e. specific filename, file state. Fields are initialized during startup from LNCf or ICF parameters. State information is updated during execution. Specific Fct entries are pre-assigned to the files accessed.

IMN Manager Process Control Block (IMNMCb)

This control block contains variables controlling access to the IMNM process i.e. IMNM process name. Fields are set during initialization from LNCf parameter and updated during execution as states change.

IMNi/IMNsc Primary Destination Control Table (PriDct)

This table contains the main DCT fields i.e. Interac node ID, message class and node name which are allocated in primary global data space. This table must be maintained in parallel with the DCT.

Interchange Logging Control Block (ILFCb)

This control block contains the ILF information which defines and controls each opened ILF log file usage. This control block is initialized during start up based on parameters in LNCf and updated during processing.

Network Definition Control Block (NetCb)

This control block contains IMNi/IMNsc configuration variables i.e. its own Interac Node ID, service type (acquirer or issuer). These variables are initialized during startup from the information in LNCF and ICF.

Process Control Block (ProcCb)

This control block contains variables that define the state of the running process, i.e. process state, process ID. Fields are set during initialization and updates as states change during operation.

SAF File Control Block (SAFCb)

This control block contains the Store-and-Forward (SAF) information which defines and controls each opened SAF file usage. This control block is initialized during start up based on information in ICF and updated as processing states change.

Security Device Control Block (Sdcb)

This is a standard structure defined a security device information used in BASE24 product.

Settlement Control Block (SetlCb)

This structure contains the variables related to current switch date and time, and cutover timer. This control block is initialized during startup based on the parameters in LNCF.

Timer Control Block (TimerCb)

This is a BASE24 standard structure defined to use in timer processing.

Timeout Table (TimeoutTbl)

This table contains the processing timeout interval values i.e. key exchange timeout, pacing abnormal timeout. This table is initialized during startup based on the information in LNCF and ICF.

IMNi/IMNsc Initialization

At initial startup, IMNi/IMNsc performs the following procedures:

1. Reads the name of its LNCf. The startup message from the Nucleus contains the name of the LNCf. This name allows the IMNi/IMNsc process to access the LNCf for the assigns and params needed for processing.
2. Opens the LNCf, reads the assigns and parameters listed in Section 2. If an error occurs, the IMNi/IMNsc process logs an error message and abends.
3. Closes the LNCf.
4. Initializes the extended memory required for this process.
5. Opens the Token File (TKN) if an TKN assign was found in the LNCf.
6. Reads the token log records into extended memory.
7. Closes the TKN File.
8. Reads the appropriate TKN records into extended memory.
9. Opens the Interchange Configuration File (ICF). If an error occurs, the IMNi/IMNsc process logs an error message and abends.
10. Reads IMN control record from the ICF and obtains the local node information. i.e. local IMN node ID.
11. Determines the posting dates for BASE24-pos/BASE24-atm by comparing the current time to the BASE24-pos/BASE24-atm cutover times taken from the LNCf.
12. Updates the ICF with the current posting dates
13. Allocates extended memory segment.
14. Opens the Interchange Log Files (ILFs) for the current switch posting day, and the next day. If an error occurs, the IMNi/IMNsc process logs an error message and abends.

15. Repeats these following steps until there are no more IMN destination records in ICF.
 - Reads IMN destination record from ICF to obtain the other remote node information.
 - Opens and reads the IMN destination record from the Key File (KEYF). If an error occurs, the IMNi/IMNsc process logs an error message and abends.
 - Builds the Destination Control Table (DCT) from ICF and KEYF records that apply to this process.
 - Opens the Store-And-Forward (SAF) File and updates DCT with SAF information.
 - Builds FIL Key Table (FKT) from this destination KEYF record.
 - Closes the ICF file.
16. Verifies that all Key values loaded in the FKT table are all unique.
17. Opens the Entry Point Control File (EPCF). If an error occurs, the IMNi/IMNsc process logs an error message and abends.
18. Reads the entry point definitions from the EPCF.
19. Builds the BIN Routing Table (BINT) for each card prefixes and its IMN Node ID from the Interac BIN table (CONFIIN).
20. Opens the IMN Admin File.
21. Opens the security modules of Transaction Security Service (TSS) subsystem defined in LNCF.
22. Opens a link to a specific CIL entry point.
23. Opens a link to the IMNi/IMNsc Manager (IMNM) process.
24. Cutovers any old records in the SAF file if it is in the cutover window.
25. Uploads SAF statistics to the IMNM process.
26. Initializes process timer and starts the timers.

27. Waits for a message from the IMN node or a local BASE24-pos/BASE24-atm Authorization process.

IMN Manager (IMNM)

Overview

The IMN Manager process is responsible for several tasks relating to the automatic management of the IMNi and IMNsc processes. These tasks include:

1. The accumulation of key usage counters;
2. The triggering of key changes on configured-driven threshold and at the end of cutover;
3. The transfer of new key information between multiple IMNi/IMNsc processes;
4. The maintenance of IMNi/IMNsc outbound pacing status;
5. The selection of alternate IMNi/IMNsc processes for each destination;
6. The performance of Base24 NCS commands for status inquiries and some SAF management;
7. The initiation of simultaneous IMNi/IMNsc warmbooting;
8. The retrieval of source and destination node ID and switch date for the IMN Admin server process; and
9. The updating of the switch date at start of cutover.

Please note that if running the IMNi and IMNsc processes in the same logical network, there must have 2 IMN Manger processes; one for the IMNi and other for IMNsc. In additional, the IMN Manager object for IMNi and IMNsc must be conformed to the IMNi and IMNsc object, respectively.

IMNM Data Control Blocks

The IMNM process maintains a number of internal tables and structures that control its processing as follows:

Destination Control Table (Dct)

This table contains the secondary DCT fields i.e. keys information, connection state, SAF state, pacing status, and number of supporting IMNi/IMNsc processes.

File Control Table (Fct)

This table contains disk files information accessed during execution i.e. filename, file number for LNCf, ICF, LMCF, and SAF template files.

IMNi/IMNsc Process Table (Ipt)

This table contains the process symbolic name, process ID ,number of I/O outstanding, link status for each IMNi/IMNsc process managed by IMNM.

Network Control Block (NetCb)

This control block contains its Interac Node ID, service type (acquirer or issuer), product ID, and FI ID.

Process Control Block (ProcCb)

This control block contains IMNM process name, cpu, process id, process state, trace bit settings and the number of entries for SAFfct, dct and ipt tables.

Primary Destination Control Table (PriDct)

This table contains the main DCT fields i.e. Interac node ID, message class and node name. This table must be maintained in parallel with the DCT.

SAF File Control Block (SAFCb)

This control block contains the all the SAF files information i.e. SAF notify percentage, SAF forward percentage and the SAF slow interval.

SAF File Control Table (SAFFct)

This table contains the SAF filename, file number, maximum number of files records, recalculation flag, and the destination index that uses this SAF file.

Settlement Control Block (SetlCb)

This table contains the cutover start and times and the cutover key exchange delay timer.

Timer Control Block (TimerCb)

This is a standard timer control block used in BASE24 product.

Incoming Messages to IMNM

The IMNM receives messages from the following processes:

1. The IMNi/IMNsc processes send the following messages to the IMNM directly (via a pro^writeread procedure call).
 - Key used messages are sent whenever the IMNi/IMNsc usage of a key has passed a defined threshold.
 - Key Change Sent messages are sent when a key exchange request message has been sent out to the associated IMN member.
 - Key Change Complete message are sent when IMNi/IMNsc receives an approval of the key exchange response message.
 - Key Change Failed messages are sent when a key exchange message has failed.
 - Key Change Required messages are sent when a key exchange is required.
 - Pacing Notify messages are sent when a change of its status has taken place for an outbound message.
 - SAF Statistics message are sent when the number of writes to a SAF has passed a defined threshold.
 - IMNi/IMNsc Up message containing pacing notification data is sent when the IMNi/IMNsc process has opened the IMNM process.
2. The IMN Admin server process sends the following messages directly to the IMNM (via PATHSEND).
 - Get Source message is sent when the IMN Admin screen is enter or to obtain the source node ID.
 - Edit Destination message is sent when the IMN Admin screen information is validated or to look up a node ID and message class in the destination control table or to obtain the switch date.

3. The NCS screen or NCPCOM sends the IMNM the following operator command messages through XPNET.
 - New MAC/MSG/PIN Key command is sent to initiate a key exchange.
 - Key Status command is sent to determine the key states.
 - Set command is sent to set the state for a connection.
 - SAF Mode command is sent to set the operator SAF mode.
 - SAF Status command is sent to determine the SAF status.
 - Warmboot command is sent to warmboot the IMNM and all its IMNi or IMNsc processes.
 - Trace command is sent to set and clear trace bits.
 - Debug command is sent to activate the IMNM in debugging mode.
 - Dump command is sent to cause IMNM to do a programmatic dump.

Outgoing Messages from IMNM

The IMNM sends the following messages directly to the IMNi/IMNsc processes (via a pro^writeread procedure call).

1. Key Change commands are sent when the IMNM determines that a new given key is required.
2. Set Key Info commands are sent to all IMNi/IMNsc processes connected to the IMN destination node when an IMNi/IMNsc has completed a key exchange for the IMN destination node or has received a key synchronization error response.
3. Alter Pacing Destination commands are sent when an IMNi/IMNsc does not have a valid alternate IMNi/IMNsc process for an IMN destination node and an alternate IMNi/IMNsc process has become available.
4. SAF State messages are sent when the IMNM process starts, or the SAF state change as a result of SAF statistics message received from IMNi/IMNsc processes.

5. Set commands are sent when an operator issues a SET command in NCPCOM or NCS to change the state of the connection.
6. Warmboot commands are sent to each IMNi/IMNsc process during processing of a NCPCOM/NCS warmboot command.

Network Control Commands

This section provides a list of the BASE24-pos/BASE24-atm commands that the operator can send to the IMNi/IMNsc process. Many of these commands can be entered by an operator via the Network Control Supervisor (NCS) screen and serve as a tool that operators can use to perform network management functions such as changing processing parameters, warmbooting and monitoring processes.

Commands sent to an IMNi/IMNsc Process

The command syntax:

>deliver <process symbolic name>, <command>

COMMAND	DESCRIPTION
9501	Warmboot alias, see description for "WARMBOOT" command.
9502	Cutover alias, see description for "CUTOVER" command.
ADX <new file extent value>	Changes the trace audit file extent parameter.
ANA <audit file A name>	Change the trace audit file A name.
ANB <audit file B name>	Change the trace audit file B name.
ATMSTATUS	For SCD only. Request the overall status includes the following items: • Current status • Total response time • Current transaction number • Average response time • Congested threshold number • Blocked threshold number • Recent weight factor number The requested information is sent to the current logging facility.

CNXNSTATUS	Request the overall connection status includes the following items for each connection: • Acquirer and issuer connection state • Connection status • Outbound and inbound entry point state The requested information is sent to the current logging facility.
CUTOVER	Message sent from BASE24-pos/BASE24-atm Settlement Initiator to report BASE24 cutover.
DEBUG	Places an IMNi/IMNsc process in the DEBUG state. The debug message places the IMNi/IMNsc process into Inspect and brings up an Inspect prompt on the home terminal (the terminal on which the network was brought up). Inspect is a HP programming tool. This command should only be used to interactively debug a running process when in a test environment. Executing this command in a production environment can result in a significant impact on system performance.
DISPLAY {DCT} {DCT SAF} {DCT KEY} {TIMERS} {EPT} {TRACEBITS}	Displays the contents of various tables, filenames, timers etc...
DUMP	Directs an IMNi/IMNsc process to request a programmatic dump. A snapshot dump is taken which can then be read using RDEBUG.
INA	Request information for the audit file names and file extent.
PACING STATE { NORMAL} { BLOCKED }	Changes the overall pacing state of the IMNi/IMNsc process.
PERFORMANCE	Displays the current status of the interchange

{INBOUND} {IN} {OUTBOUND} {OUT} {REPEAT}	including the accumulated performance information.
POSSTATUS	For IDP only. Request the overall status includes the following items • Current status • Total response time • Current transaction number • Average response time • Congested threshold number • Blocked threshold number • Recent weight factor number The requested information is sent to the current logging facility.
RELOAD EP { ALL } { <EP NAME> }	Re-reads entry point definitions for all entry points or a specified entry point. IMNi/IMNsc closes the entry points and re-opens the entry points with their new definitions.
SAF DELETE <DEST NODE ID> <MSG CLASS>	Deletes the oldest outstanding SAF request for a destination.
SHOW DCT	Shows the contents of the DCT table and CGT tables. The new DCT fields shown are different from the DISPLAY command.
START EP { ALL } { <EP NAME> }	Initiates communication on all entry points or a specified entry point.
STATUS	Requests the status of an IMNi/IMNsc process. The requested information is sent to the current logging facility.
STOP EP { ALL } { <EP NAME> }	Terminates communication on all entry points or a specified entry point.

SWA	Switch from trace audit file A to B or file B to A.
SWL <IMNM Log file name>	Switches to use a new IMNM log file.
TRACE OFF	Stops the trace function within an IMNsc process for a specific trace bit or all trace bits.
TRACE ON [<BIT>]	Starts a trace function within an IMNi/IMNsc process for a specific trace bit or all trace bits. The trace function generates log messages to report specific classes of events. The log messages display the name of the procedure from which it was written or other helpful processing information. The trace function should only be enabled in a test environment. Enabling the trace function in a production environment can result in a significant impact on system performance. <u>BIT number</u> <00> -- \$Receive IO <01> -- Timer activity <02> -- Nucleus CMD Messages <03> -- Nucleus POS Messages <04> -- Entry Point Messages <05> -- IMN Network messages <06> -- IMN Pacing Messages <07> -- Keychange messages <08> -- Open/Close activity <09> -- Initialization <10> -- SAF Activity <11> -- ILF Activity <12> -- Other Disk Activity <13> -- IMNI Manager messages <14> -- SAF Cutover Messages
WARMBOOT	Warmboots an IMNi/IMNsc process. Warmboot processing is similar to process initialization except the IMNi/IMNsc process does not re-read the EPCF. If an error is encountered, the old environment is restored.

Commands sent to the IMNM Process

The command syntax:

>deliver <IMNM symbolic name>, <command>

COMMAND	DESCRIPTION
9501	Warmboot alias, see description for “WARMBOOT” command.
DEBUG	Places an IMNM process in the DEBUG state. The debug message places the IMNM process into Inspect and brings up an Inspect prompt on the home terminal (the terminal on which the network was brought up). Inspect is a HP programming tool. This command should only be used to interactively debug a running process when in a test environment. Executing this command in a production environment can result in a significant impact on system performance.
DUMP	Directs an IMNM process to request a programmatic dump. A snapshot dump is taken which can then be read using RDEBUG.
KEYSTATUS <DEST NODE ID> <MSG CLASS>	Displays the key management status of the specified remote Issuer node.
NEWMACKEY <DEST NODE ID> <MSG CLASS> OR NEWPINKEY <DEST NODE ID> <MSG CLASS> OR NEWMSGKEY <DEST NODE ID> <MSG CLASS>	New Key — IMNM initiates a key exchange with the specified remote Issuer node. The “new msg key” command is used for IDP only.

SAFMODE <DEST NODE ID> <MSG CLASS> { FAST } { SLOW } { NORMAL } { STORE } { FORWARD }	SAF Mode — Sets the SAF mode for a destination. IMNM updates the ICF to record the new state, and relays the SET command to all IMNi/IMNsc processes that support the destination.
SAFSTATUS <DEST NODE ID> <MSG CLASS>	Displays the SAF processing status of the specified remote Issuer node.
SET <DEST NODE ID> <MSG CLASS> { DOWN } { READY } { BLOCKED }	Set destination - sets the status of an Issuer AIL relative to the remote Acquirer AIL. IMNM updates the ICF to record the new state, and relays the SET command to all IMNi/IMNsc processes that support the destination.
SWL <IMNM Log file name>	Switches to use a new IMNM log file.
TRACE ON [<BIT>]	Starts a trace function within an IMNM process for a specific trace bit or all trace bits. The trace function generates log messages to report specific classes of events. The log messages display the name of the procedure from which it was written or other helpful processing information. The trace function should only be enabled in a test environment. Enabling the trace function in a production environment can result in a significant impact on system performance. <u>BIT number</u> <00> -- \$Receive IO <01> -- Timer activity <02> -- Nucleus CMD Messages <03> -- Nucleus SCD Messages <04> -- Entry Point Messages <05> -- IMN Network messages <06> -- IMN Pacing Messages <07> -- Keychange messages <08> -- Open/Close activity

	<09> -- Initialization <10> -- SAF Activity <11> -- ILF Activity <12> -- Other Disk Activity
WARMBOOT	Warmboots the IMNM and IMNi/IMNsc processes. IMNM re-reads its initialization information, and relays the warmboot command to all IMNi/IMNsc processes.

Section 7

Dynamic Key Management

This section describes how IMNi/IMNsc handles the dynamic key management to obtain the integrity and confidentiality of the transactions.

Overview

IMNi maintains three types of security keys: Message Authentication key (MAC), PIN Encryption key (PIN) and Message Encryption key (MSG).

IMNsc maintains two types of security keys: Message Authentication key (MAC), PIN Encryption key (PIN).

For each remote issuer node, the acquirer IMNi/IMNsc process periodically generates a unique PIN Key, a unique MAC and a unique Message key (only in IMNi) for conveyance to the issuer. Once the keys successfully exchange, the acquirer IMNi/IMNsc will begin to use these keys to process requests sent to the issuer and responses received from the issuer. The issuer IMNi/IMNsc, in turn, uses the same keys to process request messages from the acquirer and responses sent to the acquirer. The issuer IMNi/IMNsc also uses these acquirer keys for issuer generated request messages since the issuer IMNi/IMNsc does not generate any keys of its own.

Security Keys

For each remote issuer node, the acquirer IMNi/IMNsc sends key exchange request messages based on the parameter driven thresholds in ICF. Each acquirer IMNi/IMNsc designates a unique time configured in LNCF parameter to carry out daily key exchange with other issuers, to help reduce congestion around peak and cutover times.

Master File Key (MFK)

The Mast File Key is used for protecting other key encrypting keys and working keys (i.e. PIN Encryption Key). This key is only used to encrypt keys for storage on a file. This key is never used to encrypt PIN or data, and will never be shared with another institution. Each institution is responsible to establish and store a MFK either as a plaintext key help under dual control with split knowledge, or as a plaintext key confined to a physical secure device.

Key Exchange Key (KEK)

Key Exchange Keys are required to encrypt new working keys sent to member institutions. The KEKs are different from the MFK in that they are held on a bilateral basis between two institutions. They are encrypted under the MFK and stored in an ISECD file.

Working Keys

Per Interac's regulation, it is a requirement to establish sets of two working keys, a current key and an alternate, for the various security functions. Each acquirer IMNi must establish three working keys for communication with each issuer IDP member. Each acquirer IMNsc must establish two working keys for communication with each issuer SCD member. These keys are as follows:

PIN Encryption Key (KPE)

The PIN encryption Key (KPE) is used to encrypt and decrypt PIN blocks. The acquirer IMNi/IMNsc maintains a KPE counter, by connection, for the number of times an acquirer KPE is used, and to generate a PIN key exchange based on the parameter driven thresholds in ICF, or daily key exchange parameter in LNCF.

Message Authentication Key (KMAC)

This key is used to generate and verify Message Authentication Codes (MAC). The acquirer IMNi/IMNsc maintains KMAC counter, by connection, for the number of times an acquirer KMAC is used, and to generate a MAC key exchange based on the parameter driven thresholds in ICF, or daily key exchange parameter in LNCF.

The same key used by the acquirer IMNi/IMNsc to generate a MAC on a request must be used by the remote issuer IMNi/IMNsc to generate the MAC for the response message.

Message Encryption Key (KME)

This key is used to encrypt certain portions of the message, when necessary. The acquirer IMNi maintains a KME counter, by connection, for the number of times an acquirer KME is used, and to generate a MSG key exchange based on the parameter driven thresholds in ICF, or daily key exchange parameter in LNCF.

Dynamic Key Exchanges

To exchange a key, IMNM selects an IMNi/IMNsc process that can send requests to the remote node, and directs that process to send a key change request (ISO message type 0800). The key change request carries a new MAC'd PIN encryption or message authentication key from an Acquirer node to

an Issuer node. The receiving node verifies the new key and returns a response (ISO message type 0810) to the Acquirer node indicating that the new key has been implemented. The Acquirer node begins using the new key when a successful key change response arrives.

Until a successful key exchange response is received, the Acquirer node continues to use the old key, with the other key index. This ensures that Acquirer and Issuer always share a key they can use for communication.

For each key used by the IMNi/IMNsc in creating the external message, a key index is included in the message. The key index indicates which of the two working keys was used to create the security data.

A cryptographic service message (CSM) contains a message number or counter (BM71) which is associated with each Key Exchange message used in the CSM. The only CSM in the network is a Key Exchange message. The acquirer IMNi/IMNsc maintains a counter for each issuer connection and the issuer IMNi/IMNsc also maintains the expected value of each acquirer connection. If the counter in the CSM is greater or equal to the expected value, the issuer IMNi/IMNsc accepts the key exchange message. If the counter in the CSM is less than the expected value, the issuer IMNi/IMNsc rejects the key exchange message.

Counters are defined and manipulated as binary numbers. The counter value is saved and stored in KEYF with the associated key exchange message, so the IMNi/IMNsc can use this counter to detect any key synchronization errors.

IMNi/IMNsc uses key management messages to exchange new security keys. The IMNM process initiates an exchange of security keys to Issuer destinations according to the key management requirements in the Interac IDP/SCD Service Regulations. Events that trigger key exchanges include:

- IMNM process startup or warmboot.
- The number of times a key has been used reaches the threshold defined on page 13 of the ICF.
- The IMN-DAILY-KEYCHANGE-TIME defined in LNCF has arrived.
- A key synchronization error has been detected.
- IMNM receives an NCS operator command to exchange a key.
- The Issuer node returns a response indicating no keys are available.
- The number of MAC failures or PIN decryption failures reaches the threshold defined on page 13 of the ICF.
- For IDP only, the number of MSG encryption failures reaches the threshold defined on page 13 of the ICF.

Double Length Keys

Double length keys are 128 bits in length (compared to single length keys which are 64 bits in length) and are a pre-requisite to using Triple-DES Encryption Algorithm for PIN and message authentication and/or message encryption between an acquirer and an issuer.

The double length key is an enhancement that will be implemented on the Interac network in the future. Please note that both IMNi/IMNsc supports single length key at this present

Dynamic Key Exchanges at Initialization

At system startup, the acquirer IMNM determines which IMNi/IMNsc process will actually perform the key exchange and directs that IMNi/IMNsc process to generate and distributes working keys. For SCD, MAC keys are exchanged first, then PIN keys. For IDP, MAC keys are exchanged first, then PIN keys followed by MSG keys.

If an Issuer node receives a request before the Acquirer node has exchanged keys with it, the Issuer node returns a response code “85” to indicate no keys are available. The Acquirer node then initiates key exchanges.

Key Exchanges Procedure for an Acquirer IMNi/IMNsc

The key exchange procedure for an acquirer IMNi/IMNsc is as follows:

1. If key index “1” is being used, IMNi/IMNsc sends a key change request message (0800) using key index “2” to the remote issuer node.
2. After the acknowledgement (0810) is received, the acquirer IMNi/IMNsc verifies the response code and determines if a key exchange is successful. If it is a successful key exchange, the IMNi/IMNsc notifies the IMNM process for a successful key exchange. Then, the IMNM broadcast to all IMNi/IMNsc process to start using key index “2”.

The issuer IMNi/IMNsc always uses the same key as indicated in the request key index, so it does not start to use key index “2”.

3. If key index “2” is being used, then key index “1” is to be changed following the procedures in the previous two steps.

Acquirer IMNi/IMNsc Initial Tasks

1. During acquirer IMNM process startup, the IMNM determines which acquirer IMNi/IMNsc process will actually perform the key exchange and send a broadcast message to that IMNi/IMNsc process with key exchange required state based on the key types. The key exchange starts with MAC, PIN and MSG key types (IMNi only).
2. When the acquirer IMNi/IMNsc receives a key exchange request from the IMNM, IMNi/IMNsc issues a key generation command to the security module, specifying a key exchange value of that issuer node, a key index, a key type of “1” (PIN Encryption key), “3” (Message Authentication key) or “2” (Message Encryption key, for IMNi only).
3. IMNi/IMNsc performs as follows:
 - a. Stores an output value of a new working key encrypted under the key exchange key in a key exchange request message (0800) key.; along with a key type, a future key index, the acquirer AIL ID, and a MAC
 - b. Uses an output value of a new working key encrypted under the Master File Key (MFK) as a MAC key. The MAC key is used to calculate the MAC of a key exchange request message before sending to the issuer node.
 - c. Notifies the IMNM that a key exchange request is sent.

Issuer IMNi/IMNsc Initial Tasks

1. Upon receipt of the key exchange request at the issuer node, IMNi/IMNsc translates the key value in a key exchange request message from encryption under Key Exchange Key (KEK) info encryption under the issuer's Master File Key (MFK).
2. IMNi/IMNsc generates the MAC of the request message by using the MAC key that encrypted under MFK and compares if the generated MAC value is as same as the MAC value presented in the request message. If it is not the same value, IMNi/IMNsc replies the key exchange response message (0810) with invalid MAC response code.
3. Then IMNi/IMNsc compares the key counter received in the key exchange request message with an expected value counter. The incoming key exchange request message is rejected with MAC key synchronization error if the counter is not equal to the expected counter value.
4. If no errors are identified, the new key and the counter in the key exchange request are loaded into the issuer's KEYF and its internal tables.
5. IMNi/IMNsc notifies the IMNM that the key exchange is completed, logs the key exchange message in the Interchange Transaction Log file, formats the a key exchange response message (0810) that contains the sequence number counter taken from the original 0800 message and sends a key exchange response (0810) to the acquire node.

Acquirer IMNi/IMNsc Final Tasks

1. When the acquirer IMNi/IMNsc receives a key exchange response message (0810), IMNi/IMNsc examines the response code. If it is a declined key exchange response, IMNi/IMNsc logs the key exchange response in ILF. If it is an approval key exchange response, IMNi/IMNsc generates the MAC value of the receiving response message and compares it with the MAC value presented in the message. If the result is not the same value, IMNi/IMNsc logs the key exchange response message (0810) with invalid MAC response code.
2. Then IMNi/IMNsc compares the key counter received in the key exchange response message with an expected value counter. If the counter is not equal to the expected counter value, IMNi/IMNsc logs the key exchange response message in ILF with key synchronization error.
3. If no errors are identified, IMNi/IMNsc loads the cryptogram key value, counter value and key index in the response message into the KEYF and its internal tables.
4. IMNi/IMNsc activates to the new key and logs the message in ILF.
5. IMNi/IMNsc notifies the IMNM for the key exchange results whether it is failed or successful.

Message Authentication

A Message Authentication Code (MAC) is used to validate the authenticity and integrity of the message and as a means to protect against message modification. All messages are protected by Message Authentication.

In the request message, the acquirer IMNi/IMNsc generates a MAC using the ASCII data of the required fields including the length portion if present and transmits this MAC along with the request. Upon receiving the request, the issuer IMNi/IMNsc computes and verifies the MAC.

Fields in the request message to be used (in BIT Map No. sequence, when present) for generating MACs are:

BIT Map No	Field Name
2	PAN
3	Processing Code
4	Transaction Amount
11	Systems Audit Trace Number
37	Retrieval Reference Number
48	Additional Data Private
71	Message Number
95	Replacement Amount

In the response message, the issuer IMNi/IMNsc generate a MAC using the same key as the request and the required fields and transmit this MAC along with the reply. Upon receiving the response, the acquirer IMNi/IMNsc computes and verifies the MAC. The acquirer IMNi/IMNsc does not verify the MAC if the response message has one of the following response codes: 63 (Invalid MAC), 85 (No keys available for use), 87 (PIN key synchronization error), 88 (MAC key synchronization error), 91 (Issuer is inoperative), 92 (IMN facility cannot be found for routing), or 96 (System Malfunction).

Fields in the response message to be used (in BIT Map No. sequence, when present) for generating MACs are:

BIT Map No	Field Name
2	PAN
3	Processing Code
4	Transaction Amount
11	Systems Audit Trace Number
37	Retrieval Reference Number
39	Response Code
48	Additional Data Private
71	Message Number
72	Message Number Last
95	Replacement Amount

Message Encryption

The Message Encryption is only applied to POS. The only message element, other than the PIN, which needs to be encrypted in the IMN is the account balance information.

A customer account balance must be encrypted in the IMN network. The issuer IMNi encrypts the account balance value in BM120 (Additional Amount) with Message Encryption Key. The acquirer IMNi decrypts the balance upon receipt.

PIN Encryption

PIN Encryption is used to protect the PIN by encrypting a formatted PIN block. The acquirer IMNi/IMNsc passes the PIN data (BM52) to the issuer IMNi/IMNsc encrypted under in the PIN Encryption Key (KPE) and formatted in the ANSI PIN block format. On receipt of a transaction request, the issuer IMNi/IMNsc translates the PIN block by using the KPE index in the message and passes it to the authorization module to verify the PIN.

Section 8

Communication Interface with IMN

This section describes the interface between the IMNi/IMNsc and the IMN.

A number of key concepts are necessary to properly understand IMN architecture:

1. Each IMN node is the logical address on the IMN of a set of Acquirer and/or Issuer functions. An IMN node is identified by an assigned six-digit value and by an equivalent 6-character name. A node is composed of a collection of modules sharing common definition tables and residing in one system image. A Member may have one node or many. Two nodes may co-reside in the same hardware system image. Each IMN node is composed of the CIL and the CCL as described below.
2. The CCL is made up of hardware and software, which meets the Internet Protocol (IP) standards for external communications networking. Functions include managing logical links, comprising connections to other IMN nodes. The CCL also consists of IMN/IP software and, operating system software and hardware to control the lower communication layers, in order to provide reliable communication among Member nodes and to the IMNCC. This structure allows the IMN to isolate communications technology issues in the CCL using TCP/IP and other IP-related facilities.
3. The CIL provides a single logical link of the multiple low-level paths between any Acquirer-Issuer pair. Its application knowledge is restricted to a standard message header, which includes connection routing data, message class, transaction type, tracing information, and timestamps. Functions include interface to CCL, interface to the IMNCC, establishment of connections (a connection is an application specific logical path between Acquirer and Issuer), transaction routing to CCL components, timing of transactions and handling of timeout conditions, statistics generation, collection and forwarding to the IMNCC, and availability testing.
4. A Connection is a logical path for a specific message class between an Acquirer IMN node and an Issuer IMN node. A connection will be defined to contain a set of two or more TCP/IP sockets, administered by the CCL, through two or more physical links. A connection will be supported by redundant software components. Therefore a connection will survive at least one logical or physical communications failure. The requesting node sends request messages and receives response messages from the responding node on the same connection. Only one connection from node A to node B for message class C may be active in the network at any one time, though other definitions may exist, awaiting activation as alternates. Each connection is defined in the Inter-Member Network

- Address table (IMNAT), which contains the necessary information for creating and sending connection requests. Each such table is a public document and will be available to both node partners. The connection from node B to node A for message class C, if exists, is defined separately and contains different sessions.
5. An entry point (EP) defines an interface between the IMNi/IMNsc known as Application Interface Layer (AIL) and an IMN node. Each entry point may serve traffic with all destination nodes and all message classes, though a Member may optionally restrict an entry point to one message class. Likewise, the IMN selects an EP for incoming messages based on matching message and EP attributes.
 6. A handshake message and response are exchanged by both entry points and sessions in order to validate the identity and supported function of the participating processes.
 7. An echo message and response may be exchanged periodically over entry points in order to verify that AIL-CIL paths exist and are functioning. Alarms (see Connection status below) are issued when each path changes from echo positive to echo negative.
 8. A pacing message is sent when a component wishes to advise the component from which it normally receives messages about its ability to accept work on this path. The pacing state may be normal, congested (please choose another path if available), or blocked (further messages may be lost).

The Transaction flow is therefore as follows:

- The requesting IMNi/IMNsc creates a request message prefixed by an IMN header and selects an eligible and adequately performing entry point to the IMN node.
- The requesting IMN node uses the IMN header information to select the connection and uses its internal knowledge paths and its partial knowledge of the responding node derived from pacing flows to route the message across a session or to return the message as a response marked undeliverable. The requesting IMN node also keeps a copy of the request in a context buffer to match against the subsequent response.
- The responding IMN node uses the IMN header information and its internal knowledge of its own component paths to route the incoming request to an eligible AIL entry point, or to return the message as a response marked undeliverable.
- The responding IMNi/IMNsc generates a response message with an IMN header and selects an IMN node entry point.

- The responding IMN node uses the IMN header information, the Connection tables, its internal knowledge of its own component paths and its partial knowledge of the requesting node derived from response messages to route the response across a session, or to declare the message undeliverable, log and drop it.
- The requesting IMN node uses the IMN header information and its internal knowledge of its own component paths to route the incoming response to an IMNi/IMNsc entry point, or to declare the message undeliverable, log and drop it. The entry point chosen must have the same AIL and subsystem ID as the original Acquirer. The response header is checked against the saved request context entry.

Communication Interface Layer (CIL)

The Communications Interface Layer (CIL) is the layer of the IMN Network immediately adjacent to IMNi/IMNsc process. IMNi/IMNsc processes communicate with CIL via a standard HP inter-process message. This inter-process messages flow in both directions: IMNi/IMNsc to IMN and IMN to IMNi/IMNsc.

As the communication between the IMNi/IMNsc and the CIL flows rapidly in both directions, the standard HP method of inter-process communication using WRITEREAD and REPLY is not sufficient. The IMNi/IMNsc and the CIL process will each open the other to allow for duplex traffic.

Entry Point Definition

An Entry Point (EP) defines an interface between the IMNi/IMNsc and its associated IMN interface node. Likewise, the IMN selects an EP for incoming messages based on matching message and EP attributes. The Entry Point selected by the CIL to receive a response must match the AIL ID and source subsystem ID of the original request IMN header.

The entry points are defined in the Entry Point Configuration Table (IMEPCT) which is maintained through the IMN Pathway environment called EPDEFN. The IMEPCT is an SQL table used by IMN application. There is also an Enscribe version for IMNi/IMNsc – EPCF. The EPCF contains the same data as IMEPCT including the entry point status. IMNi/IMNsc reads the EPCF during start up and entry point warmboot. It is recommended to ensure that the EPCF is kept up to date with any changes that may have occurred.

When the IMNi/IMNsc initializes, or activates an entry point, IMNi/IMNsc will open the CIL process. When the open has completed, IMNi/IMNsc will build and send an entry point handshake message. This consists of the IMN header with both requester and responder node ID. The message type is ISO 0800, message subtype 160. The ISO data message length is zero. The IMN header data fields

populate from EPCF file (i.e. entry point ID, AIL ID). This is to ensure that both IMNi/IMNsc and the CIL are working based on the same definition.

Also part of the handshake message is a security byte. This is a unique one-byte field generated randomly by the IMNi/IMNsc. This security byte is used in all subsequent messages to the CIL module on this entry point. Any messages not containing the correct security byte will be rejected by the CIL.

The AIL ID in the handshake message is used to verify the direction of the entry point. An AIL ID of “0” means this is used as an issuer entry point, receiving requests from and sending responses to the CIL. An AIL ID of not “0” represents an acquirer entry point which sends requests and receives responses from the CIL.

Similarly, the CIL opens the IMNi/IMNsc and sends an identically formatted handshake message. The CIL will generate its own security byte and this byte must accompany every subsequent message from the CIL. The IMNi/IMNsc will check the message to ensure that the CIL and the IMNi/IMNsc are working with the same entry point definition. If any disagreement occurs, the IMN status code is set to 84 (Protocol Error) in the appropriate reason code.

If a disagreement takes place over the entry point definition, then the entry point is closed and marked as “STOPPED” both internally and on the EPCF. From this state, the entry point can only be restarted by an operator command.

Entry Point States

An entry point can be in one of five states:

1. **Pending**
A handshake message has been attempted but not completed.
2. **Stopped**
The entry point has been stopped by an operator or handshake failure and is awaiting an operator command to re-establish the link.
3. **Available**
The handshake is completed. The entry point is now waiting for message.
4. **Unavailable**
The entry point is temporarily unusable for some other reason.
5. **Active**
The entry point is available and an ISO non-800 message has flowed successfully. “Active” is distinguished from “available” largely for the purposes of operator inquiry

Entry Point Message Processing

After an entry point has been successfully initialized, the handshake messages have been exchanged and messages can flow in both directions across the link. IMNi/IMNsc performs the following processing for any entry point message. For the IMN reason codes given below, an issuer IMNi/IMNsc will use 5xxx, and an acquirer IMNi/IMNsc will use 0xxx.

Request Messages from the CIL

- Verify that a handshake has taken place. If not, set IMN status code to 84 and IMN reason code to 5011 (No handshake in effect).
- The security byte in the IMN header must match the security ID that was passed during the handshake. If not, set the IMN reason code to 5007 (invalid security value) on the response.
- Verify that the entry point supports the message class in the IMN header. If not, set the IMN reason code to 5003 (invalid message class) on the response.
- If the destination AIL ID is invalid, set the IMN reason code to 5005 (invalid responder AIL ID) on the response.
- If the request message cannot be passed to the appropriate BASE24 modules via the XPNET, the message type is converted to a response, the ISO response code set to 91 (AIL not available) with reason code 5013 (No modules available), and the product reason code set to indicate the nature of the error.

Response Messages from the CIL

- Verify that a handshake has taken place. If not, set IMN status code to 84 and IMN reason code to 0011 (No handshake in effect), and drop the message.
- If the security byte in the IMN header does not match the security ID that was passed during the handshake, set the IMN status code to 89 (Undeliverable response), reason code to 0007 (invalid security value) and drop the message. A handshake then is initiated.
- Verify that the entry point supports the message class in the IMN header. If not, set the IMN status code to 89, reason code to 5003 (Invalid message class) and drop the message.
- If the subsystem ID is not correct, set the IMN status code to 89, reason code to 0006 (Invalid requester subsystem ID), drop the response message and initiate a handshake.
- If the response message cannot be passed to the appropriate BASE24 module via XPNET, set the ISO response code to 89, reason code 0013 (No modules available), product reason code set to indicate the nature of the error and drop the message.

Request Messages to the CIL

- Once IMNi/IMNsc successfully formats the ISO request message then IMNi/IMNsc will select an entry point to forward the message to the CIL. The selection criteria for a request message on an entry point is as follows:
 - Select an entry point with the best response times based on past performance calculations.
 - Select an entry point that matches Destination node and message class.
 - Select an entry point that supports an AIL ID other than "0".
 - Select an entry point that is in "Pacing Normal" state over the one that is in "Pacing Congested" state (even if the Congested entry point has a calculated response time faster than the Normal entry point) and a "Pacing Congested" over a "Pacing Blocked" entry point.
- If there is no entry point that meets the above criteria then an alternate IMNi/IMNsc is selected. The request is forwarded to that IMNi/IMNsc and it will attempt to send the message.
- If the alternate IMNi/IMNsc has no entry point that meets the above criteria or there is no alternate IMNi/IMNsc available, then a response message will be generated and returned with BASE24 response code 100 (Unable to process) and logged to the ILF.

Response Messages to the CIL

- Once IMNi/IMNsc successfully formats the ISO response message then IMNi/IMNsc will select an entry point to forward the message to the CIL. The selection criteria for a response message on an entry point is as follows:
 - Select an entry point with the best response time based on past performance calculations.
 - Select an entry point that matches Destination node and message class.
 - Select an entry point that supports a AIL of "0".
 - Select an entry point that is in "Pacing Normal" state over the one that is in "Pacing Congested" state (even if the Congested entry point has a calculated response time faster than the Normal entry point) and a "Pacing Congested" over a "Pacing Blocked" entry point.
- If there is no entry point that meets the above criteria then an alternate IMNi/IMNsc is selected. The response is forwarded to that IMNi/IMNsc and it will attempt to send the message.
- If the alternate IMNi/IMNsc has no entry point that meets the above criteria or there is no alternate IMNi/IMNsc available, then a reversal message will be generated (if the response was approved), returned to BASE24 and logged to the ILF.

Section 9

Interchange Log File (ILF)

The Interchange Log File (ILF) contains one record for each transaction processed by the IMNi/IMNsc process. The ILF is used as input to the settlement report programs for the IMNi/IMNsc process. A new ILF is created each business day when cutover occurs. The IMNi/IMNsc process creates and writes a log record once for each transaction when the transaction response is received. ILF only logs transaction with message type flagged with "Y" at ICF screen 14. This section provides the basic record structure for the IMNi/IMNsc ILF and the listing of all the substate values.

LNCF assign: ILF

ILF Record Structure

The fields included in an Interchange Log File (ILF) record for the IMNi/IMNsc Interface process are described below.

- | | | |
|-------|--|-------------------|
| 1-55 | 02 PRIKEY | KEYTAG PR |
| | The alternate key to this ILF record. For the IMNi/IMNsc Interface process, PRIKEY consists of the following fields: TRANTIM, PAN, REF-
NUM, TRAN-DAT, and TERM-ID or ACQ-ISTCDE. | |
| 56 | 04 BYTE | TYPE CHARACTER 55 |
| 57-58 | 02 USER-FLDB | PIC X |
| 59-60 | 02 SUBSTATE | TYPE BINARY 16 |
| | The exception indicator for an authorization record. The IMNi/IMNsc Interface process uses this field to determine what error messages should be printed on reports with the transaction detail information if the transaction did not complete as intended. Valid values are described in the next paragraph. | |
| 61-76 | 02 REC-TYP | TYPE BINARY 16 |
| | Describes the type of record. The IMNi/IMNsc Interface process uses this field to determine the type of record it is processing. | |
| | Values are: | |
| | 01 = BASE24-atm | |
| | 02 = BASE24-pos | |
| | 03 = BASE24-afs | |
| | 90 = Representation | |
| | 92 = Fee collection | |

- 93 = Funds disbursement
94 = Gross interchange value
95 = Funds transfer totals
- 77-82 02 ORIG-NET-PRO PIC X(16)
The name of the process to which the reply is to be sent if the request is from BASE24. Reversals initiated by the interchange are also sent to this process.
- 83-90 02 CUR-DAT-ADD PIC X(6)
The date (YYMMDD) the record was added to the ILF.
- 91-98 02 CUR-TIM-ADD PIC X(8)
The time (HHMMSSTT) at which the record was added to the ILF.
- 02 LAST-UPDATE TYPE BINARY 64
A Julian timestamp indicating the time the record was last updated. This information is printed on reports for reversal transactions.
- 99-102 02 SEM
The following fields are standard for all external messages and are placed here in order to allow for standard access to the information.
- 103-110 04 TRAN-TYP PIC X(4)
The transaction type from the Processing Code data element (P-3) in the external message.
- 111-114 04 RESP-CDE PIC X(8)
The response code from the Response Code data element (P-39) in the external message.
- 115-120 04 RVSL-CDE PIC X(4)
The reversal code from the Response Code data element (P-39) in the external message if the transaction is a reversal.
- 121-128 04 SWI-DAT PIC X(6)
The transaction date (YYMMDD) from the Local Transaction Date data element (P-13) in the external message.
- 129-134 04 SWI-TIM PIC X(8)
The transaction time (HHMMSSTT) from the Local Transaction Time data element (P-12) in the external message.
- 135-150 04 TRACE-NUM PIC X(6)

The trace number from the Systems Trace Audit Number data element (P-11) in the external message.

151-152 04 USER-FLD (Not Used) PIC X(16)

153-814 02 INTRN-MSG-SAVEAREA-LGTH TYPE BINARY 16
The length of the standard internal message that follows.

153-814 02 ATM TYPE ATM
Values of selected fields from the BASE24-atm Standard Internal Message (STM) for a transaction processed by BASE24-atm. For a complete description of the STM format, please refer to the appropriate BASE24 Manual.

815-818 02 POS REDEFINES ATM
TYPE POS
Values of selected fields from the BASE24-pos Standard Internal Message (PSTM) for a transaction processed by BASE24-pos. For a complete description of the PSTM format, please refer to the appropriate BASE24 Manual.

02 EXTRN-MSG-LGTH PIC 9(4)
The length of the message stored in the EXTRNMSG.SAVEAREA field.

819-2000 02 EXTRN-MSG

04 SAVEAREA PIC X
OCCURS 0 TO 3254 TIMES
DEPENDING ON EXTRN-MSG-LGTH
The external message for the transaction. The IMNi/IMNsc Interface process uses this field for storing transaction information.

Substate

IMNi/IMNsc process uses this field to determine what error messages should be printed on reports with the transaction detail information if the transaction did not complete as intended.

Substate Value	Substate Literal	Description
0	SUB-OK-L	OK. No error occurred when a transaction is processed.
1	SUB-SYSTEM-ERROR-L	System error; occurs when no other substate fits. Error on the transmission of a SEM (Standard External Message).
2	SUB-FRMT-ERR-L	Formatting error

		Error formatting a PSTM into a SEM. The transaction will not be forwarded.
3	SUB-FAIL-PRE-AUTH-L	Preauth failed IMNi/IMNsc does not use this substate.
4	SUB-RQST-DOWN-L	Requester node down Failed on sending the response message to the BASE24 authorization process.
5	SUB-TIMEOUT-L	Timeout Outbound transaction timed out.
6	SUB-RESP-DOWN-L	Responder node down Failed on attempt to resend the request message on another Entry Point or the issuer or when the acquirer receives the SAF response with the following IMN status code: "80" – Undeliverable "87" – Pacing connection blocked "91" – AIL not available
7	SUB-RVSL-NOT-FOUND-L	Reversal not found IMNi/IMNsc does not use this substate.
8	SUB-MISC-L	Miscellaneous. Error IMNi/IMNsc does not use this substate.
11	SUB-MAC-FAIL-L	MAC failure No keys are available.
12	SUB-SAF-SEM-ONLY-L	Response to a SAF transaction Only the SEM portion was available at the time when the transaction was logged. This substate occurs when an IMNI/IMNSC switches its SAF transactions to another IMNI/IMNSC process.
13	SUB-SAF-DELETED-ERROR-L	SAF record has been deleted due to a SAF response received with one of the following response code: "6" - Edit check error "12" - Invalid processing code "13" - Invalid amount "81" - Invalid PIN block "82" - PIN length error "92" - Financial institution or intermediate network facility cannot be found for routing "Q0" - Capture date error
14	SUB-SAF-DELETED-CUTOVER-L	SAF 420's deleted at cutover time with reversal reason code of timeout.
15	SUB-SAF-REQUEST-L	SAF request SAF request message is generated for message types 220, 402, 420, 6xx (Admin).
16	SUB-SAF-DENIED-L	SAF request is denied
17	SUB-ILF-NORMAL-L	ILF normal Transaction response received and successfully sent by issuer.
18	SUB-SAF-UNMATCHED-RESP-L	SAF unmatched response Cannot find original SAF request for the response.
19	SUB-SAF-RETRY-MAX-L	SAF retry reached maximum IMNi/IMNsc does not use this substate.
20	SUB-LATE-RESP-L	Late response Received late response from IMN or BASE24.

21	SUB-SECURITY-ERROR-L	Security error MAC generation or verify failure as well as any security error such as invalid security block or invalid key or key synchronized error.
22	SUB-SAF-UNACKNLDG-RVSL-L	Unacknowledged reversal SAF response with invalid capture date – SAF deleted.
23	SUB-SAF-DELETED-RESP-OK-L	SAF response deletion successful SAF response message received OK. Request message deleted from SAF.
24	SUB-SAF-DELETED-RESP-OK-L	SAF delete command SAF request deleted from operator command.
30	SUB-INVALID-DESTN-L	Destination node not found Destination node ID not found. Checked on acquirer side.
31	SUB-INVALID-ROUTE-L	PAN not in the BIN table PAN routing not found in BIN table. Checked on acquirer side.
32	SUB-POS-INVALID-L (IMNi) SUB-ATM-INVALID-L (IMNsc)	POS/ATM transaction not processed Transaction type not entered on ICF list of allowed transactions. Checked on acquirer side.
33	SUB-SWITCH-ORIG-L	Switch originated transaction; possible duplicated
34	SUB-NOT-INTERAC-L	Not an INTERAC transaction
35	SUB-INVALID-TRANS-L	Transaction type not allowed
36	SUB-INVALID-SERVICE-L	Service not allowed
37	SUB-UNABLE-TO-SEND-L	Forward only mode Unable to send transaction because destination is in forward mode only.
38	SUB-INVALID-AMT-L	Amount too large The amount is too large for switch variables.
39	SUB-ALT-ROUTE-L	Unable to alternate route transaction
40	SUB-INVALID-DATE-L	Invalid transaction date
41	SUB-INTERNAL-ERROR-L	Internal processing error IMNi/IMNsc does not use this substate.
42	SUB-NOT-PROCESSED-L	SEM message not processed due to error found in SEM
43	SUB-CAPDATE-CHANGED-L	Capture date changed for cutover

Section 10

Transaction and Response Code

AIL Response Code

IDP AIL Response Code	Description
000	Approved With Balances
001	Approved Without Balances
002	Approved With Country Club Status
003	Approved With Maybe Status (Id Required)
004	Approved With Pending Identification (Signed)
005	Approved Blind
006	Approved With VIP Status
007	Approved Administrative Transaction
008	Approved After Checking the National Negative
009	Approved With Commercial Status
050	Declined
051	Expired Card
052	Pin Tries Exceeded
053	Sharing Is Not Allowed
054	No Security Box
055	Invalid Transaction
056	No Support
057	Lost Or Stolen Card Status
058	Invalid Status
059	Restricted Status
060	Account Not On The CAF
061	PBF Record Not Found
062	PBF Update Error
063	Invalid Authorization Type
064	Bad Track Data
065	Adjustment Is Not Allowed
066	Invalid Credit Card Advance Increment
067	Invalid Transaction Date
068	PTLF Error
069	Bad Message Edit
070	No IDF
071	Invalid Routing To Authorization
072	Card Found On The National Negative File
073	Invalid Routing To Authorization Service Field
074	Unable To Authorize

075	Invalid Pan Length
076	Insufficient Funds
077	Preauthorization Limit Reached
078	Duplicate Transaction
079	Maximum Online Refund Limit Reached
080	Maximum Offline Refund Limit Reached
081	Maximum Credit Per Refund Reached
082	Maximum Number Of Times The Card Is Allowed To Be Used Has Been Reached
083	Maximum Refund Credit Limit Exceeded
084	Customer Selected Neg Reason
085	Inquiry Not Allowed
086	Over The Floor Limit
087	Maximum Number Of Refund Credits Reached
088	Place Call For Referral
089	Card Status Inactive Or Closed
090	Referral File (PRF) Full
091	Negative Card File Problem
092	Cash Advance Less Than Specified Minimum
093	Delinquent
094	Over Limit Table
095	Amount Over Maximum
096	Pin Required
097	Mod 10 Check
098	Force Post
099	Bad PDF
100	Unable To Process The Transaction
101	Issue A Call
102	Call
103	Negative Card File Problem
104	CAF File Problem
105	Card Not Supported
106	Amount Is Over The Maximum
107	Daily Usage Limit Reached
108	Capf Not Found
109	Cash Advance Less Than Specified Minimum
110	Number Of Times Used Limit Exceeded
111	Delinquent
112	Over Limit Table
113	Timeout
115	PTLF Full
120	UAF File Problem
121	Admin File Problem
122	Unable To Validate Pin
130	CRYPTOGRAM failure

131	CVR Check
132	TVR Check
133	Reason-Online Check
134	Fallback Check
150	Merchant Is Not On File
200	Invalid Account
201	Incorrect Pin
202	Cash Advance Less Than The Minimum Required
203	Transaction Requires Administrative Card
204	Enter Lesser Amount
205	Invalid Advance Amount
206	CAF Not Found
207	Invalid Transaction Date
208	Invalid Expiration Date
209	Invalid Transaction Code
210	PIN Sanity Error
212	Destination Not Available
251	Purchase With Cash Back
252	Debit Card Processing Not Supported Through Crt
400	Request CRYPTOGRAM failure
401	Security Module Parameter error
402	Security Module failure
403	KEYI Record Not Found
404	ATC Check Failure
405	CVR Decline
406	TVR Decline
407	Reason Online Code decline
408	Fallback Decline
810	Timeout
877	Invalid PIN Block
878	Bad Pin Length
898	MAC Security Error
900	Pin Tries Exceeded
901	Expired Card
902	Neg Capture Card
903	CAF Status 3
904	Cash Advance Less Than Specified Minimum
905	Number Of Times Used Limit Exceeded
906	Delinquent
907	Over Limit Table
908	Amount Is Over The Maximum
909	Capture
910	Request CRYPTOGRAM Failure
911	CVR Decline
912	TVR Decline

950	Administrative Card Not Found
951	Administrative Card Not Allowed
952	Approved Administrative Request In The Window
953	Approved Administrative Request Outside The
954	Approved Administrative Request Anytime
955	Chargeback Approved
956	Chargeback Approved
957	Chargeback Accepted—Bad Prefix
958	Chargeback Bad Response or Prefix
959	Chargeback Not Supported
960	Chargeback Approved
961	Chargeback Approved
962	Chargeback Accepted
989	KMAC sync error
999	KME sync error

SCD AIL Response Code	Description
00	Approved With Balances
01	Approved No Balances For Display
50	Unauthorized Usage
51	Expired Card
52	Invalid Card
53	Invalid Pin
54	Database Problem
55	Ineligible Transaction
56	Ineligible Account
57	Transaction Not Supported
58	Insufficient Funds No Amount 3
59	Insufficient Funds With Amount 3
60	Uses Limit Exceeded
61	Withdrawal Limit Would Be Exceeded
62	Pin Tries Exceeded
63	Withdrawal Limit Already Reached
64	Invalid Credit Card Cash Advance Amount
65	No Statement Information For The Account
66	Statement Information Not Available
67	Invalid Cash Back Amount
68	External Decline
69	No Sharing Between The Card Issuer And
70	System Error
71	Contact Card Issuer
72	Destination Not Available

73	Routing Look Up Problem
74	Message Edit Error
81	Security Module Parameter Error
82	Security Module Failure
83	KEYI Record Not Found
84	ATC Check Failure
85	CVR Decline
86	TVR Decline
87	Request Cryptogram Failure
88	Fallback Decline
90	Approved Transaction Inside Window
91	Approved Transaction Outside Window
92	Approved Transaction Balance Anytime

Please refer to BASE24-atm or BASE24-pos Transaction Processing Manual on the description of the code.

ISO Response Code

ISO Response Code	Description
00	Approved
05	Do not honour
06	Edit error
12	Invalid Processing Code
13	Invalid Amount
14	Invalid Card Number
19	TAS error. Re-enter transaction
40	Requested function not
51	Non-sufficient Funds
52	No Chequing Account
53	No Savings
54	Expired Card
55	Incorrect Personal Identification Number
56	No Card record PAN not on CIS database.
57	Transaction not Permitted to Cardholder
58	Transaction not Permitted to Terminal
61	Exceeds withdrawal amount limit
62	Restricted Card
63	Security violation
65	Exceeds withdrawal frequency limit
68	Response received too late (timeout)
75	Allowable number of PIN tries exceeded
77	Invalid Capture Date

81	Invalid PIN block
82	PIN length error
84	Counter out-of-sequence in File
85	No Keys available for use
86	Message Encryption Key synchronization error
87	PIN Key Synchronization error
88	MAC Key synchronization error
91	Issuer or switch is inoperative
92	MEMBER or inter-mediate network facility cannot be found for routing
94	Duplicate transmission
96	System malfunction
Q0	Edit error in capture date
Q7	Exceeds TAS limit
Q9	Response not listed above has been received

ISO to BASE24 response code mapping

The table below translates Interac ISO Response codes to BASE24-pos response codes.

IDP Response Code Mapping		
Description	ISO Code	→ BASE24 Code
Approval	00	001
Pick Up Card	04	909
Do Not Honour	05	058
Edit Error	06	069
Invalid Processing Code	12	209
Invalid Amount	13	069
Invalid Card Number	14	064
RFI Error. Re-enter Transaction	19	050
Unacceptable Transaction Fee	23	056
File Update Not Supported	24	056
Unable To Locate Record On File	25	056
Duplicate File Update Record	26	056
File Update Field Edit Error	27	056
File Update Locked/Not Avail	28	056
File Update Not Successful	29	056
Expired Card, Pick Up	33	901
Restricted Card, Pick Up	36	909
Pin Tries exceeded, Pick Up	38	900
No Credit Account	39	060
Requested Function Not Supported	40	056
Lost Card, Pick Up	41	909

Stolen Card, Pick Up	43	909
Non-Sufficient Funds	51	076
No Chequing Account	52	060
No Savings Account	53	060
Expired Card	54	051
Incorrect Pin	55	201
No Card Record	56	206
Transaction Not Permitted To Cardholder	57	056
Transaction Not Permitted To Terminal	58	055
Exceeds Withdrawal Amount Limit	61	095
Restricted Card	62	059
Mac Security Failure	63	898
Exceeds Withdrawal Freq Limit	65	082
Response Received Too Late:Timeout	68	810
Allowable Pin Tries Exceeded	75	052
Invalid Capture Date	77	067
Invalid Pin Block	81	877
Pin Length Error	82	878
Counter Out Of Sequence	84	074
No Keys Available For Use	85	074
KME Synchronization Error	86	074
KPE Synchronization Error	87	074
KMAC Synchronization Error	88	074
Issuer Inc Or Switch Inoperative	91	074
No Intermediate Network Facility	92	105
Duplicate Transmission	94	078
System Malfunction	96	074
Edit Error In Capture Date	Q0	207
Exceeds RFI Limit	Q7	050
	Q9	050

The table below translates Interac ISO Response codes to BASE24-atm response codes.

SCD Response Code Mapping		
Description	ISO Code	→ BASE24 Code
Approve without balances	00	001
Approve with balances	00	000
Do not honour	05	068
Edit error	06	074
Invalid processing code	12	055
Invalid amount	13	074
Invalid card number	14	052
Acquirer Error	19	070

Non-sufficient funds	51	058
No chequing account	52	056
No saving account	53	056
Expired card	54	051
Incorrect PIN	55	053
No card record	56	052
Exceeds withdraw amount limit	61	061
Restricted card	62	050
MAC value incorrect	63	070
Hard capture. Retain card	67	150
Late response (timeout)	68	070
PIN tries exceeded	75	062
No credit card account	76	056
Invalid capture date	77	074
Invalid PIN block	81	070
PIN length error	82	070
Counter out of seq. in file	84	070
No key available for use	85	070
PIN key sync error	87	070
MAC key sync error	88	070
Sender inoperative	89	072
Issuer inoperative	91	072
Routing not found	92	073
Violation of law	93	068
Duplicate Transmission	94	070
System malfunction	96	070
Edit error in capture date	Q0	074
Unknown response code	Q9	074

ISO to BASE24 reversal code mapping

The table below translates Interac ISO reversal codes to BASE24-pos reversal codes.

IDP Reversal Code Mapping		
Description	ISO Code → BASE24 Code	
Not Applicable	00	00
Timeout	01	01
Destination Not Available	02	03
Fraudulent Transaction	03	02
Edit Error In Response	06	02
Customer Cancel	08	08
Invalid Processing Code	12	02
Original Amount Incorrect	13	02

Invalid Card Number	14	02
Invalid Tran Fee	23	02
MAC Failure	30	03
MAC Key Sync Error	31	03
Restricted Card	36	02
No Credit Account	39	02
Requested Function Not Supported	40	02
Lost Card	41	02
Stolen Card	43	02
Not Sufficient Funds	51	02
No Chequing Account	52	02
No Savings Account	53	02
Expired Card	54	02
Incorrect Pin	55	02
No Card Record	56	02
Trxn Not Permitted To Cardholder	57	02
Trxn Not Permitted To Terminal	58	02
Exceeds Purchase Amt Limit	61	02
Restricted Card	62	02
Exceeds Purchase Frequency Limit	65	02
Invalid Capture Date	77	02
Amountt Error - Exceeds Original Apvl Amount	78	78
Time Expired - Auth Time Limit	79	79
Kme Sync Error	86	03
FIID Can Not Be Found For Route	92	03
Duplicate Transmission	94	02
System Malfunction	96	19
Edit Error In Capture Date	Q0	02
Exceeds RFI Limit	Q7	02
Invalid Auth Id Number	R9	R9

The table below translates Interac ISO reversal codes to BASE24-atm reversal codes.

SCD Reversal Code Mapping		
Description	ISO Code	→ BASE24 Code
Not applicable	00	00
Time out	01	01
Destination not available	02	03
Edit error in response	06	02
Customer cancellation	08	08
Hardware problem	10	10
Original amount incorrect	12	12

Terminal malfunction	13	10
Partial dispense	15	00
Duplicate transaction	16	16
Reconciliation error	17	17
Suspected Malfunction	22	20
MAC failure	30	30
MAC key sync error	31	31

BASE24 to ISO response code mapping

The table below translates BASE24 Response codes to ISO Response codes.

IDP Response Code Mapping		
Description	BASE24 CODE	→ ISO CODE
Approve – With balances	000	00
Approve - Without Balances	001	00
Approve - Country Club	002	00
Approve - Maybe More Id	003	00
Approve - Pending Id	004	00
Approve - Blind	005	00
Approve - VIP	006	00
Approve - Admin Tran	007	00
Approve - Against Nat'l Neg	008	00
Approve - Commercial	009	00
Decline - Declined	050	62
Decline - Expired Card	051	54
Decline - Exceeded Pin Tries	052	75
Decline - No Sharing	053	58
Decline - No Attalla Box	054	96
Decline - Invalid Transaction	055	40
Decline - No Support	056	40
Decline - Lost Or Stolen Card	057	62
Decline - Invalid Status	058	62
Decline - Restricted	059	62
Decline - No Accounts	060	56
Decline - No PBF	061	56
Decline - PBF Update Error	062	96
Decline - Invalid Auth Type	063	57
Decline - Bad Track2	064	14
Decline - Adjustment Not Allowed	065	40
Decline - Invalid Increment	066	05
Decline - Invalid Transaction Date	067	77
Decline - TLF Error	068	96
Decline - Msg Edit Error	069	06

Decline - No IDF Record	070	92
Decline - Invalid Route Auth	071	92
Decline - On Nat'l Neg	072	62
Decline - Inv Route Service	073	92
Decline - Unable To Authorize	074	91
Decline - Invalid Pan Length	075	14
Decline - Low Funds	076	51
Decline - Pre-Auth Full	077	61
Decline - Pre-Auth Full	077	65
Decline - Duplicate Transaction	078	94
Decline - Max Refund Online	079	61
Decline - Max Refund Offline	080	61
Decline - Max Per Refund	081	61
Decline - Num Times Used	082	65
Decline - Max Refund Total	083	61
Decline - Cust Neg Reason	084	05
Decline - Inq Not Allowed	085	40
Decline - Over Floor Limit	086	61
Decline - Max Refund Number	087	65
Decline - Place Call	088	62
Decline - CAF Status 0 Or 9	089	62
Decline - Referral File Full	090	96
Decline - Neg File Problem	091	96
Decline - Advance < Minimum	092	61
Decline - Delinquent	093	62
Decline - Over Limit Table	094	61
Decline - Amount Over Max	095	61
Decline - Pin Required	096	82
Decline - Mod Ten Check	097	14
Decline - Force Post	098	05
Decline - Bad PBF	099	14
Refer - Unable To Process	100	91
Refer - Issue Call	101	91
Refer - Call	102	62
Refer - Neg File Problem	103	96
Refer - CAF File Problem	104	96
Refer - Card Not Supported	105	92
Refer - Amount Over Max	106	61
Refer - Over Daily Limit	107	61
Refer - CAF Not Found	108	56
Refer - Advance < Minimum	109	61
Refer - Num Times Used	110	65
Refer - Delinquent	111	62
Refer - Over Limit Table	112	61
Refer - Timeout	113	68

Refer - PTLF Full	115	96
Refer - Admin File Problem	121	05
Refer - Unable To Valid Pin	122	81
Refer - Cryptogram Failure	130	96
Refer - Cvr Decline	131	57
Refer - Tvr Decline	132	58
Refer - Reason-Online Check	133	58
Refer - Fallback Check	134	62
Service - Merchant Not On File	150	06
Error - Account	200	06
Error - Incorrect Pin	201	55
Error - Advance < Minimum	202	13
Error - Admin Card Needed	203	05
Error - Enter Lesser Amount	204	61
Error - Invalid Advance Amount	205	13
Error - CAF Not Found	206	56
Error - Invalid Transaction Date	207	77
Error - Invalid Exp Date	208	06
Error - Invalid Trancode	209	12
Error - Pin Sanity Error	210	87
Error - Destination Not Available	212	91
Error - On Cash Amount	251	13
Decline - Debit Not Supported	252	40
Error - Cryptogram Failure	400	96
Error - Security Module Parm	401	96
Error - Security Module	402	96
Error - No Keyi Record	403	96
Error - Atc Check Failure	404	57
Error - Cvr Decline	405	57
Error - Tvr Decline	406	58
Error - Reason-Online Check	407	58
Error - Fallback Decline	408	62
Error - Timeout	810	68
Error - Invalid Pin Block	877	81
Error - Bad Pin Length	878	82
Error - Mac Security Error	898	63
Capture - Pin Tries Exceeded	900	75
Capture - Expired Card	901	54
Capture - Neg Capture	902	05
Capture - CAF Status 3	903	05
Capture - Advance < Minimum	904	05
Capture - Num Times Used	905	05
Capture - Delinquent	906	05
Capture - Over Limit Table	907	05
Capture - Amount Over Maximum	908	05

Capture - Capture	909	05
Capture - Cryptogram Failure	910	05
Capture - Cvr Decline	911	05
Capture - Tvr Decline	912	05
Admin - Card Not Found	950	56
Admin - Card Not Allowed	951	57
Admin - Approved In Window	952	00
Admin - Approved Out Of Window	953	00
Admin - Approved Anytime	954	00
Admin - Chrgbk/Appr Update	955	00
Admin - Chrgbk/Appr Update	956	00
Admin - Chrgbk/Incorrt Pref	957	00
Admin - Chrgbk/Incorrt Rcd	958	05
Admin - Admin Transaction Not Support	959	40
Admin - Chrgbk/Appd Not Update	960	00
Admin - Chrgbk/Appd Not Update	961	00
Admin - Chrgbk/Accepted	962	00
Error - Kmac Sync Error	989	88
Error - Kme Sync Error	999	86

SCD Response Code Mapping		
Description	BASE24 CODE → ISO CODE	
Approve - With balances	00	00
Approve - Without balances	01	00
Restricted Card	50	62
Expired Card	51	54
Invalid Card	52	56
Invalid PIN	53	55
Database Problem	54	96
Ineligible Transaction	55	12
Ineligible Account	56	52, 53 or 76
Transaction not supported	57	05
Insufficient funds Balance <= 0	58	51
Insufficient funds Balance >0	59	51
Uses limited exceeded	60	61
Withdraw limited would exceed	61	61
PIN tries exceeded	62	75
Withdraw limit exceeded	63	61
Invalid advance amount	64	05
No statement info	65	05
Statement info not available	66	05
Invalid cash back amount	67	05
External Decline	68	05

No sharing agreement	69	05
System Error	70	96
Contact card issuer	71	06
Destination not available	72	91
Routing look up problem	73	92
Message edit error	74	06
Security module parm error	81	96
Security module failure	82	96
No KEY record	83	96
ATC check Failure	84	05
CVR decline	85	05
TVR decline	86	05
Request cryptogram failure	87	05
Request cryptogram failure	87	96
Fallback decline	88	62
Approved inside window	90	05
Approved outside window	91	05
Approved anytime	92	05

BASE24 to ISO reversal code mapping

The table below translates BASE24 Reversal codes to ISO Reversal codes.

IDP Reversal Code Mapping		
Description	BASE24 CODE → ISO CODE	
Not application	00	00
Time out	01	01
Command rejected	02	06
Destination not available	03	02
Customer Canceled	08	08
Hardware error	10	96
System error	19	96
Suspect transaction	20	96
MAC failure	21	96
KMAC sync error	22	96
Message reply error	23	96
Invalid MAC	24	96
MAC failure	30	30
MAC key sync error	31	31
Amount error	78	78
Time expired	79	79
KME sync error	86	86
AUTH Id not found	R9	R9

The table below translates BASE24 Reversal codes to ISO Reversal codes.

SCD Reversal Code Mapping		
Description	BASE24 CODE → ISO CODE	
Not applicable	00	00
Time out	01	01
Invalid response	02	06
Destination not available	03	02
Customer Canceled	08	08
Hardware error	10	10
Original amount incorrect	12	12
Terminal malfunction	13	13
Partial dispense	15	15
Duplicate transaction	16	16
Reconciliation error	17	17
Suspect transaction	20	00
MAC failure	21	30
KMAC sync error	22	13
Message reply error	23	00
Invalid MAC	24	00
MAC failure	30	30
KMAC sync error	31	31

Processing Codes

Debits

Processing Codes	Description
001000	Purchased Goods and Services from Savings
002000	Purchase Goods and Services from Chequing
021000	Debit Reversal to Savings
022000	Debit Reversal to Chequing
031000	Pre-authorized request [funds guarantee] from Savings
032000	Pre-authorized request [funds guarantee] from Chequing
171000	Completion to Savings
172000	Completion to Chequing
181000	Offline funds authorization to Savings
182000	Offline funds authorization to Chequing

Credits

Processing Codes	Description
200010	Returns/adjustments to Savings
200020	Returns/adjustments to Chequing
220010	Credit Reversal to Savings
220020	Credit Reversal to Chequing

Balance Inquiries

Processing Codes	Description
30100	Available Funds Inquiry—Savings
30200	Available Funds Inquiry—Chequing

Tracing SCD Transactions

Processing Codes	Description
011000	Trace of Withdrawal from Savings
012000	Trace of Withdrawal from Chequing
013000	Trace of Cash Advance from Credit Card

Summary of the values for coding the processing code field

Position	Values
1	Type of Effect on Account
2	Type of Transaction <u>CODE Transaction TYPE</u> 00 Purchase goods and services, or Advice 02 Reversal of credit amount 03 Pre-authorized request 17 Previously authorized 18 Offline authorized 20 Return of goods or services, or Credit 22 Reversal of Debit amount 30 Available Funds inquiry
3 and 4	Type of Account for Debits and Available Funds Inquires
5 and 6	Type of Account for Credits <u>CODE ACCOUNT TYPE</u> 10 Savings account 20 Chequing account

Trace Reason Code (Request)

Trace Reason Code	Description
01	Customer inquiry
02	Customer dispute
03	Fraudulent Transaction
04	Customer denies Transaction was finalized
06	Error, Transaction failed edit check
12	Invalid Transaction. Edit error in Processing Code
13	Invalid Amount
14	Invalid Card Number
23	Unacceptable Transaction Fee
33	Expired Card
36	Restricted Card
39	No Credit Account
41	Lost Card
43	Stolen Card
51	Non-sufficient Funds
52	No Chequing Account
53	No Saving Account
54	Expired Card
55	Incorrect Personal Identification Number
57	Transaction not Permitted to Cardholder
61	Exceeds purchase amount limit
62	Restricted Card. Card temporarily disallowed from interchange Transactions
65	Exceeds purchase frequency limit
77	Invalid Capture Date
78	Amount Error. Exceeds original approval amount
79	Time Expired. Exceeds Authorization time limit
94	Duplicate transmission

Trace Action Code (Response)

Trace Action Code	Description
01	Being investigated
02	Adjustment will follow
03	Transaction is valid
04	Adjustment is applied

Network Management Information Code

Subtype	Description
101	Key change
301	Echo-test
060	Pacing message
061	Send message to Error-Handler
160	Handshake message

NOTE: Only code 101 is used in BM70. The other values are used in the IMN header Message Subtype field.

Interac Reversal Reason Code

The following codes will be used for System Generated Reversals.

Code	Description
01	Time out
02	Destination not available
06	Edit error in response
08	Customer cancellation
13	Original amount incorrect
30	MAC failure [MAC value is incorrect]
31	MAC Key Synchronization error
86	Message Encryption Key Synchronization Error
94	Duplicate transmission
96	System malfunction

The following codes will be used for System Generated Issuer Reversals.

Code	Description
03	Fraudulent Transaction
06	Error, Transaction failed edit check
12	Invalid Processing Code
13	Invalid Amount
14	Invalid Card Number
23	Unacceptable Transaction Fee
36	Restricted Card
39	No Credit Account
40	Requested function not supported
41	Lost Card
43	Stolen Card
51	Non-sufficient Funds
52	No Chequing Account
53	No Savings Account

54	Expired Card
55	Incorrect Personal Identification Number
56	No Card record
57	Transaction not Permitted to Cardholder
58	Transaction not permitted to Terminal
61	Exceeds purchase amount limit
62	Restricted Card. Card temporarily disallowed from interchange Transactions
65	Exceeds purchase frequency limit
77	Invalid Capture Date
78	Amount Error. Exceeds original approval amount
79	Time Expired. Exceeds Authorization time limit
92	MEMBER or intermediate network facility cannot be found for routing
94	Duplicate transmission
96	System malfunction
Q0	Edit error in capture date
R9	Invalid Authorization ID number

Message Class

Message Class	Description
00	All message classes
01	IDP Debit
02	Test IDP Debit
03	SCD
04	Test SCD
05	Credit Transaction
06	Test Credit

IMN Status Code

The following codes are valid for the IMN Status Code in the IMN Header

Code	Description
00	Valid transaction / Normal state
80	Undeliverable request (no reversal required) - see IMN reason codes
81	Timeout (reversal required)
82	Undefined error (reversal required)
83	Null response (not required by architecture)
84	IMN protocol error (on non-ISO message flows)
85	Pacing congested (pacing or handshake messages only) - see IMN reason codes
86	Pacing blocked (pacing or handshake messages only) - see IMN

	reason codes
87	Pacing connection blocked (pacing, or handshake messages only) - see IMN reason codes
88	Pacing connection normal (pacing, or handshake messages only)
91	AIL not available For purposes of statistical reporting only, the following additional "synthetic" IMN status code values will be used. They will not actually appear in the IMN Header passed to the AIL.
68	No context in CIL (C=1 in IMN Header)
69	Damaged & rebuilt from context (D=1 in IMN Header)
89	Undeliverable response (dropped) - see IMN reason codes

IMN Reason Code

The first digit of the Reason Code defines the component that entered the code.
First Digit Source of Code First Digit Source of Code.

Code	Description
0xxx	Originating AIL
1xxx	Originating CIL
2xxx	Originating CCL
3xxx	Responding CCL
4xxx	Responding CIL
5xxx	Responding AIL

The remaining three digits define the reason for the status condition:
Value Transaction Meaning Value Communications Meaning.

Code	Description
x000	No supplementary information needed
x001	Invalid Requester Node ID
x002	Invalid Responder Node ID
x003	Invalid message class
x004	Invalid Requester AIL ID
x005	Invalid Responder AIL ID
x006	Invalid Requester SSID
x007	Invalid security value
x008	Unsupported version
x010	Wrong connection for this message
x011	No handshake in effect
x012	Invalid path (Session, Process, Entry Point ID)
x941	Best path declared congested by remote node (deferrable)
x013	No module available (see Product Reason Code)
x942	Best path declared connection blocked by remote node
x020	Unavailable to source INC node
x021	Invalid message type

x022	Invalid IMN status code
x023	Invalid IMN flags
x024	Invalid IMN reason code
x025	Invalid AIL response code
x026	Invalid message subtype
x070	Path not defined
x071	Path stopped (e.g. no automatic recovery)
x072	Path not started
x073	Path has no active sessions
x140	Pacing - connection normal
x141	Rejected deferrable - congested
x142	Pacing - connection blocked
x150	Pacing - path normal
x151	Pacing - path congested
x152	Pacing - path blocked
X153	Pacing . connection congested
x952	Best path declared blocked by remote

Appendix A: IMN Administrative Adjustment

This appendix provides an overview of the IMN Administrative Adjustment process. For more information on this topic, please refer to the BASE24 IMNA User Guide or IDP/SCD Functional Specifications.

The design of the system relies on the Card Acceptor or Acquirer to inform the Issuer whenever a transaction is not completed, by sending a Reversal message. Any problems between the acquirer and the issuer must be immediately corrected by the system using a System Generated Reversal. If a terminal or terminal controller is not capable of generating a reversal, the acquirer could maintain a terminal "in-flight" file with a copy of outstanding requests, or problems between the acquirer and the Card Acceptor may not be detected until their daily settlement cycle has completed. In this case, the acquirer can correct problems with an Adjustment message via the BASE24 IMN Administrative (IMNA) screen.

When an issuer wants more information from an acquirer concerning a transaction performed by its Cardholder on an acquirer terminal, the issuer should submit a trace request to the acquirer.

The Administrative message is treated as a "must complete" message, and placed on a SAF file before it is forwarded.

A 600 or 602 Administrative message may also be used to submit a trace request. The Acquirer sends a 600 message, and the Issuer sends a 602. A trace response is also sent with a 0600/0602 message. A response could also be in the form of a 0620/0622 Administrative Adjustment request, which affects settlement totals between the Acquirer and the Issuer. A 620 Administrative Adjustment request can be sent by an Acquirer as a debit adjustment of a Transaction, after the Issuer has debited the Acquirer too much because of a settlement error; or as a credit adjustment of a Transaction, after the Acquirer has debited the Issuer too much because of a settlement error. The 622 may be sent by an Issuer as a credit or debit adjustment. A credit or debit adjustment is indicated in BM48, and would normally be the opposite of the action indicated in the processing code field. It is possible for two Member nodes to be out-of-balance in the following situations:

- A reversal cannot be matched by the recipient to the appropriate 200 or 220 Transaction;
- A reversal is refused by the recipient because of an edit error, or removed from the SAF by the originator because of being transmitted more than a threshold number of attempts
- A required reversal is not sent by the switch software or by the Card Acceptor

terminal after an Acquirer software or hardware failure.

Sample IMNA screen

BASE24-SWI		IMN ADMIN MESSAGES		CA60	07/08/31	15:50	01 OF 01
RECORD CREATION: (MM DD YY)		11 16 03	TIME: 16 01 13 41		SEARCH NODE:		
MSG TYPE: 0 (0=TEXT; 1 = TRACE REQUEST; 2 = TRACE RESPONSE; 3 = ADJUST)							
SOURCE NODE: 777762 ACQ NODE 777761 ISS NODE 777762 MSG-CLASS 03							
SCD TRANSACTION (Y/N): Y							
PAN:		TERMINAL LOCAL DATE (MM DD YY): 11 16 03					
TRXN AMOUNT:		.00	TERMINAL LOCAL TIME (HH MM SS): 12 00 00				
PROCESSING CODE:		NO-CODE	TERMINAL ID:				
ORIG SETTLE DATE (MM DD):		11 12	TERMINAL OWNER ID:				
REFERENCE/SYSTEM TRACE NUMBER:			TRXN FEE:		0.00		
ALIAS:		PASSWORD:					
		TRACE ACTION/REASON CODE:					
TEXT: TESTING ADMIN MESSAGE							
DESTINATION IMNI: P1A^IMSC1				RESPONSE CODE: 00			
***** BASE24 *****							
FILE DESTINATION:				NEW LOGICAL NETWORK ID:			
F7-SEND TO NET		SF7-CONFIRM	F12-HELP	F16-EXIT			
RECORD RETRIEVED FROM \ACCIS1.\$CDN60.CA60DATA.IMNADMN							0000

Accessing the IMNA Screen

The IMNA screen can be reached directly from the BASE24 Logon screen by typing IMNA in the File Destination field before logging on to the BASE24 system. To reach the IMNA screen from any other BASE24 screen, type IMNA in the File Destination field and press F16. The IMNM process for the logical network must be running in order for the source id to be set on the screen and for editing of the transaction ('FI-Edit Data' or 'F7-Send To Net') to be successful.

Appendix B: ISO 8583 message format for IDP and SCD

For the ISO 8583 message formats for IDP and SCD, please refer to the IDP Functional Specification section 5.5 and SCD Functional Specification section 4.2.

These sections detail the layouts of the IDP and SCD messages. They also list the data fields associated with each message type. The ISO standard name, description, length, format and associated bit position are provided for each data element.

Appendix C: Token

The table below identifies the various tokens used by the IMNi/IMNsc Interface process along with their token IDs. For more information on these tokens, refer to the "BASE24 Tokens Manual", "BASE24-atm EMV Support Manual" and "BASE24-pos EMV Support Manual".

Token ID/Name	IMNi	IMNsc
B0 - Switch Token	X	X
BC - TSS Index Token	X	X
BO - Encrypted Balance Token	X	
CB - POS Balance Token	X	
CI - POS Merchant Token	X	

Token B0 Switch Token

IDP

Described below are the fields in the IMNi switch token which redefines the switch token (swi-tn).

02 INTERAC-FLAG PIC X.

The following field is used to determine whether the terminal that initiated the transaction was Interac compatible. This field must be set by the Acquirer and is checked by the Acquirer IMNi. If this field is not set to "1" then the acquirer IMNi will reject the request.

Valid values:

- "0" - Non Interac compatible terminal.
- "1" - Interac compatible terminal.

02 FIL-ID PIC X.

The following field indicates the appropriate Source FIL ID. FIL ID is also known as AIL ID. This field is used by the Issuer AIL at the responder node to route a later message to a specific Acquirer AIL.

Valid values are:

- " " (blank) - Initialized Value (originator)
- "0" - Issuer FIL
- "1" - "9" - Acquirer FIL

02 IMN-NODE-ID PIC X(6).

This field will contain the Interac Issuer node ID number (CFI node * ID). This field is filled in by the IMNI and is logged to the PTLF by the Authorization process to generate the Marketing Statistics report. This field is set by the Acquirer IMNi.

02 ISO-COND-CDE PIC X(7).

The following field contains the Interac ISO Condition Code assigned to all transactions to and from Interac through the IMNI process. The IMNI process is responsible for inserting the correct value into inbound PSTMs. The Acquirer is responsible for filling in the correct value in reversals.

02 ISO-RECON-NUM PIC X(2).

The following field contains the Interac ISO Reconcile Slot Number assigned to all transactions to and from Interac through the IMNI process. The IMNI process is responsible for inserting the correct value into inbound PSTMs. The acquirer is responsible for filling in the correct value in reversals.

02 SHORT-RETL-NAM PIC X(15).

The following field contains the Interac ISO Retailer Short Name assigned to all transactions to and from Interac through the IMNI process. The IMNI process is responsible for inserting the correct value into inbound PSTMs. The acquirer is responsible for filling in the correct value in reversals.

SCD

Described below are the fields in the IMNsc switch token which redefines the switch token (swi-tn).

02 FIL-ID PIC X.

The following field indicates the appropriate Source FIL ID. This field is used by the Issuer FIL at the responder node to route a later message to a specific Acquirer FIL.

Valid values are:

- " " (blank) - Initialized Value (originator)
- "0" - Issuer FIL
- "1" - "9" - Acquirer FIL

02 IMN-NODE-ID PIC X(11).

The following field will contain the Interac Issuer node ID number (CFI node ID). This field is filled in by the IMNsc and is logged to the TLF by the Authorization process to generate the Marketing Statistics report. This field is initialized to zeros by the device handler.

Token BC TSS Index Token

The TSS Index token provides indexes of keys used by the Transaction Security Services (TSS) process. Transaction acquirers use this token to indicate that only the specified keys are used to attempt PIN, MAC, or data verification instead of repeating an attempt with more than one key.

This token is used in both IMNi and IMNsc. The fields in the TSS Index token are described below.

02 PIN-KEY-IDX	PIC X.
The index of the PIN encryption key	

02 MAC-KEY-IDX	PIC X.
The index of the MAC encryption key.	

02 DATA-KEY-IDX	PIC X.
The index of the data encryption key.	

02 USER-FLD1	PIC X(3).
This field is not used.	

Token BM Transaction Subtype Token

Transaction Subtypes are four-character values used to indicate that the processing associated with a particular transaction code should be altered. The transaction Subtype token identifies the type of transaction being processed. The fields in the Transaction Subtype token are described below.

02 TXN-SUBTYP PIC X(4).

A subtype identifier to further describe this transaction. All alphanumeric characters are valid in this field. Values P000 through RZZZ are provided for user-defined transaction subtypes. Subtypes in the ranges 0000 through OZZZ and S000 through WZZZ are reserved for use by BASE24 products. Subtypes in the ranges X000 through ZZZZ are reserved for use by distributors. Please refer to BASE24 Base Token manual for more details.

IMNi sets this field to "CI00" for Canadian Idebit transaction.

02 ACQ-PROC-CDE PIC X(6).

The acquirer's external processing code. The first two characters of the processing code indicate the type of transaction, the next two characters specify the from account for the transaction, and the last two characters specify the to account for the transaction.

02 ISS-PROC-CDE PIC X(6).

The issuer's external processing code. The first two characters of the processing code indicate the type of transaction, the next two characters specify the from account for the transaction, and the last two characters specify the to account for the transaction.

02 USER-FLD1 PIC X(20).

This field is not used.

Token BN Data Encrypted Key Token

The fields in the Data Encryption Key token are described below.

02 FLD-TYP PIC X.

The type of data being encrypted. Currently, the only valid value is 1 (balance).

02 USER-FLD PIC X.

This field is not used.

02 ENCRYPT-TYP PIC X(2).

The type of data encryption to be performed. Valid values are as follows:

03 = Electronic Code Block (ECB)

04 = Cipher Block Chaining method 16 byte binary (CBC)

14 = Cipher Block Chaining method 16 byte unpacked hex (CBC)

24 = Cipher Block Chaining method 32 byte character hex (CBC)

02 ENCTYPE-KEY PIC X(16).

The security module encrypted data encryption key. Valid values for each position in this field are 0 through 9 and A through F. This field must contain 16 valid characters.

Token BO Encrypted Balance Token – Binary Format

The Encrypted Balance token can be either binary or ASCII format, depending on the value of the DATA-TYP field included in the message. The fields in the Encrypted Balance token are described below.

02 DATA-TYP PIC X.

The type of encrypted data contained in the following fields.

Valid values are as follows:

B = Binary

U = Unpacked ASCII hexadecimal

This field must set to “B” for IMNi.

02 USER-FLD PIC X.

This field is not used.

02 BAL-1 PIC X(16).

The first encrypted balance. This value can contain the available balance. If only one balance is sent, this field will be populated.

02 BAL-2 PIC X(16).

The second encrypted balance. This value can contain the available balance. If only one balance is sent, this field will be populated. This is not used in IMNi.

Token CB POS Balance Token – Binary Format

The POS Balances token is used by BASE24 on a balance inquiry when the currency of the account is different from the currency used by the POS device. This token also can be added by one of the following methods:

- By the ACI Standard POS Device Handler when the PTS return balances flag is set to the value “Y”.
- By the BASE24-pos Authorization module when the CPF return balances flag is set to the value “1”.
- By the IMNi Interface when balances are returned by the issuer.

02 ACCT-AMT-1 TYPE BINARY 64 SIGNED.
A balance specified in the account currency.

02 TXN-AMT-1 TYPE BINARY 64 SIGNED.
A balance specified in the transaction currency.

02 ACCT-CRNCY-CDE	PIC 9(3)
The ISO numeric currency code of the account currency.	

02 TXN-CRNCY-CDE	PIC 9(3)
The ISO numeric currency code of the transaction currency.	

02 CONV-RATE	PIC 9(8).
The rate used to convert the account balances to transaction currency. This field is in ISO format.	

02 CONV-DAT PIC 9(4).
The date on which the currency conversion was performed, in MMDD format.

02 CPF-RTRN-BAL PIC X(1).
The value of the return balance flag on the CPF.

02 USER-FLD1 PIC X(5).

Token CI POS Merchant Token

The fields in the POS Merchant token are described below.

02 E-COMM-GOODS-IND PIC X.

A code indicating the type of merchandise sold. This code is passed from the acquiring terminal and can be specific to a transaction or merchant. Valid values are as follows:

- D = Digital
- P = Physical goods
- S = Services

02 EXISTING-DEBT-IND PIC X.

A code indicating whether a credit card is used to pay for an existing debt. If a credit card was not used to pay for an existing debt, this field is blank. This field is passed from the acquiring terminal. The only valid value is 9 (Payment on existing debt).

02 DEFERRED-BILLING-IND PIC X.

A code indicating whether a purchase is made with the payment deferred until a later date. This code is passed from the acquiring terminal. Valid value are as follows:

- 0 = Deferred billing is not provided.
- 1 = Deferred billing is used at the point of service.

02 RELN-PARTICIPANT-IND PIC X.

A code indicating whether the merchant or acquirer has a special relationship with the cardholder. This code is passed from the acquiring terminal. Valid values are as follows:

- 0 = Not a relationship participant or relationship not provided.
- 1 = Relationship participant.

02 DPC-NUM PIC X(4).

The DPC number of the accepting entity to which transactions are routed for POS pass through processing. This field is set from the DESTINATION DPC field on BASE24-pos Terminal Data file (PTD). Valid values are 0000 – 9999.

02 PINPAD-ID PIC X(16).

The PIN pad identifier passed from the acquiring terminal. This value is supported in a pass through mode only.

02 ACQ-TERM-ID PIC X(16).

The terminal ID passed in the message from the acquiring terminal.

02 RCNCL-ENT **PIC X.**

A code identifying the entity used to reconcile for POS pass through processing. This field is set from the RECONCILIATION ENTITY field on BASE24-pos Terminal Data File (PTD). Valid values are as follows:

- 0 = No totals
- 1 = Totals by DPC
- 2 = Totals by retailer ID
- 3 = Totals by retailer group
- 4 = Totals by terminal ID
- 5 = Totals by clerk ID

02 ACQ-TERM-POST-DAT **PIC X(6).**

The transaction posting date passed from the acquiring terminal.

02 PRE-AUTH-CHRGBK **PIC X.**

A flag indicating whether online chargebacks are generated for preauthorization hold completion transactions where the preauthorization hold has expired, is not found, or the completion amount is greater than the originally authorized hold amount. This field is set from the PREAUTH CHARGEBACKS field on BASE24-pos Terminal Date file (PTD).

- Y = Yes, chargebacks are generated.
- N = No, chargebacks are not generated (default).

02 ENHANCED-PRE-AUTH **PIC X.**

A flag indicating whether enhanced preauthorization processing is used for the acquiring terminal. Valid values are as follows:

- Y = Yes, use enhanced preauthorization processing.
- N = No, do not use enhanced preauthorization processing (default).

02 USER-FLD1 **PIC X(21).**

Reserved for future use.

Appendix D: Running IMNi and IMNsc on the same Logical Network

Typically, the IMNi and IMNsc are running in the separate logical network, however, for the customers who wish to run the IDP and SCD Interac switch interfaces in the same logical network, BASE24 6.0 IMNi and IMNsc can be configured to serve this purpose without any functionality changes.

In general, each IMNi and IMNsc processes have its own manger process in the same logical network. The IMNi and IMNsc process notify and communicate to its own IMNM by referring the IMNM process ID defined in LNCF assign (IMN-IMNM-SC for IMNsc Manager process ID and IMN-IMNM for IMNi Manager process ID). The IMNi and IMNsc share the same database (i.e. KEYF, ICF, and EPCF) except SAF and ILF files. The IMNi and IMNsc must be configured to have its own exclusive SAF and ILF files. Please note that the record size of SAF file in IMNsc is different to the SAF file in IMNi.

End of document